
TimeMorph: Dual-view LLM-based Few-Shot Time Series Classification

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Abstract

Few-shot time series classification (TSC) is a fundamental yet challenging problem in many real-world domains where labeled sequences are costly to obtain. While pretrained large language models (LLMs) have shown strong potential for data-scarce learning by reusing knowledge acquired from large-scale pretraining, their benefits remain difficult to realize for TSC: time series are continuous numerical observations rather than discrete language tokens, and accurate classification requires sensitivity to both temporal dynamics and waveform morphology. We propose **TimeMorph**, an end-to-end framework that adapts pretrained LLMs for few-shot TSC by grounding time-series evidence in a form suitable for language-model reasoning. TimeMorph first constructs a dual-view representation of each sequence, where a temporal branch captures chronological variations such as trends and local changes, and a morphology branch captures global shape patterns from a two-dimensional waveform view. The fused evidence is then projected into the LLM embedding space through task-aware pretraining, enabling the pretrained model to operate on time-series-derived tokens. For prediction, TimeMorph replaces unconstrained text generation with closed-set class-token scoring over the valid label set, and further incorporates semantic label priors when meaningful class names are available. Experiments on all 128 UCR datasets under full-label-space 1-, 2-, 5-, and 10-shot protocols show that TimeMorph achieves the best average accuracy and average rank among strong task-specific, Transformer-style, and LLM-style baselines. Additional external evaluations and ablation studies further validate the contributions of dual-view modeling, embedding-space alignment, and class-token scoring. These results suggest that pretrained LLMs can serve as effective reusable priors for few-shot TSC when numerical time-series evidence is properly represented, aligned, and decoded.

1 Introduction

Time series classification (TSC) is challenging when labeled sequences are scarce in healthcare monitoring, industrial diagnosis, sensor analytics, and related domains [Ismail Fawaz et al., 2019]. In many practical scenarios, collecting labels requires domain expertise, new categories may appear with only a few examples, and class distributions can be highly imbalanced. Reliable few-shot TSC therefore requires more than fitting a classifier on a small support set. A successful model should capture transferable evidence from both temporal dynamics and waveform morphology, reflecting how humans naturally interpret sequences by tracking numerical evolution while recognizing shape-level patterns.

Discriminative evidence in time series is inherently heterogeneous. Some classes are distinguished by chronological dynamics, such as local trends, amplitudes, phase shifts, and long-range evolution, while others depend more strongly on waveform morphology, such as recurring peaks, valleys, cycles,

motifs, and discriminative subsequences [Tang et al., 2020, Ye and Keogh, 2009b, Dempster et al., 2020, Ismail Fawaz et al., 2020, Wang and Oates, 2015a]. This dual-view observation suggests that effective few-shot TSC models should extract and combine both temporal and morphological features. However, many existing methods emphasize one dominant form of evidence. CNNs and Transformer-based models can achieve strong performance when abundant labeled training data are available, but they may overfit to unstable cues and generalize poorly under few-shot supervision [Ismail Fawaz et al., 2020, Dempster et al., 2020, Yang et al., 2024]. Prototype-based methods are more data-efficient, yet they still do not fully resolve the need to jointly preserve temporal and morphological evidence [Snell et al., 2017, Zhang et al., 2020, Tang et al., 2020]. In particular, models that mainly operate on ordered numerical trends can overlook the intuitive visual structure of the waveform, while purely shape-oriented representations may miss chronological dependencies needed for fine-grained discrimination.

Recently, pretrained large language models (LLMs) have become appealing for few-shot learning because they provide reusable pretrained computation, conditional-generation objectives, and token-level scoring interfaces [Brown et al., 2020, Raffel et al., 2020]. Time-series-specific adaptation work has explored numerical serialization, reprogramming, embedding alignment, and semantic anchors, but many of these methods focus on forecasting, representation learning, or open-ended generation rather than closed-set classification [Gruver et al., 2023, Zhou et al., 2023, Jin et al., 2024, Sun et al., 2023, Chang et al., 2025, Zhang et al., 2024, Jiang et al., 2024, Langer et al., 2025, Tan et al., 2024, Zheng et al., 2024]. Few-shot time-series classification with LLMs has only recently started to emerge [Chen et al., 2025]. These properties make LLMs a promising backbone for few-shot TSC, where supervised labels are limited but transferable computation and flexible decision interfaces are valuable. However, directly applying LLMs to time series remains challenging. Time-series observations are continuous, domain-specific, and structurally different from the discrete token sequences on which language models are pretrained. Moreover, few-shot TSC requires reliable closed-set discrimination over dataset-specific labels.

To make pretrained LLMs effective for dual-view few-shot TSC, three core challenges must be addressed. First, the representation should preserve both chronological dynamics and morphology-oriented cues, since either view may contain the decisive class signal. Second, fused continuous time-series features must be aligned with the pretrained LLM input space before target adaptation. Third, prediction should be restricted to valid class candidates instead of relying on unconstrained text generation, which may produce invalid labels and introduce tokenization or label-length bias [Raffel et al., 2020, Hokamp and Liu, 2017, Koo et al., 2024]. Without such an interface, the language model may become only a loosely connected generative component rather than a reliable few-shot classifier.

We propose **TimeMorph**, a dual-view time-series-to-language framework for few-shot TSC. For each sequence, TimeMorph constructs two complementary views: a temporal branch that models ordered local relations and long-range dynamics, and a morphology branch that transforms the waveform into a two-dimensional morphology map so that shape-level structure can be extracted by a frozen vision encoder. The two views are fused into continuous tokens and projected into a pretrained causal LLM, reducing the mismatch between time-series representations and the language-model embedding space. For downstream classification, TimeMorph replaces unconstrained decoding with supervised class-token scoring over dataset-specific class tokens, turning the generative interface into a discriminative closed-set classifier. When class names carry meaningful semantics, label verbalizers further initialize and regularize the corresponding class-token embeddings without requiring multi-token phrase decoding.

Our contributions are summarized as follows. First, we formulate LLM-based few-shot TSC as an interface-design problem that jointly considers complementary time-series evidence, continuous-to-language alignment, and closed-set prediction. Second, we introduce a dual-view time-series-to-language encoder that preserves chronological and morphology-oriented evidence before transferring continuous tokens into a pretrained causal LLM. Third, we propose supervised class-token scoring over valid labels, and extend it with label-verbalizer initialization and regularization when semantic label names are available. Across 128 UCR datasets, TimeMorph achieves the best average accuracy and average rank under 1-, 2-, 5-, and 10-shot settings, while additional external experiments and ablation studies validate the roles of dual-view modeling, embedding-space alignment, and closed-set class-token scoring.

93 2 Related Work

94 **LLMs and foundation models for time series.** Recent surveys show that foundation-model
95 research for time series has expanded from adapting pretrained language models to building native
96 temporal foundation models Zhang et al. [2024], Jiang et al. [2024], Ye et al. [2024], Shi et al. [2025].
97 Representative adaptation-based methods include TEST Sun et al. [2023], which aligns time-series
98 embeddings with the LLM embedding space through instance-, feature-, and text-prototype-aware
99 contrastive learning; Time-LLM Jin et al. [2024], which reprograms time-series patches into a frozen
100 LLM via text prototypes; and LLM4TS Chang et al. [2025], which uses a two-stage alignment-
101 and-task fine-tuning strategy. OpenTSLM Langer et al. [2025] moves toward time-series language
102 models that treat time series as a native modality through soft-prompt or cross-attention interfaces.
103 At the same time, recent ablation studies caution that LLM components do not automatically improve
104 time-series performance when the modality interface is poorly designed Tan et al. [2024], Zheng et al.
105 [2024]. The closest few-shot classification line is LLMFew Chen et al. [2025], which introduces a
106 patch-wise temporal convolution encoder and LoRA-tuned decoder for few-shot multivariate TSC.
107 In contrast, our work focuses on dual-view temporal evidence, explicit time-series-to-language
108 alignment, and class-token scoring rather than an external classifier head.

109 **Temporal and morphological evidence in TSC.** Classical and deep TSC methods support the
110 view that discriminative evidence can appear at multiple structural levels. Shapelet methods identify
111 local subsequences with high class-discriminative power Ye and Keogh [2009a], Tang et al. [2020],
112 convolutional methods such as InceptionTime and ROCKET exploit local receptive fields and kernel
113 responses at multiple scales Ismail Fawaz et al. [2020], Dempster et al. [2020], and patch-based
114 Transformers preserve local subseries information while enabling longer-context attention Nie et al.
115 [2022]. These methods primarily operate in the chronological-dependency space. In parallel, time-
116 series imaging methods transform one-dimensional signals into two-dimensional structures such
117 as Gramian angular fields and Markov transition fields, enabling vision models to capture global
118 shape and transition patterns Wang and Oates [2015b]. Recent work further shows that visualization
119 can improve LLM reasoning over time series and that frozen vision transformers applied to imaged
120 time series can produce strong classification representations Liu et al. [2025], Roschmann et al.
121 [2025]. These two lines motivate our dual-space hypothesis: few-shot TSC labels may depend on
122 both chronological dependencies and morphological structures.

123 **Generative classification and semantic-prior label spaces.** Casting classification as generation
124 is widely used in text-to-text learning, where labels are generated as target tokens under the same
125 sequence objective as other tasks Raffel et al. [2020]. However, unconstrained decoding is mismatched
126 with closed-set classification because many generated strings are irrelevant to the valid label set.
127 Prior work on constrained decoding motivates restricting generation to valid lexical or structural
128 outputs Hokamp and Liu [2017], Koo et al. [2024]. TimeMorph adopts the same closed-set principle
129 but implements the final decision as supervised class-token scoring: the model learns dataset-
130 specific class-token rows and compares their next-token logits rather than performing unconstrained
131 generation. For datasets with meaningful label names, we further use label verbalizers as semantic
132 priors over the same class-token rows. This keeps the decision aligned with the autoregressive
133 interface while avoiding the length bias, shared-prefix artifacts, and language-prior sensitivity of
134 direct phrase-likelihood classification.

135 3 Method

136 We propose **TimeMorph**, a dual-view time-series-to-language framework that adapts a pretrained
137 causal language model to few-shot time series classification (TSC). The central idea is to use
138 a pretrained LLM for numerical time-series data without serializing the raw sequence into text.
139 Given an input sequence, TimeMorph first extracts complementary temporal and morphology-based
140 evidence, then maps the resulting time-series tokens into the LLM embedding space, and finally
141 performs closed-set classification by scoring dataset-specific class tokens.

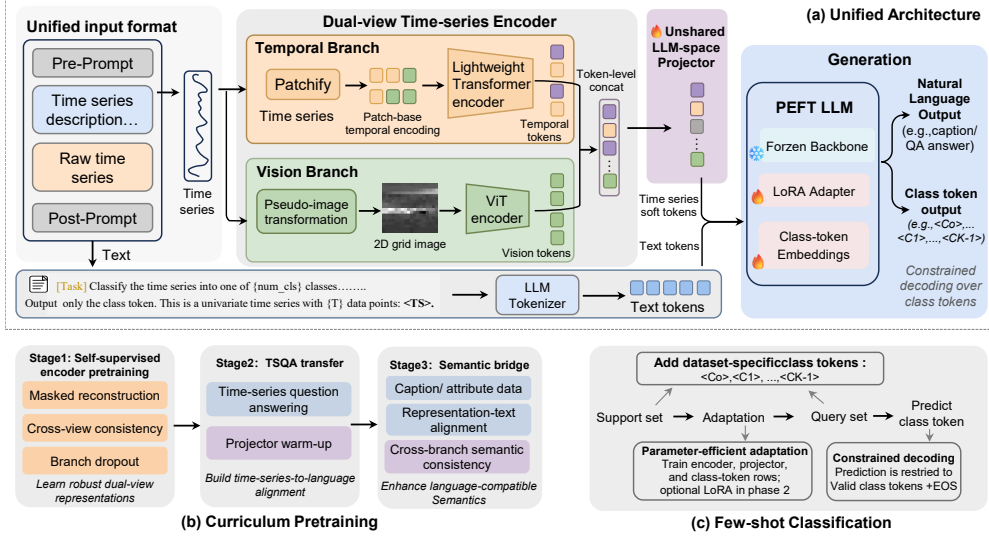


Figure 1: Overview of TimeMorph. The temporal branch extracts ordered patch-level dynamics from the raw one-dimensional sequence, while the morphology branch converts the same sequence into a two-dimensional morphology map and extracts shape-oriented tokens with a frozen vision backbone. The two token sequences are fused and projected into the embedding space of a pretrained causal LLM. Prediction is performed by closed-set class-token scoring over dataset-specific valid labels. The training schedule contains Stage I representation pretraining, Stage II embedding-space alignment pretraining, Stage III optional label-semantic calibration, and target adaptation with warm-up and joint adaptation phases.

3.1 Problem Setup

Univariate time series classification. Let D denote a target classification dataset with label set $\mathcal{Y}_D = \{0, \dots, C-1\}$. Each example consists of a univariate time series $x \in \mathbb{R}^T$ and a class label $y \in \mathcal{Y}_D$. Given a query sequence x , the goal of TSC is to predict its class label from the temporal signal. We focus on closed-set classification over the full label space: all C classes are present during adaptation and evaluation, and prediction is restricted to the valid label set $\mathcal{Y}_D$.

Few-shot learning setup. In the K -shot setting, the support set $\mathcal{S}_D^K$ contains at most K labeled examples per class:

$$\mathcal{S}_D^K = \{(x_i, y_i)\}_{i=1}^{n_D}, \quad |\{i : y_i = c\}| \leq K, \quad c \in \mathcal{Y}_D. \quad (1)$$

The model is trained on $\mathcal{S}_D^K$ and evaluated on query examples from the official test split.

3.2 Framework

Figure 1 gives an overview of TimeMorph. The architecture consists of three components: a dual-view encoder, an embedding-space projection interface, and a closed-set class-token scoring rule. The dual-view encoder G_θ extracts complementary evidence from a temporal view and a morphology view encoder. The projection interface P_ψ maps the fused time-series tokens to the hidden dimension of a pretrained causal LLM. The class-token scoring rule converts the LLM output logits into predictions over the valid labels of the target dataset. These components are learned with a staged training schedule described in Section 3.2.4.

3.2.1 Dual-View Time-Series Encoding

Time series labels can depend on different types of evidence. Some classes are distinguished by chronological dynamics, such as local trends, phase changes, amplitude variations, or long-range evolution. Others are better characterized by waveform morphology, such as peaks, valleys, cycles,

163 motifs, or discriminative subsequences. TimeMorph therefore represents each input sequence through
 164 two complementary views and processes them with two corresponding branches.

165 **Temporal encoder.** The temporal view preserves the original chronological ordering of the raw
 166 sequence. Let x^{pad} be the sequence after right replicate padding. Given patch length ℓ and stride s ,
 167 we extract overlapping temporal patches

$$R_i = x_{(i-1)s+1:(i-1)s+\ell}^{\text{pad}}, \quad i = 1, \dots, N_t, \quad N_t = \left\lfloor \frac{T + p_t - \ell}{s} \right\rfloor + 1, \quad (2)$$

168 where p_t is the padding length needed to form complete patches. Each patch is linearly embedded,
 169 combined with a learnable positional embedding, and passed through a temporal Transformer:

$$Z^t(x) = G_t([W_t R_1 + \pi_1; \dots; W_t R_{N_t} + \pi_{N_t}]). \quad (3)$$

170 The output $Z^t(x)$ is a sequence of temporal tokens that encode ordered patch-level dynamics of the
 171 original waveform.

172 **Morphology encoder.** The morphology view exposes shape-level structure by converting the same
 173 one-dimensional sequence into a two-dimensional morphology map. We first normalize x with robust
 174 per-sequence statistics:

$$\bar{x}_t = \frac{x_t - \text{median}(x)}{\text{IQR}(x) + \epsilon}. \quad (4)$$

175 The normalized sequence is then unfolded into overlapping local windows. Let $q = \max(1, \lfloor \sqrt{T} \rfloor)$
 176 be the window length and $\delta = \max(1, \lfloor \rho q \rfloor)$ be the stride. We construct the morphology map by
 177 normalizing the unfolded matrix, resizing it to the input resolution of the vision backbone, and
 178 replicating it across RGB channels:

$$I(x) = \text{RGB}(\text{Resize}(\Gamma_{\text{img}}(\text{Unfold}_{q,\delta}(\bar{x})))), \quad (5)$$

179 where $\Gamma_{\text{img}}(\cdot)$ denotes per-map normalization and contrast adjustment. Each row of $I(x)$ corresponds
 180 to a local waveform window, so the map preserves local subsequence shapes while making cross-
 181 window morphology visible to a vision encoder. This construction exposes repeated local shapes and
 182 cross-window transitions rather than treating the time series as an arbitrary natural image. A frozen
 183 vision backbone G_m extracts morphology tokens:

$$Z^m(x) = G_m(I(x)). \quad (6)$$

184 We discard the vision class token and use only the patch tokens. The vision backbone remains frozen;
 185 trainable parameters are introduced in the temporal branch, projection modules, class-token rows,
 186 and LLM adapters.

187 **Dual-view fusion.** The temporal and morphology tokens are projected to a common dimension and
 188 concatenated:

$$\tilde{Z}^t(x) = Q_t(Z^t(x)), \quad \tilde{Z}^m(x) = Q_m(Z^m(x)), \quad Z_x = [\tilde{Z}^t(x); \tilde{Z}^m(x)]. \quad (7)$$

189 The fused sequence $Z_x \in \mathbb{R}^{N_x \times d_e}$ serves as the time-series token sequence passed to the LLM
 190 interface.

191 3.2.2 Time-Series Token Projection

192 The dual-view encoder produces continuous time-series tokens, whereas a causal language model
 193 operates on embeddings of discrete text tokens. TimeMorph bridges this modality mismatch with an
 194 input-side projection interface that maps Z_x into the LLM embedding space, avoiding the need to
 195 serialize numerical observations as text.

196 Let P_ψ be a trainable projection module from encoder dimension d_e to the LLM hidden dimension,
 197 $\tau(\cdot)$ be the tokenizer, and E_{lm} be the LLM input embedding matrix. For a prompt $p = (p^{\text{pre}}, p^{\text{post}})$,
 198 the conditioned input embeddings are

$$H(x, p) = [E_{\text{lm}}(\tau(p^{\text{pre}})); P_\psi(Z_x); E_{\text{lm}}(\tau(p^{\text{post}}))]. \quad (8)$$

199 The projected time-series tokens are therefore inserted into the same embedding sequence as text
 200 tokens and can condition the causal LLM through its standard forward pass. When a training or target
 201 setting provides auxiliary textual descriptions, they are included as part of p and encoded by the same
 202 tokenizer. The learning objectives for this interface are introduced together with the staged training
 203 schedule in Section 3.2.4.

204 3.2.3 Closed-Set Class-Token Classification

205 Few-shot TSC requires prediction over a fixed set of valid labels rather than open-ended text genera-
 206 tion. For each target dataset D , TimeMorph assigns a compact set of single-token class identifiers:

$$\mathcal{C}_D = \{\langle \text{cls}_0 \rangle, \dots, \langle \text{cls}_{C-1} \rangle\}, \quad \mathcal{V}_D = \{v_0, \dots, v_{C-1}\}, \quad (9)$$

207 where v_c is the vocabulary index of class token $\langle \text{cls}_c \rangle$. We implement these identifiers as newly
 208 added special tokens, so each class corresponds to exactly one vocabulary index. Let p_D denote the
 209 classification prompt for dataset D . After the LLM conditions on $H(x, p_D)$, let $h_\Theta(x, p_D)$ be the
 210 final hidden state at the last prompt position. With output embedding matrix W_{out} , the score for class
 211 c is the next-token logit assigned to its class token:

$$s_c(x) = [W_{\text{out}} h_\Theta(x, p_D)]_{v_c}. \quad (10)$$

212 Prediction is restricted to the valid class-token set:

$$\hat{y} = \arg \max_{c \in \{0, \dots, C-1\}} s_c(x). \quad (11)$$

213 The supervised objective is the cross entropy over only the valid class tokens:

$$\mathcal{L}_{\text{ct}} = -\log \frac{\exp s_y(x)}{\sum_{c=0}^{C-1} \exp s_c(x)}. \quad (12)$$

214 Thus, TimeMorph keeps the autoregressive vocabulary head but turns the final decision into a one-
 215 forward-pass closed-set comparison over C valid labels. Within the vocabulary parameters, only the
 216 input embeddings and output rows of the new class tokens are trainable; all other vocabulary rows
 217 remain fixed.

218 **Label semantic priors.** For anonymous-label benchmarks, class tokens are learned as dataset-
 219 specific identifiers. When class names are meaningful, TimeMorph uses label verbalizers to initialize
 220 and regularize the same class-token rows. For class c , let $\mathcal{R}_c$ be a set of template-consistent verbalizers
 221 derived from its canonical label name. We compute a semantic prototype by averaging the LLM
 222 embeddings of the verbalizers:

$$p_c = \frac{1}{|\mathcal{R}_c|} \sum_{r \in \mathcal{R}_c} \frac{1}{|\tau(r)|} \sum_{u \in \tau(r)} E_{\text{lm}}(u). \quad (13)$$

223 The prototype is rescaled to match the average norm of vocabulary rows and is used to initialize the
 224 input and output rows of the corresponding class token. During adaptation, we regularize these rows
 225 toward a detached prototype $\tilde{p}_c$:

$$\mathcal{L}_{\text{sem}} = \frac{1}{C} \sum_{c=0}^{C-1} [1 - \cos(e_{v_c}, \tilde{p}_c) + 1 - \cos(w_{v_c}, \tilde{p}_c)], \quad (14)$$

226 where e_{v_c} and w_{v_c} are the input and output rows of class token v_c . The adaptation loss is

$$\mathcal{L}_{\text{adapt}} = \mathcal{L}_{\text{ct}} + \lambda_{\text{sem}} \mathcal{L}_{\text{sem}}. \quad (15)$$

227 For datasets without meaningful label names, we set $\lambda_{\text{sem}} = 0$.

228 3.2.4 Staged Pretraining and Target Adaptation

229 TimeMorph learns the time-series-to-language interface with three pretraining stages followed by
 230 target-dataset adaptation. Stage I learns reusable dual-view time-series representations, Stage II aligns
 231 projected time-series tokens with the LLM embedding space, and Stage III optionally calibrates
 232 the interface with label-related language when semantic supervision is available. The default UCR
 233 benchmark uses the Stage II checkpoint, while the semantic-label external evaluations use the Stage
 234 III calibrated checkpoint.

Stage I: Representation Pretraining. Stage I learns the dual-view encoder without class labels. In the default setting, it uses unlabeled raw series from TSQA-train [Wang et al., 2025], M4-train [Makridakis et al., 2020], and official TRAIN-split sequences from a fixed 98-dataset UCR pretraining pool [Dau et al., 2019]. No UCR TEST sequence or UCR class label is used in this stage. We combine masked temporal patch reconstruction with an invariance–variance–covariance objective over augmented views. For a minibatch $\{x_i\}_{i=1}^B$ and two augmentations $r \in \{1, 2\}$, define pooled representations for branch $b \in \{t, m, f\}$ as

$$H_b^{(r)} = [\text{Pool}_b(Z^b(x_1^{(r)})); \dots; \text{Pool}_b(Z^b(x_B^{(r)}))] \in \mathbb{R}^{B \times d_b}, \quad Z^f(x) = Z_x, \quad (16)$$

where t , m , and f denote the temporal, morphology, and fused branches. For each branch, the regularization loss is

$$\begin{aligned} \mathcal{L}_{\text{ivc}}^b = & \alpha_{\text{inv}} \frac{1}{B} \|H_b^{(1)} - H_b^{(2)}\|_F^2 + \beta_{\text{var}} \sum_{r=1}^2 \sum_j \max(0, 1 - \sigma_j(H_b^{(r)})) \\ & + \gamma_{\text{cov}} \sum_{r=1}^2 \|\text{offdiag}(\text{Cov}(H_b^{(r)}))\|_F^2. \end{aligned} \quad (17)$$

The three terms respectively match representations of two augmented views, prevent dimension-wise collapse, and penalize redundant covariance [Bardes et al., 2022].

For reconstruction, a contiguous segment is masked and the temporal branch is asked to recover the original temporal patches:

$$\mathcal{L}_{\text{rec}} = \frac{1}{2} \sum_{r=1}^2 \|R_\omega(Z^t(\mathcal{M}(x^{(r)}))) - \mathcal{P}_{\ell,s}(x^{(r)})\|_2^2, \quad (18)$$

where $\mathcal{M}$ is the masking operator, $\mathcal{P}_{\ell,s}$ extracts temporal patches using the same patch length and stride as Eq. (2), and R_ω is the reconstruction head. The representation-pretraining objective is

$$\mathcal{L}_{\text{rep}} = \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_t \mathcal{L}_{\text{ivc}}^t + \lambda_m \mathcal{L}_{\text{ivc}}^m + \lambda_f \mathcal{L}_{\text{ivc}}^f. \quad (19)$$

We apply branch dropout during this stage to prevent the fused representation from collapsing onto a single view.

Stage II: Embedding-Space Alignment Pretraining. Stage II attaches the projector P_ψ and the pretrained causal LLM to the Stage-I dual-view encoder. Let $\mathcal{D}_{\text{TSQA}}$ denote the TSQA training split from ChatTime [Wang et al., 2025]. Each example is represented as (x, p, a) , where x is a raw time series, p is a natural-language or template-based question about the series, and a is the textual answer. The projected tokens $P_\psi(Z_x)$ are inserted into the prompt embedding sequence as in Eq. (8). For answer token sequence $u_{1:M} = \tau(a)$, we apply the language-modeling loss only to the answer tokens:

$$\mathcal{L}_{\text{LM}}(a \mid x, p) = - \sum_{m=1}^M \log p_\Theta(u_m \mid H(x, p), E_{\text{lm}}(u_{<m})). \quad (20)$$

The Stage-II objective is

$$\mathcal{L}_{\text{align}} = \mathbb{E}_{(x,p,a) \sim \mathcal{D}_{\text{TSQA}}} [\mathcal{L}_{\text{LM}}(a \mid x, p)]. \quad (21)$$

This stage does not train TimeMorph to classify any target dataset. Instead, it teaches the projection interface to produce continuous tokens that the frozen LLM can condition on under its standard next-token objective. Following the implementation, we first update the projector with the dual-view encoder fixed and then jointly optimize the encoder and projector. The LLM backbone remains frozen throughout Stage II. The resulting checkpoint initializes the default UCR target adaptation.

Stage III: Optional Label-Semantic Calibration. Stage III is used when the target evaluation contains meaningful label names or textual label descriptions. It further calibrates the Stage-II interface toward label-related language using a mixture of TSQA retention examples, M4 captioning examples, and synthetic attribute supervision derived from TSQA and M4 raw series [Wang et al., 2025, Makridakis et al., 2020]. In the implementation, this corresponds to the semantic-bridge and

caption-specialization training after Stage II. The objective uses the same answer-token language-modeling form as Eq. (20), with auxiliary caption or attribute targets such as trends, periodicity, spikes, abrupt changes, and other interpretable temporal properties. Stage III is not used for the anonymous-label UCR benchmark. For semantic-label external datasets such as MIT-BIH, SleepEDF, and CinC2017 [Moody and Mark, 2001, Kemp et al., 2000, Clifford et al., 2017], target adaptation is initialized from the Stage III calibrated checkpoint and uses the label-semantic class-token prior in Eq. (14).

Target-Dataset Adaptation. For each target dataset D , adaptation proceeds in two phases. In the warm-up phase, TimeMorph updates the dual-view encoder, the projection module, and the dataset-specific class-token rows:

$$\Theta_{\text{warm}} = \{\theta, \psi, e_{\mathcal{V}_D}, w_{\mathcal{V}_D}\}, \quad \min_{\Theta_{\text{warm}}} \mathcal{L}_{\text{adapt}}. \quad (22)$$

This phase calibrates the projected time-series tokens and the closed-set decision space before updating any LLM adapters.

In the joint adaptation phase, low-rank adapters are added to the LLM:

$$\Theta_{\text{joint}} = \Theta_{\text{warm}} \cup \{\Delta\Theta_{\text{LoRA}}\}, \quad \Delta W_l = B_l A_l, \quad \text{rank}(\Delta W_l) \leq r. \quad (23)$$

The pretrained LLM weights remain frozen, and only adapter parameters are updated on the LLM side. At inference time, TimeMorph performs a single forward pass for each query sequence and ranks the valid class tokens using Eq. (10). Implementation details and staged-training data sources are provided in Appendix B.

4 Experiments

4.1 Experimental Setup

Protocol. We evaluate on all 128 datasets in the UCR Archive 2018 with a fixed full-label-space few-shot protocol. For each dataset D and $K \in \{1, 2, 5, 10\}$, all classes are included; we sample up to K labeled examples per class from the official training split and evaluate on all examples in the official test split. All methods use the same sampled support sets within each of five independent runs.

Pretraining data and default initialization. The main UCR benchmark uses the Stage II checkpoint: Stage I is trained on raw TSQA-train series, raw M4-train series, and unlabeled official TRAIN-split sequences from a fixed 98-dataset UCR pool; Stage II is trained on the TSQA training split. Stage III is not used for Table 1. No pretraining stage uses any official UCR TEST split or UCR class label. For UCR datasets included in the 98-dataset pool, TimeMorph may have seen unlabeled official TRAIN sequences before target adaptation; after this pretraining, each target run uses only the sampled support labels for supervised adaptation. For the external semantic-label datasets, target adaptation uses the Stage III calibrated checkpoint.

Baselines and metrics. We compare against CNN/prototype baselines (RESNET, TAPNET, COSCO-RESNET), Transformer-style or decomposition-based baselines (PATCHTST, DLINEAR, TIMESNET, AUTOFORMER, CROSSFORMER, FEDFORMER, INFORMER), and the LLM-style GPT4TS baseline. Accuracy is averaged over five runs and then macro-averaged over datasets. Avg averages the four shot settings, and AvgRank averages dataset-level ranks across shots; lower rank is better. Standard deviations and full per-dataset tables are in Appendix A.

Fairness and training details. All methods share the same official TRAIN support pool, TEST query split, and sampled support indices. We do not add a validation split by default; each method follows its runner’s checkpoint-selection rule, and some baselines select checkpoints using support-set performance. Appendix B.8 lists the training recipes and hyperparameters.

Implementation details. TimeMorph uses meta-llama/Llama-3.2-1B, facebook/dinov2-base, temporal patch length 16, stride 8, hidden size 128, and the Stage II checkpoint for UCR. Target adaptation uses 5 warm-up epochs, 55 joint-training epochs, and LoRA with rank 16 and alpha 32. Additional details are in Appendix B.

Table 1: Few-shot classification accuracy (%) on 128 UCR datasets, grouped by model family. We report the mean over five runs. Avg and AvgRank average the four shot-specific mean accuracies and average ranks, respectively. Best is bold and second-best is underlined.

Family	Model	1-shot	2-shot	5-shot	10-shot	Avg	AvgRank
CNN/prototype	ResNet	47.79	53.98	61.27	66.24	57.32	4.89
CNN/prototype	TapNet	45.40	50.79	59.47	64.93	55.15	5.45
CNN/prototype	COSCO-ResNet	<u>47.88</u>	<u>54.95</u>	<u>63.55</u>	<u>67.18</u>	<u>58.39</u>	<u>4.70</u>
Transformer/TS	DLinear	44.21	48.01	52.30	55.43	49.99	7.24
Transformer/TS	PatchTST	42.39	48.43	56.94	62.72	52.62	6.54
Transformer/TS	TimesNet	45.16	49.95	57.86	62.79	53.94	5.90
Transformer/TS	Autoformer	36.78	39.45	43.20	46.04	41.37	9.47
Transformer/TS	Crossformer	44.31	49.45	57.20	61.94	53.23	6.04
Transformer/TS	FEDformer	37.71	41.71	47.30	51.50	44.56	8.68
Transformer/TS	Informer	46.42	51.14	58.35	63.13	54.76	5.59
LLM/foundation	GPT4TS	35.61	38.30	42.94	46.88	40.93	9.26
Ours	ChronoMorph	48.31	55.55	64.14	69.16	59.29	4.24

Table 2: Win/Tie/Loss of ChronoMorph against selected baselines on 128 UCR datasets.

Baseline	1-shot	2-shot	5-shot	10-shot
COSCO-ResNet	68 / 0 / 60	68 / 1 / 59	58 / 0 / 70	64 / 0 / 64
PatchTST	97 / 0 / 31	97 / 0 / 31	99 / 0 / 29	94 / 2 / 32
TimesNet	88 / 0 / 40	91 / 1 / 36	86 / 0 / 42	93 / 1 / 34
GPT4TS	106 / 2 / 20	110 / 0 / 18	114 / 1 / 13	121 / 1 / 6
ResNet	64 / 1 / 63	66 / 1 / 61	67 / 0 / 61	69 / 0 / 59
TapNet	75 / 0 / 53	75 / 3 / 50	76 / 1 / 51	72 / 3 / 53

4.2 Main Results

Benchmark-level comparison. Table 1 shows that TimeMorph obtains the highest mean accuracy under all four K -shot settings: 48.31, 55.55, 64.14, and 69.16 for 1-, 2-, 5-, and 10-shot. The strongest non-TimeMorph baseline is COSCO-RESNET, with 47.88, 54.95, 63.55, and 67.18. Thus the improvement over the strongest specialized CNN/prototype baseline is modest in accuracy, especially in the lowest-shot regimes, but it is consistent across K and is also reflected in the best AvgRank (4.24 vs. 4.70 for COSCO-RESNET). The gap is more substantial against Transformer-style and LLM-style baselines: TimeMorph improves Avg by 4.53 points over INFORMER and 18.36 points over GPT4TS. These results suggest that the proposed interface is most clearly beneficial relative to generic sequence and LLM-style baselines, while remaining competitive with strong specialized CNN/prototype methods.

Dataset-level robustness. Table 2 reports Win/Tie/Loss counts against representative baselines. TimeMorph wins substantially more often than it loses against PATCHTST, TIMESNET, and GPT4TS. Against the strongest CNN/prototype baselines, the picture is closer: TimeMorph is nearly balanced with COSCO-RESNET and RESNET at several shot levels, which is consistent with the modest average-accuracy margins in Table 1.

Additional external datasets. We also evaluate MIT-BIH, SleepEDF, and CinC2017 under full-label-space 10-, 20-, and 30-shot protocols, using 10 runs for each external setting. Table 3 reports the average over these shot settings, with the full table in Appendix B.5. TimeMorph obtains the best average accuracy on MIT-BIH and CinC2017 and the second-best overall external average, but it is clearly behind RESNET and TAPNET on SleepEDF. We therefore treat these experiments as evidence of transfer beyond UCR, not as a claim of uniform dominance across external domains.

Table 3: Average few-shot accuracy (%) on three external datasets, computed over 10-, 20-, and 30-shot settings with 10 runs per setting. Best is bold and second-best is underlined.

Model	MIT-BIH	SleepEDF	CinC2017	Avg.
ChronoMorph	61.04	47.16	34.63	<u>47.61</u>
ResNet	50.23	66.84	<u>33.63</u>	50.23
TapNet	38.73	<u>66.70</u>	31.29	45.57
COSCO-ResNet	36.19	62.74	32.48	43.80
PatchTST	50.46	28.92	29.21	36.20
OneFitsAll	43.42	27.05	29.19	33.22
TimesNet	53.79	23.23	28.36	35.13
DLinear	<u>57.29</u>	19.81	24.87	33.99
Autoformer	41.64	26.48	27.45	31.86
Crossformer	56.76	24.93	25.62	35.77
FEDformer	56.93	21.79	25.05	34.59
Informer	53.25	22.45	28.39	34.70

Table 4: Compact efficiency comparison under the 10-shot UCR protocol. Runtime and memory are measured on the representative UCR runtime subset. Total parameters count the deployed model; updated parameters count downstream-adapted parameters only.

Model	Total params	Updated params	Peak GB	Infer ms/ex.
ChronoMorph	1.28B	12.3M	13.75	156.85
COSCO-ResNet	662.1K	662.1K	0.58	0.25
ResNet	629.1K	629.1K	0.33	0.26
PatchTST	665.4K	598.3K	1.54	0.75
GPT4TS	82.2M	1.1M	0.56	0.86

4.3 Computational Efficiency and Adaptation Cost

Because TimeMorph reuses a pretrained causal LLM, total parameters and adapted parameters must be reported separately. Table 4 shows the main trade-off: TimeMorph updates only 12.3M parameters per target dataset, but the frozen LLM backbone dominates memory and inference cost. TimeMorph is therefore most appropriate when transfer accuracy and reusable adaptation matter more than compact deployment. The complete efficiency table, including throughput and additional baselines, is in Appendix B.7.

4.4 Ablation Studies

Table 5 summarizes the main top-level UCR ablations on all 128 datasets. Each ablation follows the same five-run full-label-space few-shot protocol and modifies only the component under study. Focused compact views with the corresponding drops are provided in Tables 20–22. Because UCR labels are anonymous, the decision ablation tests closed-set class-token scoring rather than semantic label priors. The no-LLM ablation is reported separately in Table 23, with the full per-dataset comparison in Appendix Table 24.

Effect of staged pretraining. Removing staged pretraining drops Avg from 59.29 to 55.94, with shot-level accuracies decreasing from 48.31/55.55/64.14/69.16 to 44.70/51.81/60.65/66.60 for 1-/2-/5-/10-shot, respectively. The focused view in Appendix Table 20 reports a 3.35-point average drop, suggesting that the aggregate Stage I+II initialization helps when target supervision is too limited to learn the representation and LLM interface from scratch. This ablation removes the full pretraining pipeline at once and should not be read as isolating each substage.

Effect of dual-view time-series-to-language modeling. Temporal-only and morphology-only variants reach Avg scores of 54.21 and 55.09, below the full model’s 59.29. Appendix Table 21 shows that removing the morphology branch yields a 5.08-point average drop, while removing the temporal branch yields a 4.20-point drop. The temporal-only variant is weaker at 1-, 2-, and 5-shot, whereas

Table 5: Summary of top-level ablations on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs. Best is bold, and the best ablated variant in each column is underlined. Since UCR labels are anonymous, the label-space ablation isolates closed-set class-token scoring rather than semantic label priors.

Variant	1-shot	2-shot	5-shot	10-shot	Avg
ChronoMorph	48.31	55.55	64.14	69.16	59.29
w/o Curriculum Pretraining	44.70	<u>51.81</u>	60.65	<u>66.60</u>	<u>55.94</u>
Temporal only	43.51	50.55	58.40	64.39	54.21
Visual only	<u>45.88</u>	<u>50.02</u>	<u>60.86</u>	63.58	55.09
w/o Class-Token Label-Space Scoring	44.46	50.44	60.08	65.59	55.14

Table 6: Semantic-prior class-token ablation on external datasets. Values are average accuracy (%) over 10-, 20-, and 30-shot settings with 10 runs per setting. Δ denotes Semantic prior minus Anonymous labels.

Variant	MIT-BIH	SleepEDF	CinC2017	Avg.
Anonymous class tokens	61.04	47.16	34.63	47.61
Semantic-prior class tokens	64.18	50.83	36.27	50.43
Δ	+3.14	+3.67	+1.64	+2.82

the morphology-only variant is weaker at 2- and 10-shot. Neither single view dominates across all shots, which supports the intended complementarity between chronological and morphological evidence.

Effect of class-token scoring. Disabling class-token scoring reduces Avg from 59.29 to 55.14. As detailed in Appendix Table 22, the corresponding shot-level accuracies decrease from 48.31/55.55/64.14/69.16 to 44.46/50.44/60.08/65.59, giving a 4.15-point average drop. This indicates that, under the same adaptation budget, unconstrained generative adaptation is less stable for closed-set few-shot classification than directly scoring the dataset-specific label tokens.

Effect of semantic label priors. For external datasets with meaningful label names, Table 6 compares anonymous class-token rows with rows initialized and regularized from label-verbalizer prototypes, using the Stage III calibrated checkpoint for both variants. Semantic priors improve the 10/20/30-shot average on MIT-BIH, SleepEDF, and CinC2017, increasing the overall average from 47.61 to 50.43. The full appendix table shows that the benefit is not uniform at every shot, with a decrease on CinC2017 20-shot, but the averaged result supports the semantic-prior class-token space as a useful extension of supervised closed-set decoding.

Effect of the LLM backbone. The no-LLM variant replaces the language pathway with a lightweight linear classifier on the same dual-view features. As shown in Table 23, this ablation is evaluated on all 128 UCR datasets under the same five-run protocol. Removing the LLM backbone reduces Avg from 59.29 to 51.46, a 7.83-point drop. The decrease grows with the number of support examples: accuracy changes from 48.31 to 44.89 at 1-shot, 55.55 to 48.29 at 2-shot, 64.14 to 54.22 at 5-shot, and 69.16 to 58.43 at 10-shot. This suggests that, under our implementation and adaptation schedule, the language pathway contributes more than a lightweight classifier on the same features; however, this ablation does not isolate whether the gain comes from pretrained knowledge, additional capacity, or the token-scoring interface.

Overall, the ablations support the central design: staged pretraining improves transferability, dual-view modeling preserves more complete evidence, closed-set class-token scoring stabilizes the final decision, and semantic-prior row initialization and regularization further improve semantic-label transfer.

5 Limitations

TimeMorph improves average accuracy and average rank, but its gains over the strongest CNN/prototype baselines are modest in extremely low-shot settings. The main empirical message is therefore not that LLM-based TSC now universally dominates specialized CNN-style few-shot classifiers, but that a carefully designed time-series-to-language interface can substantially outperform previous Transformer-style and LLM-style baselines while remaining competitive with strong task-specific methods. TimeMorph also has a larger inference footprint than compact CNN baselines because it reuses a pretrained causal LLM, even though only a small fraction of parameters is updated during target adaptation. In our efficiency probe, this makes TimeMorph two to three orders of magnitude slower at inference than compact CNN/prototype baselines. The current UCR pretraining setting also uses unlabeled official TRAIN sequences from a fixed subset of UCR datasets, so it does not establish strict leave-dataset-out transfer for every UCR target dataset. The external semantic-prior results improve the averaged performance on all three semantic-label datasets, but individual shot-level effects still vary, so label semantics should be viewed as a prior over the class-token rows rather than as a replacement for supervised adaptation. TimeMorph may benefit applications where labeled time-series data are scarce, such as physiological monitoring or industrial inspection, but the present study is not a deployment-ready clinical or safety-critical system. Any deployment in such domains would require domain-specific validation, privacy review, and failure analysis beyond the scope of this paper.

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509 A Full UCR Benchmark Results

510 This appendix reports the complete per-dataset few-shot benchmark results for all 128 UCR datasets.
511 Each cell is shown as mean $\pm$ standard deviation over five runs, and the summary rows at the end of
512 each table report average accuracy, average rank, and the number of datasets attaining the best mean
513 accuracy (counting ties).

Table 7: Per-dataset 1-shot classification results on 128 shared UCR datasets. Each value is reported as mean $\pm$ standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
ACSF1	53.80 $\pm$ 4.97	42.00 $\pm$ 4.95	48.40 $\pm$ 7.30	24.00 $\pm$ 4.24	35.60 $\pm$ 3.21	27.67 $\pm$ 7.02	24.33 $\pm$ 9.71	28.33 $\pm$ 2.52	27.80 $\pm$ 7.09	31.67 $\pm$ 9.81	18.60 $\pm$ 4.51	44.00 $\pm$ 5.20
Adiac	33.81 $\pm$ 1.23	12.84 $\pm$ 3.09	33.81 $\pm$ 1.23	8.85 $\pm$ 2.36	11.97 $\pm$ 1.69	21.40 $\pm$ 4.24	34.53 $\pm$ 1.77	8.10 $\pm$ 1.92	40.82 $\pm$ 3.61	24.21 $\pm$ 3.24	3.02 $\pm$ 1.05	45.17 $\pm$ 5.28
AllGestureWiimoteX	25.71 $\pm$ 4.01	26.86 $\pm$ 2.52	24.43 $\pm$ 2.55	15.57 $\pm$ 2.95	21.86 $\pm$ 4.59	20.71 $\pm$ 2.87	11.33 $\pm$ 1.19	16.76 $\pm$ 2.52	13.43 $\pm$ 1.74	18.90 $\pm$ 1.08	21.86 $\pm$ 4.59	25.11 $\pm$ 4.65
AllGestureWiimoteY	26.37 $\pm$ 2.89	26.17 $\pm$ 2.88	22.71 $\pm$ 4.62	17.49 $\pm$ 2.02	25.17 $\pm$ 5.50	21.81 $\pm$ 2.78	12.29 $\pm$ 1.27	20.29 $\pm$ 1.98	13.06 $\pm$ 1.34	21.33 $\pm$ 2.38	25.17 $\pm$ 5.50	25.54 $\pm$ 4.42
AllGestureWiimoteZ	27.11 $\pm$ 1.95	22.66 $\pm$ 4.13	<u>25.49 $\pm$ 2.57</u>	18.26 $\pm$ 2.01	19.71 $\pm$ 2.87	19.24 $\pm$ 1.57	13.29 $\pm$ 0.43	18.76 $\pm$ 2.79	14.14 $\pm$ 1.72	19.71 $\pm$ 1.79	19.71 $\pm$ 2.87	22.26 $\pm$ 3.96
ArrowHead	47.66 $\pm$ 8.94	46.63 $\pm$ 9.45	48.00 $\pm$ 7.74	52.00 $\pm$ 6.38	49.71 $\pm$ 1.40	47.62 $\pm$ 9.46	38.29 $\pm$ 4.00	50.86 $\pm$ 1.71	35.89 $\pm$ 2.60	50.86 $\pm$ 6.05	41.49 $\pm$ 12.34	52.57 $\pm$ 3.43
BME	49.60 $\pm$ 2.85	50.80 $\pm$ 2.13	49.73 $\pm$ 3.42	56.67 $\pm$ 3.43	62.93 $\pm$ 5.35	55.33 $\pm$ 7.86	51.56 $\pm$ 3.15	61.33 $\pm$ 4.06	55.87 $\pm$ 7.71	58.89 $\pm$ 6.19	35.73 $\pm$ 3.22	66.40 $\pm$ 3.88
Beef	36.00 $\pm$ 6.41	31.33 $\pm$ 5.06	27.33 $\pm$ 7.96	38.67 $\pm$ 9.89	32.00 $\pm$ 7.67	40.00 $\pm$ 8.82	34.44 $\pm$ 6.94	41.11 $\pm$ 1.92	30.67 $\pm$ 6.83	40.00 $\pm$ 6.67	31.33 $\pm$ 9.31	34.00 $\pm$ 2.79
BeetleFly	55.00 $\pm$ 24.24	60.00 $\pm$ 25.00	61.00 $\pm$ 24.85	67.00 $\pm$ 4.47	61.00 $\pm$ 9.62	60.00 $\pm$ 13.23	55.00 $\pm$ 5.00	61.67 $\pm$ 11.55	48.00 $\pm$ 2.74	60.00 $\pm$ 13.23	70.00 $\pm$ 11.18	70.00 $\pm$ 11.73
BirdChicken	62.00 $\pm$ 8.37	71.00 $\pm$ 9.62	<u>70.00 $\pm$ 17.68</u>	56.00 $\pm$ 4.18	56.00 $\pm$ 4.18	51.67 $\pm$ 2.89	51.67 $\pm$ 12.58	53.33 $\pm$ 5.77	55.00 $\pm$ 5.00	51.67 $\pm$ 7.64	59.00 $\pm$ 9.62	68.00 $\pm$ 14.83
CBF	81.67 $\pm$ 6.47	87.62 $\pm$ 2.73	71.71 $\pm$ 7.91	54.02 $\pm$ 3.44	66.38 $\pm$ 7.40	55.85 $\pm$ 2.16	33.63 $\pm$ 1.20	58.00 $\pm$ 5.61	36.89 $\pm$ 2.37	54.78 $\pm$ 1.68	35.20 $\pm$ 3.75	72.48 $\pm$ 4.85
Car	37.00 $\pm$ 4.62	39.00 $\pm$ 4.80	41.67 $\pm$ 9.28	38.67 $\pm$ 11.63	27.33 $\pm$ 9.90	42.22 $\pm$ 5.09	24.44 $\pm$ 3.47	43.89 $\pm$ 5.85	27.00 $\pm$ 4.15	40.00 $\pm$ 7.26	30.00 $\pm$ 9.65	55.67 $\pm$ 8.94
Chinatown	66.53 $\pm$ 19.14	63.73 $\pm$ 26.98	62.51 $\pm$ 27.20	67.41 $\pm$ 23.86	67.35 $\pm$ 11.32	77.07 $\pm$ 16.93	81.83 $\pm$ 2.19	77.94 $\pm$ 9.20	77.61 $\pm$ 8.88	75.41 $\pm$ 7.92	56.68 $\pm$ 20.52	93.59 $\pm$ 3.92
ChlorineConcentration	34.61 $\pm$ 2.26	35.41 $\pm$ 4.36	34.04 $\pm$ 2.01	33.26 $\pm$ 2.96	35.54 $\pm$ 3.20	33.64 $\pm$ 3.96	32.92 $\pm$ 4.26	33.42 $\pm$ 5.09	33.78 $\pm$ 2.74	34.11 $\pm$ 3.50	39.93 $\pm$ 15.64	32.61 $\pm$ 3.30
CinCECGTorso	37.20 $\pm$ 9.73	36.03 $\pm$ 6.56	36.51 $\pm$ 8.82	40.06 $\pm$ 3.86	35.72 $\pm$ 7.09	44.86 $\pm$ 6.03	35.36 $\pm$ 5.77	50.53 $\pm$ 4.66	34.97 $\pm$ 4.68	45.68 $\pm$ 5.50	42.49 $\pm$ 6.76	48.74 $\pm$ 6.47
Coffee	78.57 $\pm$ 9.78	92.86 $\pm$ 8.38	84.29 $\pm$ 13.02	72.86 $\pm$ 14.42	74.29 $\pm$ 18.80	63.10 $\pm$ 10.91	65.48 $\pm$ 7.43	71.43 $\pm$ 16.37	76.43 $\pm$ 4.07	64.29 $\pm$ 7.14	51.43 $\pm$ 4.07	75.71 $\pm$ 13.69
Computers	57.52 $\pm$ 7.64	60.40 $\pm$ 6.13	<u>59.12 $\pm$ 4.47</u>	47.12 $\pm$ 5.87	52.64 $\pm$ 10.26	50.93 $\pm$ 1.29	48.80 $\pm$ 0.69	51.87 $\pm$ 2.57	48.96 $\pm$ 2.79	52.27 $\pm$ 3.23	45.20 $\pm$ 5.75	56.40 $\pm$ 5.77
CricketX	26.15 $\pm$ 3.29	26.10 $\pm$ 3.67	20.56 $\pm$ 4.67	17.28 $\pm$ 1.96	18.15 $\pm$ 4.50	20.34 $\pm$ 0.39	8.89 $\pm$ 0.82	17.52 $\pm$ 1.26	11.38 $\pm$ 2.35	20.68 $\pm$ 0.74	7.28 $\pm$ 1.05	26.26 $\pm$ 5.29
CricketY	26.97 $\pm$ 3.03	24.46 $\pm$ 4.25	22.72 $\pm$ 1.74	18.56 $\pm$ 4.13	17.90 $\pm$ 3.61	22.82 $\pm$ 1.56	10.34 $\pm$ 1.89	20.94 $\pm$ 3.72	11.23 $\pm$ 0.80	21.20 $\pm$ 3.98	9.54 $\pm$ 2.33	23.33 $\pm$ 5.19
CricketZ	32.51 $\pm$ 2.32	28.72 $\pm$ 2.15	26.56 $\pm$ 4.19	19.64 $\pm$ 2.73	22.26 $\pm$ 3.91	20.34 $\pm$ 4.11	11.03 $\pm$ 1.54	20.00 $\pm$ 2.31	12.26 $\pm$ 1.25	20.34 $\pm$ 2.57	7.64 $\pm$ 1.05	<u>29.28 $\pm$ 4.82</u>
Crop	25.10 $\pm$ 4.94	28.14 $\pm$ 2.90	25.10 $\pm$ 4.94	10.23 $\pm$ 2.66	17.27 $\pm$ 1.43	27.73 $\pm$ 2.89	24.10 $\pm$ 4.30	26.19 $\pm$ 2.96	27.27 $\pm$ 3.58	30.31 $\pm$ 5.00	3.80 $\pm$ 0.58	36.48 $\pm$ 3.38
DiatomSizeReduction	79.67 $\pm$ 5.51	55.36 $\pm$ 19.14	84.12 $\pm$ 4.82	89.22 $\pm$ 8.91	49.61 $\pm$ 24.86	75.71 $\pm$ 7.74	55.01 $\pm$ 11.77	56.43 $\pm$ 20.40	50.07 $\pm$ 12.66	76.47 $\pm$ 11.70	41.44 $\pm$ 13.25	85.82 $\pm$ 8.55
DistalPhalanxOutlineAgeGroup	43.02 $\pm$ 17.16	43.02 $\pm$ 18.34	43.60 $\pm$ 14.92	42.73 $\pm$ 19.01	46.76 $\pm$ 16.49	45.80 $\pm$ 19.81	46.28 $\pm$ 21.27	56.59 $\pm$ 9.36	50.94 $\pm$ 19.66	46.28 $\pm$ 20.78	35.68 $\pm$ 14.61	42.01 $\pm$ 16.44
DistalPhalanxOutlineCorrect	62.10 $\pm$ 1.32	57.32 $\pm$ 9.22	63.04 $\pm$ 2.44	57.54 $\pm$ 11.13	53.55 $\pm$ 11.63	53.99 $\pm$ 14.78	53.26 $\pm$ 12.27	53.62 $\pm$ 11.97	55.14 $\pm$ 10.52	53.26 $\pm$ 12.97	56.74 $\pm$ 7.30	60.58 $\pm$ 2.98
DistalPhalanxTW	51.22 $\pm$ 5.15	44.03 $\pm$ 11.70	53.38 $\pm$ 4.02	43.74 $\pm$ 5.17	51.22 $\pm$ 6.58	45.32 $\pm$ 9.89	42.69 $\pm$ 4.40	38.13 $\pm$ 7.51	48.20 $\pm$ 5.84	47.48 $\pm$ 5.62	18.56 $\pm$ 14.51	54.53 $\pm$ 1.72
DodgerLoopDay	32.50 $\pm$ 7.34	25.00 $\pm$ 4.05	29.75 $\pm$ 7.68	41.50 $\pm$ 2.05	26.00 $\pm$ 6.58	40.42 $\pm$ 4.73	20.42 $\pm$ 1.91	32.92 $\pm$ 3.15	21.50 $\pm$ 4.45	40.42 $\pm$ 7.94	26.00 $\pm$ 6.58	34.50 $\pm$ 4.38
DodgerLoopGame	52.17 $\pm$ 6.30	55.07 $\pm$ 17.38	49.13 $\pm$ 8.90	60.87 $\pm$ 12.35	59.86 $\pm$ 22.05	51.45 $\pm$ 9.96	51.21 $\pm$ 5.14	56.52 $\pm$ 10.82	50.43 $\pm$ 2.44	51.45 $\pm$ 9.59	59.86 $\pm$ 22.05	56.96 $\pm$ 11.69
DodgerLoopWeekend	84.49 $\pm$ 8.37	74.78 $\pm$ 12.27	83.19 $\pm$ 10.24	93.62 $\pm$ 3.81	96.23 $\pm$ 1.19	95.65 $\pm$ 1.26	63.77 $\pm$ 4.41	94.44 $\pm$ 0.42	68.41 $\pm$ 4.57	93.72 $\pm$ 5.86	96.23 $\pm$ 1.19	85.36 $\pm$ 4.30
ECG200	63.00 $\pm$ 12.61	72.00 $\pm$ 6.75	66.20 $\pm$ 8.32	71.80 $\pm$ 4.44	61.60 $\pm$ 12.82	61.33 $\pm$ 10.97	65.00 $\pm$ 11.79	63.00 $\pm$ 20.66	64.20 $\pm$ 14.29	68.67 $\pm$ 7.09	51.40 $\pm$ 12.93	60.60 $\pm$ 6.91
ECG5000	54.22 $\pm$ 21.00	50.05 $\pm$ 20.59	61.53 $\pm$ 13.66	62.80 $\pm$ 19.19	53.35 $\pm$ 20.14	69.07 $\pm$ 12.52	54.44 $\pm$ 9.79	57.20 $\pm$ 16.09	62.35 $\pm$ 12.99	<u>68.84 $\pm$ 13.99</u>	19.70 $\pm$ 25.30	67.18 $\pm$ 10.57
ECGFiveDays	58.65 $\pm$ 7.69	57.24 $\pm$ 6.51	54.80 $\pm$ 13.00	57.63 $\pm$ 2.63	62.21 $\pm$ 8.35	57.99 $\pm$ 4.84	54.86 $\pm$ 3.17	55.87 $\pm$ 1.56	54.08 $\pm$ 6.73	<u>58.46 $\pm$ 5.08</u>	51.54 $\pm$ 6.38	61.77 $\pm$ 6.61
EOGHorizontalSignal	20.11 $\pm$ 2.01	22.38 $\pm$ 1.86	22.60 $\pm$ 2.77	22.43 $\pm$ 4.01	21.22 $\pm$ 3.15	20.63 $\pm$ 4.63	15.29 $\pm$ 2.63	23.57 $\pm$ 5.45	10.39 $\pm$ 2.26	22.01 $\pm$ 4.37	27.24 $\pm$ 2.99	29.06 $\pm$ 4.16
EOGVerticalSignal	11.27 $\pm$ 3.67	14.09 $\pm$ 3.23	13.26 $\pm$ 3.09	24.20 $\pm$ 3.27	23.76 $\pm$ 3.73	23.39 $\pm$ 2.79	11.23 $\pm$ 0.70	24.86 $\pm$ 4.17	11.27 $\pm$ 3.83	23.11 $\pm$ 4.07	19.06 $\pm$ 6.08	27.35 $\pm$ 1.90
Earthquakes	50.07 $\pm$ 8.55	51.08 $\pm$ 6.33	50.94 $\pm$ 6.46	49.35 $\pm$ 5.05	45.61 $\pm$ 20.94	48.68 $\pm$ 19.16	49.40 $\pm$ 1.10	49.40 $\pm$ 14.03	47.05 $\pm$ 5.90	47.00 $\pm$ 24.65	46.04 $\pm$ 13.71	55.54 $\pm$ 11.57
ElectricDevices	28.83 $\pm$ 9.01	<u>32.82 $\pm$ 12.31</u>	28.35 $\pm$ 12.24	17.59 $\pm$ 1.67	27.21 $\pm$ 7.38	21.93 $\pm$ 4.54	17.11 $\pm$ 1.87	15.12 $\pm$ 2.81	17.04 $\pm$ 2.05	15.61 $\pm$ 7.10	17.06 $\pm$ 3.10	33.14 $\pm$ 10.96
EthanolLevel	24.44 $\pm$ 1.30	<u>25.40 $\pm$ 0.85</u>	25.28 $\pm$ 1.57	24.92 $\pm$ 1.50	23.80 $\pm$ 1.65	26.13 $\pm$ 0.58	26.53 $\pm$ 1.55	<u>26.67 $\pm$ 1.29</u>	25.32 $\pm$ 1.50	25.80 $\pm$ 1.39	24.76 $\pm$ 1.40	26.68 $\pm$ 1.86
FaceAll	42.17 $\pm$ 4.79	32.36 $\pm$ 9.47	42.17 $\pm$ 4.79	28.66 $\pm$ 9.09	19.36 $\pm$ 3.07	29.11 $\pm$ 4.19	33.12 $\pm$ 5.46	25.94 $\pm$ 1.75	34.05 $\pm$ 3.40	32.92 $\pm$ 2.10	8.64 $\pm$ 3.46	32.84 $\pm$ 6.86
FaceFour	68.64 $\pm$ 10.56	47.50 $\pm$ 11.12	56.36 $\pm$ 4.30	64.32 $\pm$ 4.73	54.09 $\pm$ 8.34	70.08 $\pm$ 4.30	52.65 $\pm$ 3.99	67.05 $\pm$ 7.10	51.36 $\pm$ 4.28	73.11 $\pm$ 3.99	42.95 $\pm$ 6.60	54.77 $\pm$ 8.25
FacesUCR	54.83 $\pm$ 5.21	37.63 $\pm$ 5.89	43.65 $\pm$ 3.88	35.69 $\pm$ 7.47	20.81 $\pm$ 1.95	36.24 $\pm$ 3.26	41.98 $\pm$ 2.95	40.05 $\pm$ 2.80	41.87 $\pm$ 3.09	42.63 $\pm$ 2.58	16.92 $\pm$ 7.54	50.37 $\pm$ 2.98
FiftyWords	13.27 $\pm$ 1.89	12.79 $\pm$ 3.45	13.27 $\pm$ 1.89	32.70 $\pm$ 4.33	19.60 $\pm$ 4.90	35.90 $\pm$ 3.31	3.44 $\pm$ 1.59	38.97 $\pm$ 6.44	11.30 $\pm$ 2.86	36.70 $\pm$ 4.41	3.74 $\pm$ 2.40	36.48 $\pm$ 2.66
Fish	49.03 $\pm$ 5.96	21.26 $\pm$ 3.60	<u>45.71 $\pm$ 1.71</u>	36.00 $\pm$ 3.98	23.20 $\pm$ 3.80	32.57 $\pm$ 4.57	16.95 $\pm$ 2.93	30.29 $\pm$ 2.62	21.71 $\pm$ 2.77	31.81 $\pm$ 4.76	20.80 $\pm$ 5.10	34.74 $\pm$ 4.91
FordA	59.23 $\pm$ 9.68	57.68 $\pm$ 10.00	59.83 $\pm$ 8.76	50.73 $\pm$ 1.01	56.70 $\pm$ 8.04	50.30 $\pm$ 0.33	49.57 $\pm$ 1.42	49.72 $\pm$ 0.63	49.68 $\pm$ 1.11	49.92 $\pm$ 0.80	50.61 $\pm$ 1.68	50.71 $\pm$ 0.98
FordB	52.54 $\pm$ 6.44	52.64 $\pm$ 6.88	52.40 $\pm$ 4.53	50.12 $\pm$ 1.83	53.60 $\pm$ 4.09	49.96 $\pm$ 0.94						

Table 7: Per-dataset 1-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
GesturePebbleZ1	49.07 ± 5.03	38.37 ± 6.03	53.26 ± 6.97	46.16 ± 13.08	23.60 ± 6.95	40.50 ± 15.60	20.93 ± 3.54	42.44 ± 7.05	18.95 ± 2.08	43.99 ± 8.75	23.60 ± 6.95	35.81 ± 4.48
GesturePebbleZ2	55.19 ± 11.38	42.91 ± 15.92	57.47 ± 8.89	48.35 ± 13.33	21.01 ± 3.48	47.68 ± 13.93	18.35 ± 4.94	43.88 ± 16.17	16.20 ± 5.17	55.06 ± 14.60	21.01 ± 3.48	33.29 ± 5.85
GunPoint	55.47 ± 4.79	58.13 ± 9.13	55.87 ± 5.32	53.20 ± 9.11	56.40 ± 7.58	64.00 ± 8.08	58.89 ± 10.59	63.33 ± 4.81	58.93 ± 8.45	62.89 ± 3.01	52.40 ± 11.44	58.00 ± 7.79
GunPointAgeSpan	49.94 ± 8.02	48.67 ± 3.53	49.11 ± 3.99	60.82 ± 22.09	58.73 ± 19.72	59.70 ± 21.56	52.11 ± 7.60	59.07 ± 20.81	52.41 ± 6.32	61.92 ± 19.97	48.10 ± 2.61	60.51 ± 10.32
GunPointMaleVersusFemale	63.99 ± 11.97	63.29 ± 11.59	64.30 ± 12.20	54.56 ± 4.47	56.65 ± 3.27	55.91 ± 6.97	50.95 ± 7.17	55.59 ± 9.92	50.19 ± 6.98	56.75 ± 8.76	58.04 ± 11.38	81.01 ± 10.12
GunPointOldVersusYoung	92.38 ± 10.44	89.97 ± 13.74	90.35 ± 13.23	53.90 ± 25.42	53.33 ± 25.91	58.31 ± 30.15	54.81 ± 4.33	57.46 ± 29.14	54.79 ± 8.98	58.10 ± 28.87	78.41 ± 20.62	57.97 ± 10.78
Ham	50.10 ± 9.32	51.24 ± 7.01	47.81 ± 11.28	55.24 ± 11.13	49.33 ± 2.81	48.89 ± 5.25	53.65 ± 3.06	51.43 ± 7.44	55.81 ± 6.79	50.16 ± 4.89	51.81 ± 10.55	52.76 ± 9.30
HandOutlines	46.81 ± 15.76	58.86 ± 12.82	55.84 ± 9.47	70.59 ± 7.20	75.68 ± 8.02	71.17 ± 7.55	51.08 ± 2.34	76.76 ± 10.41	52.81 ± 15.40	74.86 ± 11.22	61.19 ± 14.73	61.73 ± 6.31
Haptics	23.96 ± 5.22	22.66 ± 2.12	22.86 ± 4.94	26.49 ± 6.85	22.66 ± 4.38	25.11 ± 9.19	19.16 ± 1.81	23.16 ± 7.32	23.44 ± 3.46	25.43 ± 9.51	25.65 ± 5.26	25.65 ± 4.99
Herring	49.69 ± 6.39	49.69 ± 10.50	50.94 ± 11.51	48.12 ± 8.44	49.69 ± 7.44	47.40 ± 10.40	51.04 ± 2.39	49.48 ± 9.55	48.75 ± 8.44	48.44 ± 9.50	48.12 ± 10.39	50.00 ± 10.00
HouseTwenty	80.84 ± 7.67	74.62 ± 5.52	68.40 ± 16.13	50.25 ± 3.68	57.65 ± 8.25	68.63 ± 3.50	49.30 ± 7.15	61.62 ± 2.57	46.05 ± 6.81	66.67 ± 1.28	53.61 ± 8.52	62.18 ± 8.04
InlineSkate	16.40 ± 1.60	15.64 ± 0.76	18.00 ± 0.73	15.89 ± 3.32	14.69 ± 2.92	18.06 ± 2.11	15.15 ± 0.73	18.00 ± 2.36	15.02 ± 3.34	18.67 ± 3.56	14.87 ± 3.22	17.13 ± 2.24
InsectEPGRegularTrain	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	43.53 ± 11.83	50.44 ± 13.06	48.19 ± 5.92	42.03 ± 6.13	45.25 ± 3.22	46.27 ± 8.59	49.26 ± 6.69	87.15 ± 20.64	55.34 ± 14.47
InsectEPGSmallTrain	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	47.95 ± 6.69	55.10 ± 5.49	51.14 ± 8.16	39.36 ± 1.75	52.88 ± 6.50	37.67 ± 14.87	52.48 ± 11.47	85.78 ± 19.79	50.76 ± 7.86
InsectWingbeatSound	16.18 ± 3.00	14.23 ± 3.87	14.35 ± 2.82	39.88 ± 4.03	22.79 ± 2.65	43.42 ± 0.26	17.56 ± 4.92	43.16 ± 3.28	21.14 ± 2.99	45.61 ± 5.13	12.52 ± 3.37	25.27 ± 2.20
ItalyPowerDemand	61.83 ± 17.15	62.29 ± 17.84	64.74 ± 16.58	55.32 ± 7.10	60.35 ± 16.10	69.81 ± 17.24	68.35 ± 9.65	78.33 ± 24.61	68.98 ± 15.67	71.62 ± 21.13	47.31 ± 2.37	58.52 ± 15.27
LargeKitchenAppliances	48.48 ± 12.88	52.85 ± 11.74	46.72 ± 15.63	33.39 ± 3.38	54.88 ± 8.28	40.09 ± 11.55	34.93 ± 5.70	32.53 ± 3.23	31.47 ± 3.69	41.69 ± 12.56	33.81 ± 2.87	45.44 ± 5.60
Lightning2	50.49 ± 6.80	55.08 ± 10.15	51.80 ± 4.27	56.72 ± 11.39	51.80 ± 12.52	58.47 ± 11.16	48.63 ± 3.41	57.38 ± 11.36	51.48 ± 5.26	59.02 ± 8.67	55.08 ± 13.25	59.67 ± 8.87
Lightning7	44.93 ± 4.48	44.38 ± 4.60	40.82 ± 7.47	43.01 ± 4.18	42.19 ± 7.34	53.42 ± 10.70	15.07 ± 6.28	45.66 ± 6.76	18.63 ± 4.40	56.16 ± 8.33	24.11 ± 7.72	41.64 ± 7.60
Mallat	64.03 ± 14.52	50.42 ± 9.84	64.49 ± 9.43	80.37 ± 4.76	56.96 ± 10.09	82.32 ± 3.73	54.67 ± 2.15	75.96 ± 4.82	64.55 ± 4.53	75.72 ± 2.60	37.30 ± 5.46	68.24 ± 7.54
Meat	42.00 ± 9.96	44.00 ± 10.18	58.33 ± 10.99	77.67 ± 17.26	52.67 ± 11.94	46.11 ± 9.18	38.33 ± 8.66	33.33 ± 0.00	36.00 ± 2.53	66.11 ± 20.97	33.67 ± 0.75	55.67 ± 9.62
MedicalImages	27.92 ± 5.84	28.45 ± 4.75	24.95 ± 5.95	26.34 ± 8.67	24.16 ± 4.26	22.32 ± 5.35	26.84 ± 12.59	23.86 ± 6.12	28.53 ± 8.23	23.16 ± 6.10	4.21 ± 2.27	27.66 ± 4.41
MelbournePedestrian	51.29 ± 5.58	44.27 ± 4.81	47.47 ± 6.50	9.85 ± 0.23	31.99 ± 4.95	21.74 ± 7.37	46.44 ± 8.26	40.62 ± 4.65	43.40 ± 7.11	43.73 ± 5.33	31.99 ± 4.95	39.16 ± 5.22
MiddlePhalanxOutlineAgeGroup	38.83 ± 7.74	37.53 ± 10.57	38.05 ± 7.83	34.03 ± 9.03	35.97 ± 8.21	37.45 ± 11.78	29.87 ± 4.55	38.53 ± 15.64	33.90 ± 7.76	32.90 ± 7.86	41.69 ± 18.94	39.48 ± 7.40
MiddlePhalanxOutlineCorrect	53.68 ± 13.63	52.10 ± 14.21	54.43 ± 13.33	53.75 ± 17.89	56.49 ± 9.58	58.42 ± 17.26	55.56 ± 15.16	54.07 ± 23.81	54.50 ± 10.22	57.50 ± 21.09	52.85 ± 12.08	51.68 ± 5.00
MiddlePhalanxTW	34.03 ± 16.27	44.68 ± 7.76	32.47 ± 17.97	30.52 ± 10.75	32.34 ± 10.78	29.00 ± 15.96	38.96 ± 3.37	46.10 ± 3.62	35.45 ± 13.57	42.21 ± 5.07	15.19 ± 11.64	37.14 ± 16.84
MixedShapesRegularTrain	45.36 ± 5.29	40.59 ± 8.80	43.58 ± 10.39	47.60 ± 10.74	39.59 ± 5.62	43.05 ± 15.67	17.31 ± 1.15	45.53 ± 11.76	25.71 ± 3.77	42.76 ± 16.81	45.03 ± 12.47	56.06 ± 5.34
MixedShapesSmallTrain	52.78 ± 8.02	39.55 ± 6.53	48.84 ± 9.40	53.90 ± 7.66	45.57 ± 7.91	52.81 ± 6.78	17.32 ± 0.61	53.84 ± 4.25	24.17 ± 3.24	53.31 ± 5.69	48.73 ± 8.44	63.30 ± 3.34
MoteStrain	62.99 ± 13.96	60.67 ± 12.58	64.35 ± 11.70	63.40 ± 15.50	63.27 ± 13.76	68.45 ± 11.94	56.26 ± 12.55	63.82 ± 16.98	61.65 ± 9.57	68.66 ± 11.89	57.99 ± 9.98	54.36 ± 12.77
NonInvasiveFetalECGThorax1	19.05 ± 4.42	8.04 ± 2.52	19.05 ± 4.42	38.23 ± 1.81	12.07 ± 7.41	36.76 ± 3.30	12.57 ± 2.38	28.62 ± 3.97	14.44 ± 2.07	42.58 ± 4.62	5.58 ± 2.48	39.16 ± 2.28
NonInvasiveFetalECGThorax2	20.69 ± 5.30	9.71 ± 2.65	20.69 ± 5.30	47.48 ± 2.72	14.81 ± 5.33	46.67 ± 0.18	20.97 ± 3.22	34.59 ± 7.78	23.69 ± 1.73	49.89 ± 4.96	8.30 ± 1.72	47.48 ± 2.06
OSULeaf	45.62 ± 7.45	30.41 ± 8.35	38.35 ± 12.00	24.05 ± 2.14	22.15 ± 5.63	27.82 ± 3.13	17.77 ± 1.80	26.03 ± 1.89	20.00 ± 5.09	26.17 ± 1.56	17.60 ± 2.44	29.83 ± 5.82
OliveOil	28.00 ± 23.64	26.67 ± 11.06	47.33 ± 13.21	65.33 ± 15.20	36.00 ± 22.66	32.22 ± 16.44	44.44 ± 10.72	28.89 ± 11.71	32.00 ± 9.60	61.11 ± 15.40	27.33 ± 11.88	28.00 ± 14.26
PLAID	15.46 ± 6.16	13.97 ± 4.45	16.95 ± 4.17	17.77 ± 3.34	21.79 ± 0.97	21.54 ± 5.77	13.78 ± 3.08	19.93 ± 5.41	14.41 ± 2.63	20.30 ± 4.75	21.79 ± 0.97	32.10 ± 5.72
PhalangesOutlinesCorrect	54.90 ± 6.98	51.28 ± 10.93	54.41 ± 7.38	53.40 ± 8.33	52.54 ± 10.38	48.87 ± 4.48	51.01 ± 10.34	53.38 ± 8.94	52.73 ± 9.70	52.49 ± 9.30	54.83 ± 6.32	51.93 ± 9.54
Phoneme	10.43 ± 1.42	8.46 ± 1.00	10.43 ± 1.42	4.21 ± 0.54	5.61 ± 2.38	6.14 ± 1.81	2.39 ± 1.01	5.87 ± 1.54	3.20 ± 2.06	4.98 ± 3.20	3.42 ± 0.41	6.87 ± 1.15
PickupGestureWiimoteZ	35.60 ± 7.80	44.80 ± 4.82	42.00 ± 5.10	40.80 ± 5.40	36.00 ± 3.16	46.67 ± 9.87	17.33 ± 5.03	39.33 ± 6.43	20.40 ± 4.98	48.67 ± 8.33	36.00 ± 3.16	47.20 ± 7.29
PigAirwayPressure	23.94 ± 3.76	9.13 ± 2.66	38.65 ± 2.92	3.65 ± 1.47	3.75 ± 1.33	5.29 ± 0.83	4.01 ± 0.73	4.01 ± 1.11	3.27 ± 0.92	4.81 ± 0.96	6.15 ± 1.64	5.48 ± 2.19
PigArtPressure	40.58 ± 5.14	10.10 ± 2.36	60.10 ± 2.28	9.90 ± 2.42	2.21 ± 0.80	10.58 ± 2.10	8.01 ± 2.00	10.58 ± 1.73	4.81 ± 0.76	9.94 ± 2.27	15.58 ± 3.05	15.38 ± 4.07
PigCVP	22.50 ± 3.47	14.33 ± 2.79	50.29 ± 2.90	5.00 ± 1.25	3.27 ± 0.71	5.93 ± 0.73	3.53 ± 1.39	6.89 ± 0.56	2.02 ± 0.22	4.33 ± 0.83	8.94 ± 1.94	11.44 ± 2.08
Plane	98.86 ± 2.56	91.62 ± 6.26	95.43 ± 3.65	87.24 ± 8.67	85.14 ± 9.49	82.86 ± 4.15	78.73 ± 5.25	88.57 ± 9.09	86.48 ± 4.39	88.57 ± 6.67	26.10 ± 13.02	93.90 ± 6.92
PowerCons	76.89 ± 9.68	80.22 ± 9.41	78.11 ± 13.88	60.22 ± 11.82	61.00 ± 7.74	62.96 ± 3.70	49.44 ± 7.47	58.33 ± 2.78	51.33 ± 4.56	62.78 ± 6.41	55.22 ± 21.33	60.89 ± 11.09
ProximalPhalanxOutlineAgeGroup	48.49 ± 34.51	48.20 ± 31.13	48.68 ± 30.82	39.61 ± 19.82	39.12 ± 18.60	43.74 ± 32.89	49.59 ± 25.02	60.81 ± 40.85	38.15 ± 22.87	46.50 ± 27.14	32.68 ± 16.57	48.98 ± 23.27
ProximalPhalanxOutlineCorrect	54.43 ± 13.05	53.40 ± 15.85	51.20 ± 16.57	56.22 ± 5.47	53.26 ± 10.68	44.67 ± 13.09	46.96 ± 12.63	58.76 ± 6.63	52.78 ± 12.27	43.99 ± 6.30	57.59 ± 13.02	52.23 ± 12.03
ProximalPhalanxTW	54.63 ± 10.31	57.56 ± 9.33	58.15 ± 6.64	54.34 ± 11.16	49.76 ± 11.67	60.81 ± 11.71	55.93 ± 9.23	38.37 ± 0.75	55.80 ± 7.24	57.07 ± 10.70	17.37 ± 16.23	55.22 ± 6.63
RefrigerationDevices	36.05 ± 6.49	35.68 ± 5.16	37.44 ± 5.37	34.99 ± 2.77	33.97 ± 4.76	41.16 ± 5.10	34.58 ± 1.47	34.93 ± 0.92	34.72 ± 0.99	38.84 ± 5.04	32.75 ± 2.24	29.76 ± 11.02
Rock	29.20 ± 8.07	34.40 ± 7.92	28.40 ± 6.54	41.60 ± 16.52	26.40 ± 8.53	38.67 ± 25.32	57.33 ± 5.03	44.67 ± 18.90	36.80 ± 8.07	40.00 ± 24.25	36.40 ± 14.93	39.20 ± 13.75
ScreenType	33.92 ± 4.73	34.19 ± 7.39	34.29 ± 6.61	33.87 ± 2.72	33.87 ± 4.61	32.44 ± 5.27	32.18 ± 3.99	28.00 ± 3.40	30.99 ± 3.01	33.60 ± 4.19	30.61 ± 2.66	32.53 ± 4.84
SemgHandGenderCh2	50.87 ± 2.35	55.37 ± 5.38	53.17 ± 4.35	50.17 ± 11.45	61.00 ± 15.57	42.78 ± 13.47	52.78 ± 1.84	39.89 ± 8.47	44.67 ± 12.14	35.61 ± 1.06	55.73 ± 4.74	51.90 ± 14.92
SemgHandMovementCh2	28.89 ± 4.22	29.87 ± 4.86	29.60 ± 5.24	25.29 ± 1.13	24.89 ± 3.57	20.37 ± 0.13	21.70 ± 2.31	22.52 ± 2.76	18.71 ± 1.91	20.07 ± 3.46	27.02 ± 5.10	28.13 ± 4.95

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Table 7: Per-dataset 1-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
SemgHandSubjectCh2	28.89 $\pm$ 5.77	29.07 $\pm$ 4.60	28.00 $\pm$ 2.65	35.60 $\pm$ 4.19	29.91 $\pm$ 10.77	26.37 $\pm$ 1.89	23.11 $\pm$ 2.89	26.67 $\pm$ 4.87	21.56 $\pm$ 1.94	26.96 $\pm$ 3.68	27.91 $\pm$ 3.11	37.42 $\pm$ 8.82
ShakeGestureWiimoteZ	63.20 $\pm$ 7.69	58.00 $\pm$ 11.83	67.60 $\pm$ 11.61	40.40 $\pm$ 11.26	43.20 $\pm$ 9.55	45.33 $\pm$ 3.06	16.00 $\pm$ 7.21	40.00 $\pm$ 3.46	24.40 $\pm$ 4.98	46.00 $\pm$ 6.00	43.20 $\pm$ 9.55	60.40 $\pm$ 7.92
ShapeletSim	66.33 $\pm$ 12.25	87.00 $\pm$ 16.91	73.89 $\pm$ 23.85	50.11 $\pm$ 2.27	75.22 $\pm$ 8.30	53.89 $\pm$ 1.67	52.96 $\pm$ 1.40	48.89 $\pm$ 5.64	50.44 $\pm$ 1.82	51.11 $\pm$ 0.96	47.78 $\pm$ 3.07	62.44 $\pm$ 7.44
ShapesAll	23.63 $\pm$ 4.60	15.60 $\pm$ 1.98	23.63 $\pm$ 4.60	45.70 $\pm$ 2.59	19.27 $\pm$ 4.58	44.72 $\pm$ 2.58	6.00 $\pm$ 0.58	43.94 $\pm$ 2.14	8.07 $\pm$ 3.90	44.06 $\pm$ 1.11	8.13 $\pm$ 2.75	50.20 $\pm$ 3.03
SmallKitchenAppliances	40.37 $\pm$ 15.51	46.24 $\pm$ 15.42	44.32 $\pm$ 15.50	42.19 $\pm$ 3.16	38.13 $\pm$ 22.42	32.98 $\pm$ 5.23	32.80 $\pm$ 3.67	35.82 $\pm$ 4.27	35.41 $\pm$ 3.13	36.89 $\pm$ 7.74	38.88 $\pm$ 7.80	42.88 $\pm$ 15.27
SmoothSubspace	74.93 $\pm$ 5.59	74.40 $\pm$ 5.13	<u>74.80 $\pm$ 8.48</u>	31.73 $\pm$ 3.96	50.00 $\pm$ 10.52	46.67 $\pm$ 1.76	53.33 $\pm$ 9.45	50.22 $\pm$ 10.96	62.27 $\pm$ 6.56	60.67 $\pm$ 10.35	56.67 $\pm$ 5.19	68.53 $\pm$ 8.95
SonyAIBORobotSurface1	77.57 $\pm$ 10.30	80.80 $\pm$ 8.99	<u>80.37 $\pm$ 5.89</u>	59.80 $\pm$ 6.73	64.53 $\pm$ 18.83	52.86 $\pm$ 10.38	56.96 $\pm$ 5.82	55.35 $\pm$ 9.37	56.07 $\pm$ 4.43	57.68 $\pm$ 9.32	55.61 $\pm$ 9.09	57.20 $\pm$ 8.62
SonyAIBORobotSurface2	80.29 $\pm$ 6.43	84.09 $\pm$ 5.18	<u>80.90 $\pm$ 6.99</u>	67.39 $\pm$ 19.08	73.03 $\pm$ 13.78	66.18 $\pm$ 11.61	60.37 $\pm$ 27.05	61.35 $\pm$ 27.10	69.49 $\pm$ 15.13	68.07 $\pm$ 16.06	57.31 $\pm$ 15.03	66.09 $\pm$ 10.78
StarLightCurves	63.75 $\pm$ 3.45	62.30 $\pm$ 9.88	<u>57.59 $\pm$ 9.08</u>	57.39 $\pm$ 13.40	57.55 $\pm$ 15.28	67.54 $\pm$ 17.11	37.78 $\pm$ 3.53	57.92 $\pm$ 15.55	27.99 $\pm$ 0.00	67.36 $\pm$ 17.40	64.81 $\pm$ 14.61	63.03 $\pm$ 14.23
Strawberry	52.27 $\pm$ 4.26	51.84 $\pm$ 3.86	54.43 $\pm$ 4.33	54.86 $\pm$ 4.34	53.62 $\pm$ 3.45	<u>58.29 $\pm$ 0.95</u>	54.86 $\pm$ 13.21	60.18 $\pm$ 0.68	56.11 $\pm$ 9.26	<u>57.93 $\pm$ 0.83</u>	45.89 $\pm$ 11.15	51.46 $\pm$ 5.21
SwedishLeaf	65.95 $\pm$ 6.01	46.08 $\pm$ 6.36	65.95 $\pm$ 6.01	27.42 $\pm$ 3.44	40.83 $\pm$ 4.40	36.69 $\pm$ 1.99	27.20 $\pm$ 2.42	24.53 $\pm$ 4.08	31.65 $\pm$ 2.48	38.19 $\pm$ 0.81	8.10 $\pm$ 1.23	<u>57.98 $\pm$ 2.72</u>
Symbols	85.77 $\pm$ 4.83	70.17 $\pm$ 4.91	<u>76.68 $\pm$ 7.53</u>	79.60 $\pm$ 7.65	72.50 $\pm$ 6.12	81.54 $\pm$ 3.92	37.15 $\pm$ 0.97	82.18 $\pm$ 0.42	46.13 $\pm$ 9.06	80.57 $\pm$ 0.95	51.54 $\pm$ 6.72	80.94 $\pm$ 7.54
SyntheticControl	72.53 $\pm$ 6.61	76.40 $\pm$ 5.24	<u>68.67 $\pm$ 10.59</u>	38.47 $\pm$ 6.28	80.33 $\pm$ 9.12	47.78 $\pm$ 5.87	44.89 $\pm$ 3.20	<u>48.22 $\pm$ 0.38</u>	42.13 $\pm$ 2.87	50.00 $\pm$ 4.48	24.53 $\pm$ 7.68	84.20 $\pm$ 6.54
ToeSegmentation1	69.82 $\pm$ 7.39	69.65 $\pm$ 4.27	65.09 $\pm$ 8.05	51.67 $\pm$ 1.47	<u>55.79 $\pm$ 6.85</u>	49.27 $\pm$ 1.10	49.71 $\pm$ 3.29	46.64 $\pm$ 4.73	52.46 $\pm$ 2.82	48.39 $\pm$ 3.23	49.47 $\pm$ 2.47	58.16 $\pm$ 3.70
ToeSegmentation2	68.92 $\pm$ 8.63	70.31 $\pm$ 3.38	<u>70.92 $\pm$ 10.76</u>	50.31 $\pm$ 5.37	65.54 $\pm$ 9.84	50.26 $\pm$ 2.91	35.90 $\pm$ 10.81	50.51 $\pm$ 4.24	33.23 $\pm$ 16.47	52.05 $\pm$ 6.54	41.54 $\pm$ 14.95	75.23 $\pm$ 12.13
Trace	61.00 $\pm$ 4.64	65.20 $\pm$ 9.09	<u>65.80 $\pm$ 6.61</u>	49.60 $\pm$ 3.65	<u>74.40 $\pm$ 6.15</u>	51.33 $\pm$ 5.51	42.33 $\pm$ 4.51	47.67 $\pm$ 0.58	44.80 $\pm$ 7.69	51.33 $\pm$ 4.93	37.80 $\pm$ 6.42	99.20 $\pm$ 1.79
TwoLeadECG	69.48 $\pm$ 12.74	64.97 $\pm$ 10.26	<u>66.51 $\pm$ 13.23</u>	54.03 $\pm$ 2.33	<u>57.82 $\pm$ 5.04</u>	59.32 $\pm$ 3.05	53.32 $\pm$ 7.30	52.62 $\pm$ 3.32	54.47 $\pm$ 5.41	60.32 $\pm$ 1.06	50.73 $\pm$ 1.02	64.88 $\pm$ 11.28
TwoPatterns	48.77 $\pm$ 2.41	61.51 $\pm$ 6.85	<u>46.17 $\pm$ 11.65</u>	31.86 $\pm$ 3.24	36.86 $\pm$ 6.10	29.76 $\pm$ 2.56	25.52 $\pm$ 1.93	28.23 $\pm$ 0.74	26.06 $\pm$ 1.27	29.87 $\pm$ 1.68	26.32 $\pm$ 2.53	45.74 $\pm$ 2.64
UMD	43.06 $\pm$ 4.02	54.58 $\pm$ 9.53	45.56 $\pm$ 6.14	46.81 $\pm$ 7.70	50.14 $\pm$ 6.28	46.30 $\pm$ 5.02	45.60 $\pm$ 3.21	47.69 $\pm$ 4.19	48.61 $\pm$ 2.55	47.45 $\pm$ 6.56	31.39 $\pm$ 4.35	45.47 $\pm$ 8.73
UWaveGestureLibraryAll	29.36 $\pm$ 4.20	24.46 $\pm$ 5.19	28.34 $\pm$ 3.82	66.77 $\pm$ 5.89	41.63 $\pm$ 5.40	<u>69.19 $\pm$ 5.01</u>	25.78 $\pm$ 1.68	67.23 $\pm$ 2.91	26.67 $\pm$ 3.53	69.21 $\pm$ 6.24	50.70 $\pm$ 5.42	33.37 $\pm$ 0.26
UWaveGestureLibraryX	26.45 $\pm$ 4.19	26.60 $\pm$ 3.44	24.71 $\pm$ 1.83	47.27 $\pm$ 6.01	35.36 $\pm$ 5.04	49.30 $\pm$ 7.17	22.37 $\pm$ 0.14	47.22 $\pm$ 8.02	24.34 $\pm$ 5.30	49.17 $\pm$ 7.61	17.28 $\pm$ 8.20	30.33 $\pm$ 0.23
UWaveGestureLibraryY	25.37 $\pm$ 4.63	26.13 $\pm$ 4.45	24.62 $\pm$ 4.93	43.95 $\pm$ 4.51	38.35 $\pm$ 7.26	47.05 $\pm$ 4.02	24.58 $\pm$ 1.89	46.37 $\pm$ 4.93	26.81 $\pm$ 2.91	47.18 $\pm$ 5.20	20.37 $\pm$ 4.71	40.36 $\pm$ 0.25
UWaveGestureLibraryZ	24.77 $\pm$ 4.74	27.03 $\pm$ 3.18	24.45 $\pm$ 3.15	42.90 $\pm$ 1.96	36.37 $\pm$ 3.72	<u>44.61 $\pm$ 1.78</u>	22.47 $\pm$ 2.82	44.70 $\pm$ 2.64	25.10 $\pm$ 2.80	<u>44.75 $\pm$ 1.63</u>	22.77 $\pm$ 3.47	47.44 $\pm$ 2.61
Wafer	50.39 $\pm$ 4.56	44.94 $\pm$ 10.31	49.91 $\pm$ 13.57	67.87 $\pm$ 4.81	51.56 $\pm$ 17.59	63.90 $\pm$ 3.73	64.15 $\pm$ 3.76	63.99 $\pm$ 9.60	63.82 $\pm$ 9.82	<u>65.48 $\pm$ 1.03</u>	43.02 $\pm$ 26.21	46.27 $\pm$ 7.11
Wine	57.41 $\pm$ 14.70	54.07 $\pm$ 4.61	<u>54.81 $\pm$ 11.31</u>	52.96 $\pm$ 8.34	50.74 $\pm$ 6.23	48.77 $\pm$ 3.85	50.00 $\pm$ 0.00	50.00 $\pm$ 0.00	52.22 $\pm$ 4.97	<u>37.65 $\pm$ 19.80</u>	51.48 $\pm$ 3.31	51.11 $\pm$ 7.48
WordSynonyms	14.92 $\pm$ 3.03	13.70 $\pm$ 5.79	<u>14.92 $\pm$ 3.03</u>	19.78 $\pm$ 2.70	16.96 $\pm$ 4.46	19.75 $\pm$ 1.90	6.79 $\pm$ 1.46	21.16 $\pm$ 1.50	10.06 $\pm$ 3.54	20.53 $\pm$ 2.41	4.98 $\pm$ 1.34	<u>20.91 $\pm$ 2.73</u>
Worms	36.10 $\pm$ 9.87	33.25 $\pm$ 6.66	<u>35.06 $\pm$ 11.43</u>	17.14 $\pm$ 4.04	23.90 $\pm$ 9.39	19.91 $\pm$ 6.54	23.38 $\pm$ 4.50	18.61 $\pm$ 5.86	17.66 $\pm$ 7.03	19.05 $\pm$ 5.25	17.66 $\pm$ 2.36	24.68 $\pm$ 8.95
WormsTwoClass	44.79 $\pm$ 3.48	49.09 $\pm$ 6.96	<u>52.99 $\pm$ 6.71</u>	49.09 $\pm$ 3.72	50.39 $\pm$ 3.23	49.35 $\pm$ 3.44	48.92 $\pm$ 10.50	52.38 $\pm$ 1.98	53.51 $\pm$ 4.81	53.68 $\pm$ 3.75	52.99 $\pm$ 2.82	47.53 $\pm$ 10.69
Yoga	55.60 $\pm$ 4.57	50.68 $\pm$ 5.18	<u>53.54 $\pm$ 4.28</u>	49.99 $\pm$ 1.44	49.75 $\pm$ 4.01	49.48 $\pm$ 0.91	51.11 $\pm$ 3.62	49.62 $\pm$ 0.72	51.12 $\pm$ 3.30	48.62 $\pm$ 0.52	49.35 $\pm$ 2.85	50.42 $\pm$ 4.51
Avg. Acc.	47.79	45.40	<u>47.88</u>	44.21	42.39	45.16	36.78	44.31	37.71	46.42	35.61	48.31
Avg. Rank	<u>5.11</u>	5.80	5.19	6.45	6.92	6.04	9.11	6.17	8.51	5.30	8.82	4.58
#Best	<u>25</u>	16	14	7	5	11	2	11	2	15	4	26

Table 8: Per-dataset 2-shot classification results on 128 shared UCR datasets. Each value is reported as mean $\pm$ standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
ACSF1	61.60 $\pm$ 7.86	47.20 $\pm$ 6.30	59.20 $\pm$ 5.26	25.80 $\pm$ 4.38	42.00 $\pm$ 3.16	40.33 $\pm$ 2.31	30.67 $\pm$ 2.52	44.00 $\pm$ 6.24	35.80 $\pm$ 3.19	40.67 $\pm$ 5.69	24.80 $\pm$ 4.66	54.33 $\pm$ 4.16
Adiac	35.45 $\pm$ 2.60	14.27 $\pm$ 3.74	35.45 $\pm$ 2.60	18.87 $\pm$ 2.86	17.80 $\pm$ 5.08	27.79 $\pm$ 2.14	44.67 $\pm$ 0.39	16.62 $\pm$ 1.77	55.24 $\pm$ 4.67	32.91 $\pm$ 3.49	3.48 $\pm$ 1.27	56.01 $\pm$ 1.25
AllGestureWiimoteX	32.43 $\pm$ 2.94	27.69 $\pm$ 3.48	32.09 $\pm$ 3.58	19.46 $\pm$ 2.47	23.77 $\pm$ 3.63	24.43 $\pm$ 4.71	11.86 $\pm$ 1.00	22.67 $\pm$ 2.29	14.09 $\pm$ 1.28	21.10 $\pm$ 5.46	23.77 $\pm$ 3.63	26.51 $\pm$ 4.36
AllGestureWiimoteY	35.11 $\pm$ 5.43	28.57 $\pm$ 4.31	28.91 $\pm$ 5.81	20.54 $\pm$ 2.69	33.31 $\pm$ 2.65	24.33 $\pm$ 2.50	12.00 $\pm$ 1.38	21.90 $\pm$ 1.33	16.14 $\pm$ 1.67	22.57 $\pm$ 1.00	33.31 $\pm$ 2.65	29.43 $\pm$ 1.69
AllGestureWiimoteZ	29.80 $\pm$ 2.53	26.94 $\pm$ 2.84	29.31 $\pm$ 4.56	16.77 $\pm$ 2.02	22.97 $\pm$ 5.35	19.86 $\pm$ 4.12	12.76 $\pm$ 0.36	19.57 $\pm$ 2.48	14.97 $\pm$ 0.98	20.29 $\pm$ 3.48	22.97 $\pm$ 5.35	27.91 $\pm$ 5.21
ArrowHead	50.17 $\pm$ 5.13	44.80 $\pm$ 3.82	45.14 $\pm$ 5.25	57.03 $\pm$ 4.92	49.14 $\pm$ 13.02	53.14 $\pm$ 6.02	34.10 $\pm$ 6.27	50.67 $\pm$ 2.01	42.29 $\pm$ 4.92	55.43 $\pm$ 6.44	43.77 $\pm$ 8.76	57.03 $\pm$ 2.81
BME	48.40 $\pm$ 4.91	55.33 $\pm$ 7.02	54.40 $\pm$ 8.12	50.80 $\pm$ 5.11	67.33 $\pm$ 7.90	48.89 $\pm$ 2.14	64.00 $\pm$ 5.21	58.89 $\pm$ 3.91	67.33 $\pm$ 7.44	54.89 $\pm$ 9.20	50.80 $\pm$ 8.05	79.47 $\pm$ 11.40
Beef	42.00 $\pm$ 2.98	36.67 $\pm$ 8.50	36.70 $\pm$ 9.60	49.33 $\pm$ 7.23	39.33 $\pm$ 6.41	48.89 $\pm$ 5.09	40.00 $\pm$ 3.33	34.44 $\pm$ 5.09	42.67 $\pm$ 4.94	51.11 $\pm$ 7.70	32.67 $\pm$ 7.23	43.33 $\pm$ 5.27
BeetleFly	50.00 $\pm$ 9.35	67.00 $\pm$ 18.23	55.00 $\pm$ 12.75	52.00 $\pm$ 11.51	57.00 $\pm$ 16.43	45.00 $\pm$ 10.00	48.33 $\pm$ 12.58	38.33 $\pm$ 7.64	45.00 $\pm$ 9.35	56.67 $\pm$ 7.64	57.00 $\pm$ 4.47	75.00 $\pm$ 18.03
BirdChicken	71.00 $\pm$ 6.52	<u>72.00 $\pm$ 4.47</u>	76.00 $\pm$ 7.42	52.00 $\pm$ 14.40	65.00 $\pm$ 10.61	45.00 $\pm$ 8.66	51.67 $\pm$ 7.64	46.67 $\pm$ 5.77	53.00 $\pm$ 2.74	50.00 $\pm$ 13.23	49.00 $\pm$ 2.24	69.00 $\pm$ 22.19
CBF	85.76 $\pm$ 5.83	90.62 $\pm$ 7.16	82.18 $\pm$ 10.67	64.78 $\pm$ 6.67	82.69 $\pm$ 10.89	68.07 $\pm$ 6.00	34.89 $\pm$ 2.41	73.78 $\pm$ 6.88	36.78 $\pm$ 1.75	70.00 $\pm$ 6.65	42.84 $\pm$ 7.09	85.91 $\pm$ 4.23
Car	42.00 $\pm$ 15.43	41.67 $\pm$ 13.49	52.67 $\pm$ 7.87	55.33 $\pm$ 10.63	43.00 $\pm$ 11.45	54.44 $\pm$ 6.74	32.78 $\pm$ 0.96	52.78 $\pm$ 10.84	37.00 $\pm$ 7.76	55.56 $\pm$ 4.19	31.00 $\pm$ 10.71	63.33 $\pm$ 7.45
Chinatown	95.74 $\pm$ 1.26	96.73 $\pm$ 1.19	96.73 $\pm$ 0.81	54.52 $\pm$ 24.75	91.08 $\pm$ 5.39	85.33 $\pm$ 8.50	86.10 $\pm$ 2.71	84.55 $\pm$ 3.36	80.70 $\pm$ 5.41	82.80 $\pm$ 4.37	45.01 $\pm$ 24.65	91.84 $\pm$ 4.85
ChlorineConcentration	34.61 $\pm$ 4.95	35.24 $\pm$ 5.79	35.19 $\pm$ 5.79	34.54 $\pm$ 2.93	34.28 $\pm$ 2.22	31.22 $\pm$ 6.21	33.24 $\pm$ 2.95	33.84 $\pm$ 2.69	33.91 $\pm$ 2.07	33.69 $\pm$ 2.59	39.08 $\pm$ 9.47	32.47 $\pm$ 3.03
CinCECGTorso	46.52 $\pm$ 12.75	48.35 $\pm$ 9.58	51.06 $\pm$ 10.06	39.77 $\pm$ 4.46	<u>54.91 $\pm$ 12.73</u>	48.14 $\pm$ 2.63	36.18 $\pm$ 3.15	53.94 $\pm$ 2.51	50.30 $\pm$ 12.26	48.89 $\pm$ 2.81	49.00 $\pm$ 5.40	62.95 $\pm$ 6.66
Coffee	94.29 $\pm$ 7.41	88.57 $\pm$ 12.73	94.29 $\pm$ 3.19	92.14 $\pm$ 3.91	90.71 $\pm$ 10.89	73.81 $\pm$ 8.99	63.10 $\pm$ 7.43	85.71 $\pm$ 3.57	84.29 $\pm$ 10.29	77.38 $\pm$ 4.12	65.71 $\pm$ 15.07	88.57 $\pm$ 4.66
Computers	52.24 $\pm$ 10.15	54.40 $\pm$ 10.81	52.72 $\pm$ 12.14	48.08 $\pm$ 7.70	53.28 $\pm$ 11.25	48.00 $\pm$ 10.51	48.67 $\pm$ 4.60	48.00 $\pm$ 11.45	50.40 $\pm$ 1.72	46.67 $\pm$ 11.22	50.16 $\pm$ 7.53	57.73 $\pm$ 17.44
CricketX	36.51 $\pm$ 3.27	34.10 $\pm$ 3.98	33.03 $\pm$ 3.86	20.62 $\pm$ 2.07	30.56 $\pm$ 3.87	27.35 $\pm$ 2.06	8.12 $\pm$ 1.41	26.15 $\pm$ 1.85	12.15 $\pm$ 1.24	27.44 $\pm$ 2.96	11.54 $\pm$ 1.86	35.98 $\pm$ 3.13
CricketY	37.54 $\pm$ 5.48	30.62 $\pm$ 2.69	33.28 $\pm$ 3.86	22.56 $\pm$ 4.44	24.21 $\pm$ 2.69	26.50 $\pm$ 7.27	9.06 $\pm$ 0.39	25.90 $\pm$ 6.29	12.67 $\pm$ 0.94	27.52 $\pm$ 6.80	10.00 $\pm$ 3.24	30.43 $\pm$ 5.23
CricketZ	40.41 $\pm$ 4.15	<u>35.49 $\pm$ 2.72</u>	33.69 $\pm$ 5.21	20.36 $\pm$ 1.13	24.92 $\pm$ 3.09	26.75 $\pm$ 2.72	8.12 $\pm$ 1.50	24.44 $\pm$ 0.78	12.97 $\pm$ 2.47	26.50 $\pm$ 2.74	11.18 $\pm$ 2.42	33.33 $\pm$ 3.02
Crop	31.31 $\pm$ 3.87	37.82 $\pm$ 2.09	31.31 $\pm$ 3.87	14.67 $\pm$ 0.98	24.66 $\pm$ 2.54	36.33 $\pm$ 2.63	35.06 $\pm$ 3.15	37.89 $\pm$ 1.28	35.83 $\pm$ 2.39	38.66 $\pm$ 2.99	4.63 $\pm$ 0.75	43.94 $\pm$ 2.11
DiatomSizeReduction	78.43 $\pm$ 9.98	84.44 $\pm$ 4.38	89.74 $\pm$ 8.40	83.73 $\pm$ 5.41	63.01 $\pm$ 13.25	84.42 $\pm$ 6.89	59.91 $\pm$ 2.78	84.97 $\pm$ 5.38	59.28 $\pm$ 18.44	82.35 $\pm$ 7.47	35.10 $\pm$ 12.79	90.39 $\pm$ 5.62
DistalPhalanxOutlineAgeGroup	66.91 $\pm$ 6.08	67.77 $\pm$ 3.31	65.76 $\pm$ 6.53	55.54 $\pm$ 12.17	58.42 $\pm$ 11.15	<u>57.07 $\pm$ 19.57</u>	50.84 $\pm$ 14.56	56.12 $\pm$ 13.32	56.26 $\pm$ 12.92	57.07 $\pm$ 14.47	38.99 $\pm$ 8.80	67.77 $\pm$ 3.39
DistalPhalanxOutlineCorrect	52.32 $\pm$ 12.15	51.59 $\pm$ 11.49	52.54 $\pm$ 12.44	49.57 $\pm$ 11.83	51.45 $\pm$ 15.36	57.49 $\pm$ 13.09	56.76 $\pm$ 10.57	55.80 $\pm$ 11.93	51.38 $\pm$ 12.34	56.64 $\pm$ 12.97	50.65 $\pm$ 9.31	55.80 $\pm$ 14.95
DistalPhalanxTW	49.50 $\pm$ 8.83	55.54 $\pm$ 4.21	54.10 $\pm$ 4.92	39.28 $\pm$ 8.61	53.96 $\pm$ 7.56	41.97 $\pm$ 11.56	42.45 $\pm$ 10.74	39.81 $\pm$ 18.49	47.05 $\pm$ 8.79	45.80 $\pm$ 16.17	22.45 $\pm$ 8.85	51.37 $\pm$ 8.97
DodgerLoopDay	28.75 $\pm$ 2.34	23.75 $\pm$ 2.50	<u>28.00 $\pm$ 5.70</u>	44.00 $\pm$ 4.09	38.50 $\pm$ 7.42	39.58 $\pm$ 4.02	23.75 $\pm$ 2.50	42.50 $\pm$ 7.60	19.75 $\pm$ 4.09	42.50 $\pm$ 3.31	38.50 $\pm$ 7.42	35.50 $\pm$ 8.78
DodgerLoopGame	54.93 $\pm$ 6.45	63.04 $\pm$ 8.02	58.41 $\pm$ 11.09	53.62 $\pm$ 8.51	65.65 $\pm$ 18.08	67.87 $\pm$ 7.26	48.55 $\pm$ 4.53	71.50 $\pm$ 0.84	50.14 $\pm$ 3.81	<u>68.36 $\pm$ 3.42</u>	65.65 $\pm$ 18.08	61.01 $\pm$ 13.03
DodgerLoopWeekend	92.75 $\pm$ 2.40	87.68 $\pm$ 5.59	92.03 $\pm$ 3.69	94.64 $\pm$ 2.74	96.09 $\pm$ 1.41	96.38 $\pm$ 1.26	64.98 $\pm$ 4.71	97.34 $\pm$ 0.42	65.22 $\pm$ 8.45	96.14 $\pm$ 0.84	96.09 $\pm$ 1.41	84.64 $\pm$ 2.73
ECG200	69.40 $\pm$ 7.77	75.20 $\pm$ 7.05	71.80 $\pm$ 5.54	76.80 $\pm$ 3.56	71.80 $\pm$ 4.38	78.33 $\pm$ 1.53	74.33 $\pm$ 5.69	75.67 $\pm$ 3.21	73.60 $\pm$ 3.65	75.33 $\pm$ 0.58	52.20 $\pm$ 18.20	73.00 $\pm$ 3.46
ECG5000	61.90 $\pm$ 13.09	69.05 $\pm$ 11.38	61.86 $\pm$ 10.65	74.23 $\pm$ 10.96	58.61 $\pm$ 14.82	72.42 $\pm$ 4.03	65.61 $\pm$ 11.76	69.40 $\pm$ 4.84	67.93 $\pm$ 11.62	66.36 $\pm$ 9.81	30.14 $\pm$ 19.40	67.68 $\pm$ 9.76
ECGFiveDays	68.78 $\pm$ 9.46	56.79 $\pm$ 4.91	71.73 $\pm$ 8.76	57.12 $\pm$ 2.23	67.29 $\pm$ 3.58	66.55 $\pm$ 6.20	<u>74.64 $\pm$ 13.99</u>	75.18 $\pm$ 2.37	68.76 $\pm$ 10.24	68.52 $\pm$ 6.54	57.79 $\pm$ 7.24	69.12 $\pm$ 9.82
EOGHorizontalSignal	20.66 $\pm$ 1.70	23.43 $\pm$ 5.48	21.38 $\pm$ 4.60	25.86 $\pm$ 4.54	28.07 $\pm$ 2.90	30.66 $\pm$ 2.91	12.06 $\pm$ 1.96	31.68 $\pm$ 2.61	9.56 $\pm$ 1.50	29.01 $\pm$ 2.49	29.83 $\pm$ 3.23	37.35 $\pm$ 4.07
EOGVerticalSignal	13.37 $\pm$ 1.32	16.57 $\pm$ 3.25	14.86 $\pm$ 2.77	26.80 $\pm$ 1.98	29.56 $\pm$ 4.11	26.98 $\pm$ 4.28	11.88 $\pm$ 1.54	26.34 $\pm$ 4.16	10.83 $\pm$ 1.45	26.61 $\pm$ 6.09	20.88 $\pm$ 2.94	31.82 $\pm$ 3.47
Earthquakes	52.95 $\pm$ 7.17	57.84 $\pm$ 11.11	49.35 $\pm$ 10.38	53.24 $\pm$ 4.93	63.88 $\pm$ 6.42	63.31 $\pm$ 10.45	49.16 $\pm$ 1.10	54.92 $\pm$ 17.17	56.55 $\pm$ 8.34	60.43 $\pm$ 21.88	57.12 $\pm$ 15.34	55.25 $\pm$ 14.33
ElectricDevices	37.63 $\pm$ 9.44	41.80 $\pm$ 2.08	34.70 $\pm$ 5.95	21.27 $\pm$ 5.73	37.69 $\pm$ 10.00	<u>28.63 $\pm$ 3.21</u>	22.56 $\pm$ 2.12	20.63 $\pm$ 2.87	22.76 $\pm$ 4.51	29.88 $\pm$ 3.14	14.36 $\pm$ 4.97	47.14 $\pm$ 3.99
EthanolLevel	25.32 $\pm$ 0.64	<u>25.52 $\pm$ 1.56</u>	25.12 $\pm$ 2.09	25.76 $\pm$ 1.71	26.96 $\pm$ 0.82	26.87 $\pm$ 2.61	29.20 $\pm$ 1.91	25.67 $\pm$ 3.18	25.24 $\pm$ 0.98	26.47 $\pm$ 2.08	25.12 $\pm$ 0.81	<u>28.00 $\pm$ 3.11</u>
FaceAll	55.98 $\pm$ 6.19	45.22 $\pm$ 8.18	55.98 $\pm$ 6.19	40.27 $\pm$ 5.76	24.12 $\pm$ 4.85	38.56 $\pm$ 3.11	38.84 $\pm$ 3.81	36.71 $\pm$ 2.55	38.89 $\pm$ 3.64	41.40 $\pm$ 3.79	11.35 $\pm$ 5.01	39.09 $\pm$ 1.28
FaceFour	77.73 $\pm$ 4.99	52.05 $\pm$ 10.37	81.82 $\pm$ 7.14	74.09 $\pm$ 5.98	59.32 $\pm$ 5.76	79.55 $\pm$ 9.02	57.58 $\pm$ 7.31	80.68 $\pm$ 3.41	62.50 $\pm$ 7.14	84.85 $\pm$ 2.37	40.45 $\pm$ 14.16	68.55 $\pm$ 4.91
FacesUCR	67.20 $\pm$ 6.42	51.90 $\pm$ 6.85	<u>62.04 $\pm$ 8.47</u>	41.67 $\pm$ 3.36	25.42 $\pm$ 6.93	44.55 $\pm$ 2.52	45.95 $\pm$ 4.27	49.07 $\pm$ 5.45	48.56 $\pm$ 3.79	49.27 $\pm$ 6.47	10.96 $\pm$ 5.26	55.08 $\pm$ 2.32
FiftyWords	19.03 $\pm$ 1.51	13.49 $\pm$ 1.95	19.03 $\pm$ 1.51	40.09 $\pm$ 3.22	28.35 $\pm$ 3.05	44.69 $\pm$ 2.97	5.05 $\pm$ 0.44	46.74 $\pm$ 2.43	16.53 $\pm$ 1.75	44.18 $\pm$ 3.70	15.16 $\pm$ 3.75	47.30 $\pm$ 2.89
Fish	50.97 $\pm$ 10.77	26.97 $\pm$ 9.88	<u>50.51 $\pm$ 16.65</u>	50.17 $\pm$ 6.36	31.54 $\pm$ 5.18	45.90 $\pm$ 8.88	20.00 $\pm$ 4.12	39.05 $\pm$ 5.64	22.97 $\pm$ 3.62	45.71 $\pm$ 10.47	22.86 $\pm$ 2.83	49.91 $\pm$ 10.25
FordA	64.98 $\pm$ 8.59	62.27 $\pm$ 9.82	<u>62.82 $\pm$ 9.33</u>	49.92 $\pm$ 1.84	60.48 $\pm$ 7.29	50.30 $\pm$ 1.31	50.66 $\pm$ 0.90	50.15 $\pm$ 0.79	50.27 $\pm$ 1.01	50.58 $\pm$ 1.43	50.76 $\pm$ 2.11	53.35 $\pm$ 2.84
FordB	61.19 $\pm$ 3.67	58.10 $\pm$ 1.40	<u>58.67 $\pm$ 2.63</u>	49.09 $\pm$ 1.35	56.89 $\pm$ 2.99							

Table 8: Per-dataset 2-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
GesturePebbleZ1	62.44 ± 5.24	41.40 ± 5.40	64.88 ± 2.42	53.26 ± 4.18	29.53 ± 10.46	58.91 ± 3.41	16.86 ± 3.54	57.17 ± 0.89	24.19 ± 4.00	61.82 ± 3.41	29.53 ± 10.46	37.91 ± 6.35
GesturePebbleZ2	66.58 ± 5.17	44.05 ± 14.70	61.70 ± 6.13	51.90 ± 9.19	35.32 ± 9.94	55.06 ± 7.46	15.82 ± 0.63	46.84 ± 8.79	20.76 ± 3.15	56.75 ± 4.12	35.32 ± 9.94	41.39 ± 4.24
GunPoint	67.07 ± 8.15	65.20 ± 5.95	66.00 ± 5.89	60.13 ± 7.74	65.47 ± 7.68	55.33 ± 6.57	64.44 ± 5.18	58.44 ± 9.46	68.13 ± 5.80	57.11 ± 7.95	54.00 ± 6.53	79.33 ± 16.25
GunPointAgeSpan	65.38 ± 2.37	63.73 ± 4.38	68.61 ± 1.37	67.41 ± 13.19	61.08 ± 13.41	61.08 ± 8.21	52.22 ± 8.74	55.49 ± 10.60	52.41 ± 8.13	58.76 ± 12.22	47.41 ± 14.13	64.49 ± 7.90
GunPointMaleVersusFemale	79.30 ± 6.81	84.62 ± 8.74	81.65 ± 8.32	57.15 ± 7.06	65.95 ± 9.57	60.65 ± 4.86	50.11 ± 5.57	60.76 ± 10.27	52.59 ± 5.35	65.82 ± 3.12	65.00 ± 5.79	87.66 ± 5.66
GunPointOldVersusYoung	91.94 ± 11.58	89.97 ± 13.74	91.37 ± 12.08	72.89 ± 4.20	73.14 ± 5.24	73.54 ± 7.07	56.51 ± 2.29	72.49 ± 6.74	59.30 ± 3.02	75.87 ± 9.07	78.03 ± 17.26	70.92 ± 5.05
Ham	45.71 ± 9.69	50.86 ± 8.18	48.00 ± 10.27	54.86 ± 11.40	53.33 ± 7.28	54.92 ± 1.98	58.73 ± 5.25	50.48 ± 13.83	55.24 ± 4.62	57.78 ± 9.77	52.57 ± 10.93	50.86 ± 8.07
HandOutlines	57.19 ± 12.29	63.30 ± 2.05	60.65 ± 3.52	69.19 ± 14.14	63.57 ± 17.16	67.21 ± 16.82	52.16 ± 12.39	68.11 ± 20.52	41.57 ± 12.57	67.48 ± 15.69	69.89 ± 13.52	64.59 ± 10.14
Haptics	29.16 ± 3.65	24.35 ± 4.42	25.32 ± 2.75	28.31 ± 7.72	28.25 ± 7.08	29.33 ± 7.26	23.81 ± 0.68	29.22 ± 2.66	20.97 ± 3.07	29.00 ± 7.31	26.95 ± 8.77	26.62 ± 4.64
Herring	52.81 ± 11.18	56.88 ± 7.13	52.81 ± 11.23	51.25 ± 5.89	52.19 ± 12.13	47.92 ± 6.51	46.35 ± 3.61	45.83 ± 3.25	50.00 ± 4.69	48.96 ± 4.51	45.00 ± 9.84	55.00 ± 4.34
HouseTwenty	76.30 ± 9.04	69.24 ± 11.83	67.39 ± 13.31	60.00 ± 7.10	63.03 ± 13.56	68.91 ± 7.47	47.34 ± 6.31	59.94 ± 1.75	52.10 ± 8.21	71.15 ± 7.81	60.34 ± 11.20	71.93 ± 10.30
InlineSkate	18.58 ± 2.99	19.56 ± 1.81	19.93 ± 2.98	18.29 ± 2.68	17.05 ± 3.64	16.85 ± 1.00	15.76 ± 0.93	15.82 ± 1.62	16.29 ± 1.31	17.03 ± 0.46	16.36 ± 2.46	18.98 ± 2.54
InsectEPGRegularTrain	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	51.97 ± 7.58	58.39 ± 6.76	58.37 ± 4.44	37.08 ± 10.70	54.89 ± 2.35	37.99 ± 13.89	59.71 ± 4.91	81.69 ± 17.66	68.76 ± 9.60
InsectEPGSmallTrain	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	51.33 ± 4.40	58.47 ± 6.25	57.56 ± 4.71	41.37 ± 7.24	58.23 ± 4.08	42.65 ± 10.05	58.10 ± 4.04	80.40 ± 17.17	67.79 ± 2.45
InsectWingbeatSound	20.38 ± 3.08	16.62 ± 4.40	18.70 ± 3.47	47.87 ± 2.61	33.18 ± 5.98	50.44 ± 4.17	20.69 ± 3.83	49.41 ± 1.90	27.56 ± 2.55	51.03 ± 4.83	16.68 ± 7.51	31.49 ± 3.72
ItalyPowerDemand	68.88 ± 12.96	75.63 ± 13.47	73.61 ± 12.06	50.03 ± 0.16	79.18 ± 15.47	54.49 ± 7.77	75.87 ± 16.00	76.35 ± 22.01	83.15 ± 15.39	73.57 ± 21.06	50.07 ± 0.34	65.03 ± 13.70
LargeKitchenAppliances	45.65 ± 8.51	54.19 ± 10.41	50.93 ± 13.42	33.60 ± 7.89	44.85 ± 7.66	37.96 ± 12.60	32.00 ± 3.23	34.40 ± 7.77	32.96 ± 7.01	36.62 ± 11.05	32.00 ± 6.67	42.56 ± 5.00
Lightning2	57.05 ± 8.63	53.77 ± 7.90	59.34 ± 2.07	54.10 ± 10.50	51.80 ± 7.11	57.38 ± 4.34	49.73 ± 9.88	51.91 ± 4.13	52.13 ± 11.90	55.19 ± 6.21	50.16 ± 8.80	63.28 ± 8.25
Lightning7	45.48 ± 6.31	43.01 ± 5.94	42.74 ± 6.38	38.90 ± 8.91	47.95 ± 8.16	59.36 ± 11.07	19.63 ± 3.45	50.23 ± 12.43	27.40 ± 4.94	56.16 ± 10.34	20.82 ± 11.60	46.03 ± 6.17
Mallat	73.35 ± 15.31	47.53 ± 14.01	80.84 ± 4.13	86.79 ± 1.54	65.16 ± 19.15	91.78 ± 0.99	63.31 ± 2.66	78.79 ± 4.89	70.90 ± 5.59	87.65 ± 2.43	49.21 ± 6.11	80.67 ± 4.06
Meat	54.33 ± 7.42	42.00 ± 7.76	73.00 ± 16.39	89.67 ± 7.85	52.00 ± 12.16	53.33 ± 6.01	43.33 ± 8.33	41.11 ± 7.52	51.33 ± 18.61	57.22 ± 7.88	36.33 ± 5.82	57.67 ± 17.30
MedicalImages	33.45 ± 8.28	25.74 ± 5.92	33.61 ± 5.62	24.45 ± 6.71	27.45 ± 4.51	28.46 ± 3.05	27.85 ± 6.61	28.55 ± 2.80	34.53 ± 1.25	30.00 ± 1.71	29.82 ± 16.29	31.63 ± 5.59
MelbournePedestrian	63.62 ± 4.54	57.50 ± 8.49	63.00 ± 2.07	14.05 ± 1.77	42.42 ± 4.22	40.29 ± 3.90	51.77 ± 5.29	50.24 ± 2.74	54.32 ± 5.40	46.37 ± 2.70	42.42 ± 4.22	49.78 ± 4.68
MiddlePhalanxOutlineAgeGroup	47.01 ± 9.52	44.55 ± 12.64	45.45 ± 9.45	50.39 ± 7.58	44.29 ± 10.12	37.45 ± 10.11	46.75 ± 13.26	45.24 ± 16.76	48.57 ± 12.10	48.92 ± 14.26	22.86 ± 4.12	42.99 ± 12.95
MiddlePhalanxOutlineCorrect	49.42 ± 7.12	51.13 ± 7.83	49.42 ± 8.07	53.75 ± 12.47	44.54 ± 7.47	52.92 ± 7.99	47.54 ± 10.52	54.07 ± 11.30	48.59 ± 8.66	53.26 ± 15.07	50.72 ± 11.13	48.87 ± 3.56
MiddlePhalanxTW	37.92 ± 11.48	44.68 ± 3.99	42.73 ± 9.27	38.96 ± 14.59	39.22 ± 13.51	45.45 ± 5.95	43.07 ± 3.27	36.36 ± 13.61	46.36 ± 2.32	45.24 ± 3.58	16.36 ± 7.24	44.81 ± 7.78
MixedShapesRegularTrain	55.18 ± 4.21	33.76 ± 13.30	55.82 ± 8.68	58.38 ± 6.27	54.48 ± 12.34	62.60 ± 3.23	19.42 ± 0.39	63.00 ± 3.88	22.36 ± 6.85	63.64 ± 2.51	54.05 ± 4.44	67.98 ± 4.06
MixedShapesSmallTrain	45.73 ± 9.51	37.76 ± 12.10	51.77 ± 7.21	61.69 ± 3.51	54.28 ± 10.55	63.00 ± 2.46	19.05 ± 0.77	64.04 ± 1.32	24.01 ± 6.26	62.64 ± 3.39	57.53 ± 4.77	68.35 ± 3.26
MoteStrain	81.95 ± 4.07	75.34 ± 9.74	78.95 ± 3.25	70.97 ± 10.57	75.59 ± 8.13	78.41 ± 9.87	68.53 ± 3.31	82.48 ± 1.96	68.63 ± 10.07	83.65 ± 4.68	55.75 ± 11.79	58.56 ± 15.12
NonInvasiveFetalECGThorax1	25.55 ± 2.02	14.44 ± 3.90	25.55 ± 2.02	55.96 ± 3.30	18.44 ± 9.38	50.77 ± 4.84	18.37 ± 2.16	51.65 ± 2.21	21.42 ± 2.01	53.89 ± 1.99	13.55 ± 0.51	50.58 ± 2.13
NonInvasiveFetalECGThorax2	27.07 ± 5.56	22.30 ± 6.32	27.07 ± 5.56	62.09 ± 2.37	28.27 ± 8.71	55.69 ± 4.28	27.23 ± 2.03	56.05 ± 0.94	30.73 ± 3.23	58.63 ± 2.42	18.29 ± 1.75	59.72 ± 2.76
OSULeaf	55.70 ± 9.28	38.35 ± 6.31	56.03 ± 5.89	25.79 ± 2.77	25.29 ± 5.19	29.89 ± 3.34	18.18 ± 1.43	28.10 ± 2.07	21.90 ± 3.33	28.51 ± 4.37	23.31 ± 3.85	36.28 ± 6.97
OliveOil	64.67 ± 10.70	32.67 ± 5.48	71.33 ± 10.70	80.67 ± 3.65	17.33 ± 10.90	32.22 ± 11.71	38.89 ± 7.70	25.56 ± 12.62	24.00 ± 10.90	58.89 ± 22.69	22.00 ± 13.04	23.33 ± 9.72
PLAID	19.29 ± 3.99	16.65 ± 2.54	20.89 ± 4.30	22.98 ± 3.80	35.53 ± 7.89	27.56 ± 4.82	17.44 ± 2.97	25.88 ± 4.40	19.81 ± 3.24	28.68 ± 3.91	35.53 ± 7.89	40.19 ± 5.06
PhalangesOutlinesCorrect	52.91 ± 9.24	53.03 ± 10.79	52.61 ± 9.16	56.64 ± 11.15	54.13 ± 11.04	60.64 ± 3.81	58.70 ± 2.16	60.37 ± 4.20	52.26 ± 8.88	60.06 ± 3.62	49.04 ± 14.25	50.84 ± 2.93
Phoneme	12.22 ± 1.88	10.75 ± 2.12	11.22 ± 1.88	5.41 ± 0.40	7.31 ± 0.69	7.77 ± 0.84	4.13 ± 1.74	7.00 ± 0.45	6.19 ± 2.34	5.29 ± 0.61	5.20 ± 0.75	10.02 ± 0.59
PickupGestureWiimoteZ	43.20 ± 11.37	53.20 ± 9.55	62.40 ± 1.67	43.20 ± 6.72	40.40 ± 8.29	52.67 ± 10.07	21.33 ± 1.15	53.33 ± 4.16	27.60 ± 3.58	51.33 ± 6.43	40.40 ± 8.29	58.80 ± 3.03
PigAirwayPressure	28.37 ± 3.10	13.08 ± 1.90	49.52 ± 0.76	5.00 ± 0.26	3.65 ± 0.87	4.49 ± 0.28	3.21 ± 1.21	7.05 ± 0.73	4.62 ± 1.05	6.09 ± 0.73	5.10 ± 0.80	9.81 ± 0.43
PigArtPressure	50.58 ± 2.62	23.65 ± 3.56	69.42 ± 3.86	7.88 ± 0.26	5.29 ± 1.63	10.10 ± 0.48	8.01 ± 1.39	9.78 ± 1.11	7.12 ± 1.24	8.33 ± 0.56	19.13 ± 1.04	30.67 ± 1.38
PigCVP	31.54 ± 0.94	18.08 ± 2.27	61.92 ± 1.78	7.02 ± 0.43	3.94 ± 1.10	6.73 ± 0.00	4.01 ± 0.56	7.85 ± 1.00	4.13 ± 0.80	5.29 ± 0.83	10.10 ± 1.40	19.33 ± 0.99
Plane	100.00 ± 0.00	94.29 ± 3.75	99.05 ± 1.17	86.10 ± 6.69	91.62 ± 3.39	86.98 ± 7.02	84.44 ± 7.02	81.90 ± 1.90	87.24 ± 4.98	86.98 ± 6.34	21.71 ± 8.89	99.24 ± 0.80
PowerCons	82.11 ± 6.61	83.78 ± 3.63	82.33 ± 7.91	72.11 ± 2.98	70.78 ± 14.65	69.81 ± 4.10	52.22 ± 1.47	70.93 ± 7.46	50.67 ± 4.38	70.00 ± 8.73	65.89 ± 16.73	66.78 ± 15.52
ProximalPhalanxOutlineAgeGroup	68.10 ± 13.07	64.29 ± 28.01	64.78 ± 22.07	58.44 ± 28.63	68.00 ± 16.32	53.17 ± 31.21	56.26 ± 31.37	56.91 ± 37.90	59.22 ± 23.54	51.38 ± 35.67	33.95 ± 22.63	66.24 ± 17.89
ProximalPhalanxOutlineCorrect	65.43 ± 5.55	60.07 ± 16.64	63.85 ± 6.55	63.37 ± 8.02	62.96 ± 6.50	54.75 ± 14.29	54.30 ± 16.96	60.94 ± 6.65	59.86 ± 17.09	55.21 ± 16.86	44.88 ± 18.17	59.93 ± 12.82
ProximalPhalanxTW	50.83 ± 4.70	56.00 ± 9.92	67.12 ± 8.12	49.76 ± 11.43	53.37 ± 7.43	58.70 ± 5.35	53.98 ± 16.76	42.28 ± 24.98	56.00 ± 8.06	48.29 ± 19.01	16.98 ± 12.37	57.76 ± 7.59
RefrigerationDevices	41.39 ± 9.49	40.37 ± 10.57	41.23 ± 9.97	33.92 ± 0.58	35.31 ± 3.72	36.71 ± 1.61	33.87 ± 4.03	35.64 ± 1.78	33.81 ± 1.50	33.60 ± 3.03	33.07 ± 2.39	33.39 ± 7.13
Rock	32.00 ± 5.48	38.40 ± 8.65	37.80 ± 13.31	67.60 ± 10.53	44.80 ± 18.25	58.67 ± 13.61	70.00 ± 11.14	66.67 ± 8.08	47.60 ± 12.28	60.00 ± 14.00	44.40 ± 14.24	64.00 ± 7.07
ScreenType	36.91 ± 4.95	40.21 ± 6.72	38.40 ± 6.31	35.25 ± 3.07	36.75 ± 2.52	39.56 ± 4.47	33.78 ± 3.74	38.67 ± 2.63	33.71 ± 2.21	39.11 ± 4.89	33.81 ± 3.79	35.63 ± 4.40
SemgHandGenderCh2	55.13 ± 5.29	58.23 ± 11.97	54.27 ± 6.20	53.27 ± 13.91	62.87 ± 8.65	49.22 ± 10.20	55.56 ± 1.55	62.44 ± 3.53	43.87 ± 10.45	56.22 ± 13.04	53.70 ± 4.90	59.73 ± 10.87
SemgHandMovementCh2	33.51 ± 7.12	31.29 ± 5.18	36.00 ± 6.56	29.78 ± 3.16	23.56 ± 5.26	24.44 ± 2.52	18.89 ± 0.59	24.44 ± 1.39	18.71 ± 1.13	24.07 ± 5.35	27.91 ± 3.59	31.16 ± 4.70

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Table 8: Per-dataset 2-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
SemgHandSubjectCh2	32.00 $\pm$ 3.88	30.71 $\pm$ 4.13	33.56 $\pm$ 5.04	45.91 $\pm$ 5.66	40.67 $\pm$ 4.39	29.56 $\pm$ 4.02	23.56 $\pm$ 2.47	32.52 $\pm$ 7.62	24.09 $\pm$ 3.07	30.81 $\pm$ 0.84	29.47 $\pm$ 4.28	43.20 $\pm$ 6.72
ShakeGestureWiimoteZ	76.00 $\pm$ 6.63	76.80 $\pm$ 7.43	81.60 $\pm$ 0.89	47.20 $\pm$ 7.95	58.40 $\pm$ 5.18	54.00 $\pm$ 6.00	26.00 $\pm$ 4.00	54.67 $\pm$ 4.62	32.40 $\pm$ 2.61	56.00 $\pm$ 5.29	58.40 $\pm$ 5.18	81.60 $\pm$ 2.61
ShapeletSim	78.89 $\pm$ 14.16	98.33 $\pm$ 2.08	92.11 $\pm$ 5.26	50.11 $\pm$ 2.87	80.22 $\pm$ 16.23	47.59 $\pm$ 3.16	50.74 $\pm$ 3.06	48.15 $\pm$ 1.79	49.22 $\pm$ 2.93	46.48 $\pm$ 3.06	49.67 $\pm$ 1.50	67.44 $\pm$ 10.92
ShapesAll	36.80 $\pm$ 3.60	28.17 $\pm$ 3.75	36.80 $\pm$ 3.60	50.73 $\pm$ 1.71	32.23 $\pm$ 5.06	55.78 $\pm$ 1.50	6.11 $\pm$ 0.63	53.17 $\pm$ 1.00	18.33 $\pm$ 5.95	53.33 $\pm$ 2.62	27.63 $\pm$ 3.10	63.03 $\pm$ 2.22
SmallKitchenAppliances	48.85 $\pm$ 7.91	50.40 $\pm$ 8.87	50.19 $\pm$ 7.71	41.49 $\pm$ 3.33	47.15 $\pm$ 6.04	49.60 $\pm$ 6.54	35.20 $\pm$ 2.93	40.09 $\pm$ 2.14	42.03 $\pm$ 4.49	44.09 $\pm$ 8.36	34.61 $\pm$ 4.49	55.68 $\pm$ 8.74
SmoothSubspace	78.93 $\pm$ 10.17	77.87 $\pm$ 5.55	80.13 $\pm$ 5.80	33.33 $\pm$ 0.00	51.20 $\pm$ 5.65	58.22 $\pm$ 4.73	68.89 $\pm$ 7.31	59.11 $\pm$ 4.23	70.67 $\pm$ 4.88	81.33 $\pm$ 6.36	48.27 $\pm$ 9.05	85.07 $\pm$ 5.20
SonyAIBORobotSurface1	90.42 $\pm$ 4.34	94.08 $\pm$ 1.77	89.38 $\pm$ 5.35	64.83 $\pm$ 12.29	70.05 $\pm$ 3.77	62.90 $\pm$ 13.91	61.90 $\pm$ 7.89	66.61 $\pm$ 10.89	59.10 $\pm$ 5.18	62.73 $\pm$ 9.45	52.65 $\pm$ 8.42	71.85 $\pm$ 8.28
SonyAIBORobotSurface2	88.00 $\pm$ 2.82	84.87 $\pm$ 6.58	83.67 $\pm$ 9.41	74.59 $\pm$ 3.28	78.70 $\pm$ 2.68	69.60 $\pm$ 7.04	77.26 $\pm$ 1.79	78.49 $\pm$ 0.86	78.30 $\pm$ 1.49	79.29 $\pm$ 1.17	56.81 $\pm$ 17.39	68.21 $\pm$ 5.28
StarLightCurves	76.49 $\pm$ 11.97	66.10 $\pm$ 9.91	66.22 $\pm$ 10.70	67.92 $\pm$ 13.52	71.57 $\pm$ 12.97	63.90 $\pm$ 14.70	35.37 $\pm$ 11.93	65.31 $\pm$ 14.87	34.69 $\pm$ 14.98	63.78 $\pm$ 14.15	68.45 $\pm$ 12.08	74.47 $\pm$ 15.08
Strawberry	62.81 $\pm$ 6.83	57.14 $\pm$ 6.89	60.22 $\pm$ 9.39	51.84 $\pm$ 8.28	56.59 $\pm$ 2.41	53.33 $\pm$ 7.10	66.58 $\pm$ 2.89	52.97 $\pm$ 8.18	59.62 $\pm$ 11.05	54.59 $\pm$ 4.47	51.03 $\pm$ 11.13	59.41 $\pm$ 6.38
SwedishLeaf	72.45 $\pm$ 3.37	54.30 $\pm$ 6.89	72.45 $\pm$ 3.37	34.02 $\pm$ 3.00	43.26 $\pm$ 7.26	48.91 $\pm$ 2.97	33.28 $\pm$ 3.94	46.24 $\pm$ 1.25	38.85 $\pm$ 4.16	47.52 $\pm$ 2.78	7.94 $\pm$ 0.74	64.99 $\pm$ 3.83
Symbols	90.49 $\pm$ 1.42	75.88 $\pm$ 7.72	88.48 $\pm$ 4.41	81.23 $\pm$ 0.99	76.18 $\pm$ 6.87	81.37 $\pm$ 1.84	37.42 $\pm$ 3.12	82.04 $\pm$ 0.58	58.95 $\pm$ 6.86	80.00 $\pm$ 1.74	51.50 $\pm$ 7.93	82.69 $\pm$ 3.43
SyntheticControl	79.67 $\pm$ 4.05	85.07 $\pm$ 4.63	82.60 $\pm$ 6.01	41.40 $\pm$ 6.37	83.80 $\pm$ 5.24	55.00 $\pm$ 3.38	44.11 $\pm$ 1.39	59.89 $\pm$ 3.15	47.93 $\pm$ 4.58	62.89 $\pm$ 3.15	20.80 $\pm$ 4.95	87.93 $\pm$ 3.95
ToeSegmentation1	68.42 $\pm$ 7.02	60.79 $\pm$ 8.14	61.58 $\pm$ 7.54	53.95 $\pm$ 2.19	58.60 $\pm$ 10.97	51.90 $\pm$ 4.30	53.80 $\pm$ 2.21	53.95 $\pm$ 0.44	52.37 $\pm$ 4.44	51.61 $\pm$ 3.08	50.70 $\pm$ 2.53	60.26 $\pm$ 8.54
ToeSegmentation2	76.15 $\pm$ 9.44	79.69 $\pm$ 9.24	72.15 $\pm$ 7.68	48.92 $\pm$ 5.64	68.62 $\pm$ 12.68	66.41 $\pm$ 15.71	51.54 $\pm$ 5.38	63.08 $\pm$ 15.78	47.08 $\pm$ 8.33	63.59 $\pm$ 15.79	54.31 $\pm$ 14.96	76.15 $\pm$ 12.44
Trace	76.00 $\pm$ 3.00	76.40 $\pm$ 3.05	73.80 $\pm$ 2.95	52.40 $\pm$ 4.04	76.60 $\pm$ 4.39	47.33 $\pm$ 3.21	51.00 $\pm$ 4.00	49.00 $\pm$ 3.00	55.60 $\pm$ 4.62	49.00 $\pm$ 5.00	46.00 $\pm$ 6.78	98.40 $\pm$ 3.05
TwoLeadECG	82.53 $\pm$ 6.47	79.19 $\pm$ 7.45	70.63 $\pm$ 4.92	55.68 $\pm$ 3.84	59.95 $\pm$ 3.74	62.36 $\pm$ 5.86	56.60 $\pm$ 1.27	55.90 $\pm$ 1.72	55.49 $\pm$ 2.80	60.14 $\pm$ 7.03	48.48 $\pm$ 5.21	74.63 $\pm$ 8.09
TwoPatterns	52.71 $\pm$ 4.44	71.75 $\pm$ 5.17	65.30 $\pm$ 11.45	32.62 $\pm$ 1.84	44.61 $\pm$ 9.45	31.66 $\pm$ 1.59	24.88 $\pm$ 1.30	28.57 $\pm$ 1.14	25.14 $\pm$ 0.61	29.88 $\pm$ 2.34	27.30 $\pm$ 2.15	78.23 $\pm$ 0.41
UMD	54.86 $\pm$ 7.95	71.67 $\pm$ 9.36	68.75 $\pm$ 16.74	54.17 $\pm$ 4.55	62.50 $\pm$ 12.00	52.31 $\pm$ 6.22	47.69 $\pm$ 9.93	60.19 $\pm$ 1.45	53.06 $\pm$ 4.83	56.25 $\pm$ 6.05	47.50 $\pm$ 4.16	78.86 $\pm$ 14.11
UWaveGestureLibraryAll	32.50 $\pm$ 3.38	28.30 $\pm$ 5.01	31.51 $\pm$ 4.14	69.44 $\pm$ 6.55	43.65 $\pm$ 6.39	66.54 $\pm$ 6.96	26.72 $\pm$ 0.50	66.58 $\pm$ 7.48	30.82 $\pm$ 2.10	66.61 $\pm$ 7.59	54.22 $\pm$ 7.86	42.54 $\pm$ 0.28
UWaveGestureLibraryX	31.94 $\pm$ 2.72	28.97 $\pm$ 3.44	30.99 $\pm$ 3.77	50.42 $\pm$ 3.39	42.01 $\pm$ 2.19	51.69 $\pm$ 1.72	21.35 $\pm$ 1.20	51.05 $\pm$ 1.76	28.72 $\pm$ 2.91	50.93 $\pm$ 1.76	22.53 $\pm$ 5.80	42.54 $\pm$ 0.28
UWaveGestureLibraryY	28.00 $\pm$ 4.17	26.42 $\pm$ 2.97	27.96 $\pm$ 5.26	43.36 $\pm$ 3.42	39.43 $\pm$ 5.20	42.54 $\pm$ 7.86	19.88 $\pm$ 2.66	43.61 $\pm$ 6.37	28.02 $\pm$ 2.78	41.75 $\pm$ 7.40	28.22 $\pm$ 7.42	42.54 $\pm$ 0.28
UWaveGestureLibraryZ	31.19 $\pm$ 2.60	30.19 $\pm$ 2.10	28.42 $\pm$ 4.12	44.60 $\pm$ 3.60	41.59 $\pm$ 3.78	42.17 $\pm$ 5.87	23.93 $\pm$ 2.40	43.23 $\pm$ 7.30	29.59 $\pm$ 1.71	40.53 $\pm$ 6.44	23.38 $\pm$ 7.02	59.88 $\pm$ 2.03
Wafer	47.08 $\pm$ 18.17	57.95 $\pm$ 22.06	52.16 $\pm$ 16.40	56.77 $\pm$ 13.38	49.59 $\pm$ 17.01	67.22 $\pm$ 1.00	55.02 $\pm$ 6.93	66.32 $\pm$ 1.28	56.21 $\pm$ 9.73	67.64 $\pm$ 1.29	54.99 $\pm$ 20.43	55.01 $\pm$ 15.62
Wine	60.74 $\pm$ 8.43	50.74 $\pm$ 7.92	60.48 $\pm$ 12.52	53.70 $\pm$ 8.88	47.41 $\pm$ 11.54	53.70 $\pm$ 3.21	53.09 $\pm$ 5.35	50.00 $\pm$ 0.00	50.74 $\pm$ 1.66	54.94 $\pm$ 26.28	50.74 $\pm$ 1.66	48.89 $\pm$ 5.34
WordSynonyms	20.63 $\pm$ 4.03	16.05 $\pm$ 1.67	20.63 $\pm$ 4.03	21.35 $\pm$ 4.96	20.06 $\pm$ 3.25	29.05 $\pm$ 0.77	6.11 $\pm$ 1.90	29.73 $\pm$ 1.73	13.92 $\pm$ 2.48	29.57 $\pm$ 1.41	6.43 $\pm$ 4.20	34.92 $\pm$ 2.64
Worms	54.03 $\pm$ 11.78	44.94 $\pm$ 11.42	52.73 $\pm$ 7.32	22.60 $\pm$ 2.99	28.57 $\pm$ 6.43	19.48 $\pm$ 1.30	17.32 $\pm$ 5.41	17.75 $\pm$ 5.25	15.58 $\pm$ 4.00	20.35 $\pm$ 3.75	18.70 $\pm$ 5.00	38.70 $\pm$ 4.63
WormsTwoClass	55.32 $\pm$ 9.03	52.47 $\pm$ 8.50	55.84 $\pm$ 3.31	48.57 $\pm$ 2.69	49.09 $\pm$ 4.63	49.78 $\pm$ 2.70	51.52 $\pm$ 8.84	47.19 $\pm$ 3.97	47.27 $\pm$ 4.37	46.32 $\pm$ 1.50	48.83 $\pm$ 7.82	54.03 $\pm$ 7.72
Yoga	52.99 $\pm$ 4.56	50.68 $\pm$ 2.33	50.48 $\pm$ 3.16	50.86 $\pm$ 0.91	51.49 $\pm$ 3.75	50.12 $\pm$ 2.91	51.48 $\pm$ 2.93	50.70 $\pm$ 2.90	51.46 $\pm$ 3.58	50.58 $\pm$ 2.12	51.69 $\pm$ 4.15	50.64 $\pm$ 4.90
Avg. Acc.	53.98	50.79	<u>54.95</u>	48.01	48.43	49.95	39.45	49.45	41.71	51.14	38.30	55.55
Avg. Rank	4.84	5.63	<u>4.83</u>	6.78	6.42	6.13	9.36	6.29	8.63	5.64	9.17	4.28
#Best	33	13	19	10	2	9	4	6	3	7	2	<u>32</u>

Table 9: Per-dataset 5-shot classification results on 128 shared UCR datasets. Each value is reported as mean $\pm$ standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
ACSF1	76.00 $\pm$ 2.92	64.60 $\pm$ 5.18	72.80 $\pm$ 1.64	36.20 $\pm$ 3.49	50.40 $\pm$ 2.61	60.00 $\pm$ 2.00	40.00 $\pm$ 1.00	62.67 $\pm$ 2.52	46.80 $\pm$ 3.70	58.67 $\pm$ 1.53	30.60 $\pm$ 4.83	68.33 $\pm$ 2.89
Adiac	56.21 $\pm$ 4.18	42.81 $\pm$ 4.46	56.21 $\pm$ 4.18	35.29 $\pm$ 1.32	34.58 $\pm$ 2.33	43.82 $\pm$ 1.54	46.97 $\pm$ 3.07	38.62 $\pm$ 1.68	64.50 $\pm$ 5.47	43.99 $\pm$ 3.86	7.11 $\pm$ 1.15	69.51 $\pm$ 2.42
AllGestureWiimoteX	40.46 $\pm$ 4.50	34.14 $\pm$ 3.58	38.71 $\pm$ 5.08	23.63 $\pm$ 2.19	35.40 $\pm$ 4.31	32.24 $\pm$ 3.43	13.57 $\pm$ 1.57	29.95 $\pm$ 4.86	18.17 $\pm$ 3.28	29.38 $\pm$ 3.89	35.40 $\pm$ 4.31	39.40 $\pm$ 2.86
AllGestureWiimoteY	43.20 $\pm$ 1.70	40.60 $\pm$ 1.51	36.20 $\pm$ 3.42	27.89 $\pm$ 2.07	40.97 $\pm$ 3.06	36.76 $\pm$ 2.07	15.38 $\pm$ 0.81	34.43 $\pm$ 2.02	19.49 $\pm$ 2.37	35.43 $\pm$ 1.55	40.97 $\pm$ 3.06	48.11 $\pm$ 2.89
AllGestureWiimoteZ	41.46 $\pm$ 4.33	35.71 $\pm$ 5.12	36.46 $\pm$ 5.30	20.20 $\pm$ 1.22	27.11 $\pm$ 2.51	26.57 $\pm$ 0.25	13.86 $\pm$ 1.71	26.67 $\pm$ 2.65	17.37 $\pm$ 2.23	26.86 $\pm$ 0.99	27.11 $\pm$ 2.51	36.14 $\pm$ 0.97
ArrowHead	66.40 $\pm$ 5.44	61.03 $\pm$ 5.46	64.23 $\pm$ 7.62	59.43 $\pm$ 5.27	58.86 $\pm$ 12.66	60.19 $\pm$ 7.46	42.67 $\pm$ 0.87	60.76 $\pm$ 1.19	48.11 $\pm$ 2.87	63.43 $\pm$ 8.42	54.06 $\pm$ 9.79	73.60 $\pm$ 3.71
BME	55.60 $\pm$ 7.80	56.93 $\pm$ 3.90	68.00 $\pm$ 9.90	58.53 $\pm$ 7.20	89.60 $\pm$ 10.65	58.89 $\pm$ 5.55	73.33 $\pm$ 8.11	76.89 $\pm$ 7.34	80.00 $\pm$ 6.70	73.33 $\pm$ 5.93	53.07 $\pm$ 5.63	97.73 $\pm$ 1.53
Beef	54.67 $\pm$ 4.47	47.33 $\pm$ 2.79	55.33 $\pm$ 2.98	52.67 $\pm$ 7.60	43.33 $\pm$ 4.71	64.44 $\pm$ 3.85	50.00 $\pm$ 3.33	52.22 $\pm$ 8.39	62.00 $\pm$ 8.03	84.44 $\pm$ 1.92	35.33 $\pm$ 4.47	51.33 $\pm$ 6.91
BeetleFly	72.00 $\pm$ 7.58	60.00 $\pm$ 11.73	79.00 $\pm$ 10.84	71.00 $\pm$ 14.32	57.00 $\pm$ 8.37	70.00 $\pm$ 8.66	51.67 $\pm$ 2.89	66.67 $\pm$ 16.07	49.00 $\pm$ 6.52	70.00 $\pm$ 10.00	59.00 $\pm$ 18.51	91.67 $\pm$ 2.89
BirdChicken	69.00 $\pm$ 12.94	77.00 $\pm$ 9.75	90.00 $\pm$ 3.54	57.00 $\pm$ 5.70	61.00 $\pm$ 16.73	63.33 $\pm$ 7.64	36.67 $\pm$ 2.89	65.00 $\pm$ 10.00	54.00 $\pm$ 8.94	68.33 $\pm$ 15.28	54.00 $\pm$ 4.18	85.00 $\pm$ 11.73
CBF	95.87 $\pm$ 2.59	95.58 $\pm$ 3.67	95.89 $\pm$ 3.27	61.47 $\pm$ 7.78	95.67 $\pm$ 4.48	74.52 $\pm$ 5.66	39.04 $\pm$ 0.80	80.11 $\pm$ 4.97	42.29 $\pm$ 1.09	80.59 $\pm$ 4.09	41.20 $\pm$ 4.62	93.76 $\pm$ 3.90
Car	59.33 $\pm$ 9.02	51.00 $\pm$ 7.96	65.67 $\pm$ 5.73	52.33 $\pm$ 6.41	45.67 $\pm$ 2.79	68.89 $\pm$ 6.31	31.67 $\pm$ 5.00	48.89 $\pm$ 4.19	41.00 $\pm$ 12.56	57.78 $\pm$ 5.85	36.00 $\pm$ 4.50	69.67 $\pm$ 3.98
Chinatown	96.15 $\pm$ 2.16	96.97 $\pm$ 0.49	97.43 $\pm$ 0.38	50.96 $\pm$ 33.10	96.03 $\pm$ 2.24	84.94 $\pm$ 15.12	91.06 $\pm$ 1.94	93.00 $\pm$ 5.85	89.56 $\pm$ 2.00	87.95 $\pm$ 8.75	73.00 $\pm$ 0.76	93.46 $\pm$ 10.56
ChlorineConcentration	35.97 $\pm$ 2.09	35.63 $\pm$ 3.63	36.15 $\pm$ 2.53	35.57 $\pm$ 3.02	35.80 $\pm$ 3.16	35.02 $\pm$ 2.94	36.15 $\pm$ 1.24	35.49 $\pm$ 4.56	37.02 $\pm$ 1.05	35.96 $\pm$ 1.50	38.60 $\pm$ 5.59	39.58 $\pm$ 3.20
CinCECGTorso	49.41 $\pm$ 5.43	57.77 $\pm$ 5.88	56.17 $\pm$ 3.91	43.81 $\pm$ 1.84	69.26 $\pm$ 4.67	63.41 $\pm$ 2.73	37.80 $\pm$ 1.42	64.95 $\pm$ 2.40	64.83 $\pm$ 11.83	68.41 $\pm$ 5.18	54.58 $\pm$ 3.46	82.56 $\pm$ 7.22
Coffee	86.43 $\pm$ 14.81	97.86 $\pm$ 1.96	100.00 $\pm$ 0.00	92.86 $\pm$ 3.57	97.86 $\pm$ 1.96	86.90 $\pm$ 2.06	76.19 $\pm$ 2.06	82.14 $\pm$ 9.45	92.86 $\pm$ 10.41	85.71 $\pm$ 0.00	50.00 $\pm$ 6.19	94.29 $\pm$ 1.96
Computers	58.80 $\pm$ 7.81	61.60 $\pm$ 5.76	60.96 $\pm$ 6.58	46.64 $\pm$ 4.37	54.48 $\pm$ 7.54	48.97 $\pm$ 5.33	50.53 $\pm$ 1.29	48.00 $\pm$ 8.43	49.04 $\pm$ 4.09	48.67 $\pm$ 4.31	46.56 $\pm$ 5.23	62.67 $\pm$ 4.74
CricketX	45.90 $\pm$ 4.41	43.08 $\pm$ 3.97	43.08 $\pm$ 2.15	25.23 $\pm$ 1.80	36.10 $\pm$ 5.19	34.87 $\pm$ 4.89	11.79 $\pm$ 1.43	30.77 $\pm$ 0.51	16.92 $\pm$ 1.32	33.68 $\pm$ 3.44	11.33 $\pm$ 1.41	49.06 $\pm$ 3.21
CricketY	42.67 $\pm$ 4.49	43.08 $\pm$ 3.40	43.28 $\pm$ 6.01	30.41 $\pm$ 3.78	36.92 $\pm$ 6.92	37.86 $\pm$ 2.42	9.06 $\pm$ 1.21	36.58 $\pm$ 2.06	15.23 $\pm$ 1.85	37.26 $\pm$ 2.44	12.05 $\pm$ 2.18	49.66 $\pm$ 1.41
CricketZ	51.59 $\pm$ 3.02	49.64 $\pm$ 3.11	47.79 $\pm$ 6.26	26.31 $\pm$ 2.67	36.31 $\pm$ 4.35	37.18 $\pm$ 1.18	10.77 $\pm$ 0.51	34.87 $\pm$ 1.43	15.44 $\pm$ 1.54	37.18 $\pm$ 0.77	10.92 $\pm$ 1.59	53.68 $\pm$ 2.06
Crop	49.93 $\pm$ 1.45	46.93 $\pm$ 2.27	49.93 $\pm$ 1.45	31.91 $\pm$ 2.21	38.13 $\pm$ 2.76	46.92 $\pm$ 2.69	44.47 $\pm$ 0.97	47.82 $\pm$ 2.55	47.76 $\pm$ 1.51	48.69 $\pm$ 2.69	8.15 $\pm$ 3.22	52.18 $\pm$ 2.24
DiatomSizeReduction	85.75 $\pm$ 6.98	70.26 $\pm$ 16.84	95.95 $\pm$ 1.26	61.57 $\pm$ 2.74	65.49 $\pm$ 10.17	87.69 $\pm$ 0.50	68.63 $\pm$ 2.55	66.34 $\pm$ 8.52	73.53 $\pm$ 9.37	87.25 $\pm$ 0.00	41.83 $\pm$ 13.83	93.66 $\pm$ 1.97
DistalPhalanxOutlineAgeGroup	65.76 $\pm$ 4.54	68.63 $\pm$ 3.96	67.19 $\pm$ 5.75	58.99 $\pm$ 14.81	63.31 $\pm$ 4.46	64.75 $\pm$ 3.74	67.63 $\pm$ 2.16	69.30 $\pm$ 3.24	68.20 $\pm$ 2.35	68.59 $\pm$ 2.31	53.67 $\pm$ 12.57	69.35 $\pm$ 1.20
DistalPhalanxOutlineCorrect	66.81 $\pm$ 3.92	61.96 $\pm$ 7.85	68.62 $\pm$ 6.57	56.23 $\pm$ 10.81	60.07 $\pm$ 4.07	62.44 $\pm$ 3.65	62.68 $\pm$ 3.32	58.94 $\pm$ 3.37	63.41 $\pm$ 7.40	59.30 $\pm$ 3.08	55.87 $\pm$ 8.58	63.77 $\pm$ 11.63
DistalPhalanxTW	58.99 $\pm$ 2.69	57.12 $\pm$ 1.40	58.56 $\pm$ 3.40	49.64 $\pm$ 7.01	60.58 $\pm$ 2.61	57.31 $\pm$ 0.42	51.32 $\pm$ 3.62	59.47 $\pm$ 4.40	54.10 $\pm$ 4.24	56.59 $\pm$ 5.05	31.94 $\pm$ 15.91	56.40 $\pm$ 4.87
DodgerLoopDay	37.75 $\pm$ 3.35	40.50 $\pm$ 4.11	37.00 $\pm$ 3.26	46.75 $\pm$ 4.01	44.25 $\pm$ 2.09	52.50 $\pm$ 5.45	22.08 $\pm$ 0.72	45.42 $\pm$ 2.60	24.25 $\pm$ 3.49	50.83 $\pm$ 4.73	44.25 $\pm$ 2.09	45.75 $\pm$ 3.26
DodgerLoopGame	66.52 $\pm$ 4.02	82.32 $\pm$ 3.38	72.17 $\pm$ 3.64	76.09 $\pm$ 5.15	84.49 $\pm$ 1.89	75.36 $\pm$ 5.02	50.00 $\pm$ 2.61	78.99 $\pm$ 7.63	53.33 $\pm$ 2.09	77.29 $\pm$ 4.82	84.49 $\pm$ 1.89	72.61 $\pm$ 10.25
DodgerLoopWeekend	95.51 $\pm$ 1.65	92.46 $\pm$ 3.46	94.78 $\pm$ 1.88	97.39 $\pm$ 0.40	97.25 $\pm$ 0.61	96.62 $\pm$ 0.84	66.43 $\pm$ 4.25	97.83 $\pm$ 0.00	72.90 $\pm$ 1.82	96.86 $\pm$ 0.42	97.25 $\pm$ 0.61	93.62 $\pm$ 2.37
ECG200	76.40 $\pm$ 3.58	77.80 $\pm$ 6.26	76.20 $\pm$ 4.60	75.00 $\pm$ 1.58	62.80 $\pm$ 16.36	77.67 $\pm$ 3.06	76.67 $\pm$ 3.79	78.00 $\pm$ 2.00	77.60 $\pm$ 3.21	79.33 $\pm$ 4.93	69.60 $\pm$ 4.51	67.40 $\pm$ 8.91
ECG5000	78.85 $\pm$ 9.03	75.51 $\pm$ 9.29	75.56 $\pm$ 6.88	84.88 $\pm$ 3.51	71.69 $\pm$ 7.99	82.01 $\pm$ 5.12	76.33 $\pm$ 7.90	83.48 $\pm$ 5.73	74.58 $\pm$ 8.80	82.23 $\pm$ 5.84	18.64 $\pm$ 16.84	73.89 $\pm$ 8.21
ECGFiveDays	71.54 $\pm$ 13.30	70.27 $\pm$ 8.41	70.22 $\pm$ 9.52	69.38 $\pm$ 6.83	77.54 $\pm$ 3.65	72.94 $\pm$ 9.66	87.65 $\pm$ 8.42	75.22 $\pm$ 2.60	83.30 $\pm$ 11.51	75.61 $\pm$ 4.26	60.49 $\pm$ 6.59	70.96 $\pm$ 3.91
EOGHorizontalSignal	24.97 $\pm$ 1.88	32.71 $\pm$ 1.62	27.35 $\pm$ 3.43	29.56 $\pm$ 3.95	37.73 $\pm$ 6.23	36.92 $\pm$ 1.62	10.96 $\pm$ 1.15	38.40 $\pm$ 2.19	9.94 $\pm$ 1.90	35.17 $\pm$ 4.76	31.99 $\pm$ 1.89	46.35 $\pm$ 2.27
EOGVerticalSignal	18.18 $\pm$ 3.62	22.10 $\pm$ 3.22	18.78 $\pm$ 4.22	29.01 $\pm$ 2.45	38.23 $\pm$ 4.43	31.77 $\pm$ 5.00	8.47 $\pm$ 1.80	31.95 $\pm$ 3.43	10.39 $\pm$ 1.59	31.22 $\pm$ 4.29	22.82 $\pm$ 2.72	36.74 $\pm$ 2.24
Earthquakes	50.94 $\pm$ 12.41	50.22 $\pm$ 8.85	52.23 $\pm$ 11.61	56.83 $\pm$ 2.92	52.09 $\pm$ 5.05	56.35 $\pm$ 11.87	51.32 $\pm$ 9.11	56.12 $\pm$ 3.14	53.38 $\pm$ 4.56	54.68 $\pm$ 7.61	52.81 $\pm$ 9.99	54.68 $\pm$ 9.31
ElectricDevices	36.69 $\pm$ 5.52	36.63 $\pm$ 6.97	36.08 $\pm$ 5.74	23.65 $\pm$ 1.86	38.77 $\pm$ 8.33	38.73 $\pm$ 7.16	22.89 $\pm$ 3.05	25.09 $\pm$ 5.42	31.29 $\pm$ 6.08	38.61 $\pm$ 5.70	11.47 $\pm$ 4.02	47.06 $\pm$ 4.18
EthanolLevel	27.52 $\pm$ 1.92	26.36 $\pm$ 0.86	26.60 $\pm$ 1.66	27.88 $\pm$ 1.25	24.88 $\pm$ 0.18	28.40 $\pm$ 1.73	28.60 $\pm$ 2.78	26.53 $\pm$ 0.12	25.40 $\pm$ 0.75	29.53 $\pm$ 0.64	26.32 $\pm$ 1.07	25.92 $\pm$ 1.40
FaceAll	67.63 $\pm$ 4.66	67.75 $\pm$ 7.28	67.63 $\pm$ 4.66	53.96 $\pm$ 2.14	41.22 $\pm$ 3.29	55.76 $\pm$ 2.36	53.79 $\pm$ 5.19	55.44 $\pm$ 2.51	51.51 $\pm$ 5.08	54.30 $\pm$ 2.41	17.64 $\pm$ 1.73	54.54 $\pm$ 1.26
FaceFour	88.18 $\pm$ 2.85	61.82 $\pm$ 13.74	89.77 $\pm$ 4.62	78.86 $\pm$ 3.17	57.50 $\pm$ 8.82	86.36 $\pm$ 1.14	64.77 $\pm$ 2.27	87.88 $\pm$ 1.74	71.82 $\pm$ 1.24	85.61 $\pm$ 1.74	41.14 $\pm$ 6.88	75.68 $\pm$ 4.37
FacesUCR	78.67 $\pm$ 2.69	73.00 $\pm$ 4.70	76.25 $\pm$ 3.16	50.94 $\pm$ 2.11	43.00 $\pm$ 7.53	68.33 $\pm$ 3.31	58.46 $\pm$ 2.45	68.99 $\pm$ 1.59	60.62 $\pm$ 1.49	67.61 $\pm$ 2.78	27.20 $\pm$ 3.34	68.58 $\pm$ 2.57
FiftyWords	30.07 $\pm$ 3.14	27.91 $\pm$ 3.50	30.07 $\pm$ 3.14	47.03 $\pm$ 2.26	37.71 $\pm$ 3.50	52.23 $\pm$ 1.46	9.45 $\pm$ 3.91	56.19 $\pm$ 3.63	25.10 $\pm$ 2.19	49.38 $\pm$ 1.41	40.84 $\pm$ 2.35	61.58 $\pm$ 2.55
Fish	61.26 $\pm$ 15.63	48.69 $\pm$ 6.91	67.66 $\pm$ 6.55	56.80 $\pm$ 6.22	42.97 $\pm$ 5.32	68.76 $\pm$ 3.49	25.52 $\pm$ 2.93	56.57 $\pm$ 5.51	40.80 $\pm$ 3.78	67.05 $\pm$ 1.44	26.40 $\pm$ 8.24	64.57 $\pm$ 5.92
FordA	71.41 $\pm$ 10.22	71.94 $\pm$ 9.09	70.55 $\pm$ 10.29	50.33 $\pm$ 1.09	63.62 $\pm$ 6.82	50.28 $\pm$ 2.11	51.67 $\pm$ 1.61	49.67 $\pm$ 1.72	51.11 $\pm$ 2.04	49.82 $\pm$ 1.42	50.91 $\pm$ 1.25	49.82 $\pm$ 1.25
FordB	61.26 $\pm$ 2.75	58.96 $\pm$ 1.47	59.04 $\pm$ 3.16	50.00 $\pm$ 1.41	55.88 $\pm$ 4.39	50.86 $\pm$ 0.77	49.51 $\pm$ 0.81	48.81 $\pm$ 1.26	49.56 $\pm$ 0.80	50.21 $\pm$ 1.36	50.27 $\$	

Table 9: Per-dataset 5-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
GesturePebbleZ1	71.74 $\pm$ 3.41	64.53 $\pm$ 3.05	73.60 $\pm$ 2.12	65.35 $\pm$ 3.15	63.95 $\pm$ 8.94	65.89 $\pm$ 3.50	23.45 $\pm$ 1.21	73.64 $\pm$ 2.20	28.95 $\pm$ 4.06	76.16 $\pm$ 6.07	63.95 $\pm$ 8.94	63.84 $\pm$ 8.99
GesturePebbleZ2	67.34 $\pm$ 3.97	52.53 $\pm$ 17.40	73.04 $\pm$ 4.07	58.35 $\pm$ 6.18	48.10 $\pm$ 7.12	62.03 $\pm$ 7.30	19.41 $\pm$ 4.45	64.98 $\pm$ 9.05	24.81 $\pm$ 3.02	60.34 $\pm$ 8.06	48.10 $\pm$ 7.12	60.89 $\pm$ 8.06
GunPoint	82.53 $\pm$ 12.46	77.33 $\pm$ 12.19	79.87 $\pm$ 13.86	59.60 $\pm$ 4.87	73.07 $\pm$ 2.03	66.00 $\pm$ 6.36	55.11 $\pm$ 6.19	70.22 $\pm$ 2.52	65.87 $\pm$ 7.58	66.00 $\pm$ 2.91	62.40 $\pm$ 5.79	90.13 $\pm$ 10.49
GunPointAgeSpan	75.70 $\pm$ 7.16	69.81 $\pm$ 6.17	76.46 $\pm$ 8.05	75.51 $\pm$ 2.81	73.67 $\pm$ 12.74	76.79 $\pm$ 5.43	49.26 $\pm$ 6.34	75.74 $\pm$ 4.37	60.38 $\pm$ 3.16	69.20 $\pm$ 7.84	65.70 $\pm$ 10.14	81.52 $\pm$ 9.45
GunPointMaleVersusFemale	91.20 $\pm$ 8.64	96.65 $\pm$ 4.17	90.32 $\pm$ 9.07	61.14 $\pm$ 7.95	87.53 $\pm$ 12.91	74.37 $\pm$ 13.69	55.17 $\pm$ 0.48	75.00 $\pm$ 13.22	61.20 $\pm$ 4.73	77.95 $\pm$ 14.78	63.92 $\pm$ 13.57	94.62 $\pm$ 6.24
GunPointOldVersusYoung	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	75.87 $\pm$ 2.49	70.29 $\pm$ 8.33	70.79 $\pm$ 4.48	56.93 $\pm$ 3.97	73.23 $\pm$ 4.31	58.98 $\pm$ 5.43	70.48 $\pm$ 8.59	92.38 $\pm$ 8.67	80.44 $\pm$ 8.63
Ham	60.95 $\pm$ 6.21	60.00 $\pm$ 6.77	62.10 $\pm$ 4.59	62.86 $\pm$ 11.45	58.86 $\pm$ 7.48	60.00 $\pm$ 11.70	63.49 $\pm$ 0.55	56.51 $\pm$ 8.59	63.62 $\pm$ 3.26	66.35 $\pm$ 16.28	56.76 $\pm$ 9.56	60.57 $\pm$ 9.91
HandOutlines	46.86 $\pm$ 11.04	52.86 $\pm$ 13.07	67.24 $\pm$ 3.85	80.76 $\pm$ 4.40	74.49 $\pm$ 7.28	83.06 $\pm$ 3.92	47.84 $\pm$ 8.25	81.62 $\pm$ 4.95	35.95 $\pm$ 0.00	82.34 $\pm$ 4.91	75.57 $\pm$ 5.75	68.54 $\pm$ 6.21
Haptics	32.14 $\pm$ 2.79	28.25 $\pm$ 2.44	31.49 $\pm$ 1.92	35.39 $\pm$ 2.49	28.18 $\pm$ 3.44	36.58 $\pm$ 0.19	21.32 $\pm$ 3.65	37.12 $\pm$ 1.14	24.68 $\pm$ 3.46	37.12 $\pm$ 0.94	31.82 $\pm$ 7.39	26.95 $\pm$ 4.88
Herring	50.94 $\pm$ 5.70	55.00 $\pm$ 8.93	52.50 $\pm$ 7.46	51.88 $\pm$ 6.76	50.94 $\pm$ 6.68	48.96 $\pm$ 5.02	57.29 $\pm$ 4.51	51.56 $\pm$ 7.16	46.88 $\pm$ 13.58	49.48 $\pm$ 2.39	49.06 $\pm$ 7.13	45.00 $\pm$ 4.87
HouseTwenty	83.70 $\pm$ 6.31	81.18 $\pm$ 9.08	81.85 $\pm$ 9.95	58.99 $\pm$ 5.13	76.81 $\pm$ 7.55	73.67 $\pm$ 7.34	54.90 $\pm$ 2.57	66.95 $\pm$ 4.78	59.66 $\pm$ 6.85	72.83 $\pm$ 4.23	63.03 $\pm$ 4.28	84.37 $\pm$ 7.51
InlineSkate	20.15 $\pm$ 4.70	22.95 $\pm$ 2.93	19.75 $\pm$ 2.66	21.05 $\pm$ 2.12	18.98 $\pm$ 2.35	24.85 $\pm$ 1.07	16.00 $\pm$ 1.31	24.06 $\pm$ 0.93	16.51 $\pm$ 0.15	24.06 $\pm$ 0.90	19.20 $\pm$ 1.41	28.07 $\pm$ 1.19
InsectEPGRegularTrain	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	56.47 $\pm$ 1.83	78.96 $\pm$ 9.81	59.57 $\pm$ 2.67	38.96 $\pm$ 2.13	57.30 $\pm$ 1.41	53.17 $\pm$ 6.32	61.31 $\pm$ 3.42	89.88 $\pm$ 9.24	80.24 $\pm$ 3.80
InsectEPGSmallTrain	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	53.98 $\pm$ 1.90	77.03 $\pm$ 8.38	59.44 $\pm$ 2.44	52.61 $\pm$ 2.78	60.51 $\pm$ 1.98	55.66 $\pm$ 1.65	62.12 $\pm$ 1.62	83.13 $\pm$ 0.00	83.05 $\pm$ 2.80
InsectWingbeatSound	25.39 $\pm$ 1.91	24.63 $\pm$ 1.27	26.10 $\pm$ 5.93	53.74 $\pm$ 0.54	42.18 $\pm$ 1.58	56.25 $\pm$ 3.65	25.84 $\pm$ 1.75	58.96 $\pm$ 0.98	38.75 $\pm$ 2.42	57.51 $\pm$ 1.69	20.02 $\pm$ 5.68	41.44 $\pm$ 2.25
ItalyPowerDemand	87.66 $\pm$ 4.88	89.29 $\pm$ 4.25	88.36 $\pm$ 4.24	62.66 $\pm$ 11.58	93.39 $\pm$ 3.05	58.50 $\pm$ 7.42	89.57 $\pm$ 5.31	93.78 $\pm$ 2.95	95.72 $\pm$ 1.14	85.46 $\pm$ 2.44	56.64 $\pm$ 14.68	84.26 $\pm$ 5.94
LargeKitchenAppliances	60.00 $\pm$ 4.27	58.51 $\pm$ 9.39	62.67 $\pm$ 10.64	37.55 $\pm$ 2.23	60.48 $\pm$ 4.88	44.09 $\pm$ 3.07	36.98 $\pm$ 2.46	40.44 $\pm$ 3.60	37.71 $\pm$ 3.74	44.09 $\pm$ 6.61	37.17 $\pm$ 1.99	50.13 $\pm$ 3.07
Lightning2	64.92 $\pm$ 6.82	62.30 $\pm$ 7.42	67.87 $\pm$ 5.51	60.33 $\pm$ 3.91	64.92 $\pm$ 4.86	69.95 $\pm$ 2.50	50.82 $\pm$ 5.68	68.85 $\pm$ 5.68	61.64 $\pm$ 7.91	68.31 $\pm$ 3.41	59.02 $\pm$ 6.86	69.51 $\pm$ 10.47
Lightning7	58.90 $\pm$ 7.12	54.25 $\pm$ 7.09	60.55 $\pm$ 3.80	51.23 $\pm$ 2.84	50.96 $\pm$ 8.07	67.58 $\pm$ 3.45	18.26 $\pm$ 3.95	61.64 $\pm$ 3.62	30.41 $\pm$ 0.61	70.32 $\pm$ 2.85	22.19 $\pm$ 2.63	66.30 $\pm$ 3.83
Mallat	71.03 $\pm$ 13.07	70.33 $\pm$ 5.20	93.72 $\pm$ 2.08	88.67 $\pm$ 2.40	77.56 $\pm$ 9.50	94.66 $\pm$ 0.33	65.67 $\pm$ 5.01	87.51 $\pm$ 2.78	79.72 $\pm$ 5.50	90.06 $\pm$ 1.99	56.21 $\pm$ 5.78	90.88 $\pm$ 1.15
Meat	64.33 $\pm$ 14.22	50.00 $\pm$ 9.93	81.00 $\pm$ 8.47	85.33 $\pm$ 10.70	49.33 $\pm$ 15.48	76.67 $\pm$ 10.41	47.78 $\pm$ 15.03	37.22 $\pm$ 5.36	50.00 $\pm$ 11.61	88.89 $\pm$ 2.55	34.00 $\pm$ 1.49	64.00 $\pm$ 11.82
MedicalImages	45.61 $\pm$ 3.94	43.95 $\pm$ 4.74	42.63 $\pm$ 4.22	27.26 $\pm$ 7.75	32.26 $\pm$ 2.93	37.41 $\pm$ 0.59	36.84 $\pm$ 3.23	38.03 $\pm$ 3.53	40.16 $\pm$ 6.60	38.20 $\pm$ 1.38	5.24 $\pm$ 3.29	42.03 $\pm$ 2.75
MelbournePedestrian	76.79 $\pm$ 2.31	74.96 $\pm$ 2.28	77.84 $\pm$ 3.11	11.65 $\pm$ 1.29	58.25 $\pm$ 3.82	62.66 $\pm$ 1.83	70.73 $\pm$ 4.35	71.76 $\pm$ 3.72	70.23 $\pm$ 2.95	67.90 $\pm$ 5.27	58.25 $\pm$ 3.82	65.81 $\pm$ 3.77
MiddlePhalanxOutlineAgeGroup	47.01 $\pm$ 8.99	54.03 $\pm$ 8.97	50.78 $\pm$ 7.01	52.08 $\pm$ 8.63	49.09 $\pm$ 6.85	50.22 $\pm$ 4.61	51.73 $\pm$ 7.75	53.90 $\pm$ 7.87	55.06 $\pm$ 4.76	54.11 $\pm$ 9.46	31.82 $\pm$ 16.75	49.74 $\pm$ 4.72
MiddlePhalanxOutlineCorrect	53.20 $\pm$ 7.83	52.71 $\pm$ 15.62	58.14 $\pm$ 12.45	46.94 $\pm$ 12.76	48.93 $\pm$ 10.44	49.60 $\pm$ 3.67	58.65 $\pm$ 6.95	55.44 $\pm$ 15.07	51.68 $\pm$ 10.94	49.60 $\pm$ 9.23	50.93 $\pm$ 8.48	48.45 $\pm$ 7.93
MiddlePhalanxTW	45.84 $\pm$ 9.64	45.84 $\pm$ 3.45	45.71 $\pm$ 4.74	45.84 $\pm$ 4.10	43.12 $\pm$ 4.67	44.37 $\pm$ 0.37	40.04 $\pm$ 2.62	43.72 $\pm$ 4.56	44.16 $\pm$ 4.95	44.59 $\pm$ 2.28	22.86 $\pm$ 6.13	43.64 $\pm$ 5.71
MixedShapesRegularTrain	63.93 $\pm$ 9.77	61.94 $\pm$ 4.99	68.21 $\pm$ 8.35	68.04 $\pm$ 1.34	68.63 $\pm$ 3.58	67.15 $\pm$ 2.55	20.78 $\pm$ 0.21	70.05 $\pm$ 0.70	30.67 $\pm$ 6.49	67.15 $\pm$ 1.60	64.11 $\pm$ 1.37	76.68 $\pm$ 5.11
MixedShapesSmallTrain	66.24 $\pm$ 12.23	57.21 $\pm$ 4.26	72.37 $\pm$ 7.91	66.37 $\pm$ 2.62	66.13 $\pm$ 3.66	68.30 $\pm$ 1.42	22.61 $\pm$ 0.88	67.60 $\pm$ 3.67	25.23 $\pm$ 4.04	66.78 $\pm$ 1.26	60.00 $\pm$ 5.54	78.71 $\pm$ 1.70
MoteStrain	84.74 $\pm$ 4.04	83.85 $\pm$ 4.63	85.10 $\pm$ 2.03	80.19 $\pm$ 8.46	77.96 $\pm$ 3.58	85.73 $\pm$ 4.20	72.68 $\pm$ 2.25	80.09 $\pm$ 3.68	77.11 $\pm$ 6.04	85.36 $\pm$ 1.87	60.30 $\pm$ 15.64	80.37 $\pm$ 4.50
NonInvasiveFetalECGThorax1	40.80 $\pm$ 1.72	31.63 $\pm$ 3.55	40.80 $\pm$ 1.72	72.10 $\pm$ 1.96	51.79 $\pm$ 9.05	70.09 $\pm$ 3.02	22.04 $\pm$ 0.40	70.72 $\pm$ 1.13	32.89 $\pm$ 2.08	69.19 $\pm$ 0.16	28.56 $\pm$ 4.30	67.42 $\pm$ 0.56
NonInvasiveFetalECGThorax2	41.15 $\pm$ 5.46	43.13 $\pm$ 8.61	41.15 $\pm$ 5.46	78.55 $\pm$ 0.96	60.71 $\pm$ 3.55	75.84 $\pm$ 1.88	37.61 $\pm$ 1.37	75.40 $\pm$ 1.50	50.66 $\pm$ 2.46	74.93 $\pm$ 2.49	37.30 $\pm$ 2.25	75.31 $\pm$ 0.82
OSULeaf	57.85 $\pm$ 5.97	50.74 $\pm$ 10.99	72.23 $\pm$ 4.84	32.40 $\pm$ 1.86	35.37 $\pm$ 3.13	38.84 $\pm$ 1.09	16.80 $\pm$ 1.04	32.64 $\pm$ 2.71	24.05 $\pm$ 2.60	36.91 $\pm$ 1.04	23.06 $\pm$ 4.07	49.92 $\pm$ 6.46
OliveOil	59.33 $\pm$ 13.21	28.67 $\pm$ 13.66	74.00 $\pm$ 6.83	31.33 $\pm$ 9.60	26.67 $\pm$ 10.00	42.22 $\pm$ 25.02	40.00 $\pm$ 10.00	32.22 $\pm$ 13.47	40.00 $\pm$ 21.73	32.22 $\pm$ 22.19	26.67 $\pm$ 10.00	28.67 $\pm$ 11.69
PLAID	24.02 $\pm$ 4.96	24.51 $\pm$ 3.44	22.87 $\pm$ 2.95	30.06 $\pm$ 4.34	47.93 $\pm$ 3.93	36.56 $\pm$ 2.69	26.88 $\pm$ 2.01	40.16 $\pm$ 2.24	27.26 $\pm$ 3.48	37.93 $\pm$ 4.09	47.93 $\pm$ 3.93	52.96 $\pm$ 5.39
PhalangesOutlinesCorrect	56.22 $\pm$ 7.89	55.22 $\pm$ 9.82	56.85 $\pm$ 6.96	52.75 $\pm$ 11.57	53.15 $\pm$ 7.53	51.83 $\pm$ 11.55	50.35 $\pm$ 9.47	51.20 $\pm$ 13.21	53.05 $\pm$ 9.33	50.85 $\pm$ 11.66	53.61 $\pm$ 9.97	53.61 $\pm$ 8.69
Phoneme	19.05 $\pm$ 2.97	18.15 $\pm$ 2.47	19.05 $\pm$ 2.97	6.61 $\pm$ 0.25	9.00 $\pm$ 1.08	10.58 $\pm$ 0.95	6.87 $\pm$ 0.83	9.00 $\pm$ 0.29	9.61 $\pm$ 1.88	9.30 $\pm$ 1.34	8.21 $\pm$ 0.75	15.63 $\pm$ 1.56
PickupGestureWiimoteZ	56.00 $\pm$ 10.20	66.80 $\pm$ 5.93	73.60 $\pm$ 4.56	52.40 $\pm$ 1.67	56.80 $\pm$ 9.55	57.33 $\pm$ 3.06	27.33 $\pm$ 1.15	58.00 $\pm$ 3.46	42.00 $\pm$ 2.00	58.67 $\pm$ 1.15	56.80 $\pm$ 9.55	68.00 $\pm$ 2.45
PigAirwayPressure	25.00 $\pm$ 2.61	13.37 $\pm$ 2.05	48.17 $\pm$ 0.40	5.10 $\pm$ 0.26	4.62 $\pm$ 0.55	4.81 $\pm$ 0.00	4.17 $\pm$ 2.22	6.73 $\pm$ 0.48	4.04 $\pm$ 0.87	5.93 $\pm$ 0.56	4.42 $\pm$ 0.86	9.81 $\pm$ 0.73
PigArtPressure	50.96 $\pm$ 5.81	20.77 $\pm$ 3.29	68.17 $\pm$ 4.63	8.46 $\pm$ 0.65	5.10 $\pm$ 1.82	8.97 $\pm$ 1.21	6.57 $\pm$ 2.00	8.97 $\pm$ 0.73	6.92 $\pm$ 1.65	9.29 $\pm$ 1.11	18.17 $\pm$ 2.24	33.08 $\pm$ 1.33
PigCVP	29.33 $\pm$ 2.90	18.75 $\pm$ 2.40	62.98 $\pm$ 1.73	6.83 $\pm$ 0.22	3.37 $\pm$ 1.32	7.53 $\pm$ 0.56	4.65 $\pm$ 1.11	7.53 $\pm$ 0.28	3.75 $\pm$ 1.24	4.81 $\pm$ 0.48	10.38 $\pm$ 1.21	19.52 $\pm$ 0.55
Plane	100.00 $\pm$ 0.00	98.10 $\pm$ 2.43	99.62 $\pm$ 0.52	88.95 $\pm$ 7.93	95.24 $\pm$ 3.16	97.78 $\pm$ 0.55	96.51 $\pm$ 1.98	95.87 $\pm$ 1.45	97.14 $\pm$ 1.51	95.87 $\pm$ 0.55	23.05 $\pm$ 11.32	99.24 $\pm$ 1.24
PowerCons	80.56 $\pm$ 3.36	84.44 $\pm$ 5.02	84.00 $\pm$ 6.30	70.11 $\pm$ 5.26	81.11 $\pm$ 10.74	73.37 $\pm$ 5.45	54.26 $\pm$ 3.90	80.93 $\pm$ 10.72	52.22 $\pm$ 4.39	78.52 $\pm$ 10.32	68.89 $\pm$ 15.97	76.11 $\pm$ 6.55
ProximalPhalanxOutlineAgeGroup	73.37 $\pm$ 8.19	81.76 $\pm$ 3.12	79.90 $\pm$ 3.95	65.56 $\pm$ 9.78	75.41 $\pm$ 5.61	79.02 $\pm$ 7.19	79.02 $\pm$					

Table 9: Per-dataset 5-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
SemgHandSubjectCh2	37.11 $\pm$ 4.58	44.09 $\pm$ 1.20	35.60 $\pm$ 3.13	50.76 $\pm$ 1.90	49.02 $\pm$ 0.83	46.67 $\pm$ 2.12	27.78 $\pm$ 0.59	51.19 $\pm$ 2.74	30.62 $\pm$ 1.93	41.48 $\pm$ 6.03	36.62 $\pm$ 4.30	54.31 $\pm$ 3.37
ShakeGestureWiimoteZ	80.40 $\pm$ 4.34	91.60 $\pm$ 2.61	89.20 $\pm$ 1.10	56.40 $\pm$ 1.67	72.80 $\pm$ 5.22	62.00 $\pm$ 0.00	35.33 $\pm$ 1.15	60.67 $\pm$ 3.06	44.00 $\pm$ 2.45	55.33 $\pm$ 2.31	72.80 $\pm$ 5.22	92.00 $\pm$ 0.00
ShapeletSim	82.78 $\pm$ 10.96	99.67 $\pm$ 0.75	92.56 $\pm$ 6.55	49.44 $\pm$ 2.69	88.22 $\pm$ 10.06	55.93 $\pm$ 2.31	49.81 $\pm$ 2.80	49.63 $\pm$ 4.21	50.67 $\pm$ 1.82	49.44 $\pm$ 3.33	49.67 $\pm$ 3.98	92.00 $\pm$ 3.88
ShapesAll	57.57 $\pm$ 3.68	48.33 $\pm$ 5.58	57.57 $\pm$ 3.68	58.13 $\pm$ 1.68	51.50 $\pm$ 3.69	64.67 $\pm$ 1.64	9.22 $\pm$ 0.79	65.83 $\pm$ 3.88	13.83 $\pm$ 3.47	59.22 $\pm$ 2.28	56.47 $\pm$ 1.46	76.20 $\pm$ 1.83
SmallKitchenAppliances	59.47 $\pm$ 8.51	57.39 $\pm$ 6.38	62.35 $\pm$ 6.85	39.04 $\pm$ 4.11	57.55 $\pm$ 2.52	57.51 $\pm$ 6.62	38.67 $\pm$ 2.54	39.56 $\pm$ 5.26	46.67 $\pm$ 3.53	56.62 $\pm$ 3.96	44.00 $\pm$ 7.15	51.63 $\pm$ 12.25
SmoothSubspace	86.80 $\pm$ 2.13	82.00 $\pm$ 2.83	88.67 $\pm$ 2.36	33.87 $\pm$ 1.19	57.07 $\pm$ 4.58	65.78 $\pm$ 2.04	69.33 $\pm$ 3.71	61.11 $\pm$ 0.77	77.07 $\pm$ 5.51	84.00 $\pm$ 4.81	37.07 $\pm$ 6.74	88.00 $\pm$ 2.87
SonyAIBORobotSurface1	93.94 $\pm$ 2.22	95.71 $\pm$ 1.41	92.85 $\pm$ 2.46	61.83 $\pm$ 10.72	84.73 $\pm$ 2.02	71.21 $\pm$ 3.16	67.39 $\pm$ 1.20	71.60 $\pm$ 6.11	64.96 $\pm$ 2.40	64.89 $\pm$ 3.03	54.54 $\pm$ 12.34	84.86 $\pm$ 2.67
SonyAIBORobotSurface2	92.51 $\pm$ 3.32	91.14 $\pm$ 0.89	91.54 $\pm$ 4.19	78.15 $\pm$ 2.59	85.83 $\pm$ 4.10	79.96 $\pm$ 1.73	83.11 $\pm$ 2.73	81.78 $\pm$ 2.73	80.76 $\pm$ 4.57	83.21 $\pm$ 2.88	61.15 $\pm$ 14.84	77.21 $\pm$ 3.67
StarLightCurves	81.52 $\pm$ 8.85	60.67 $\pm$ 16.70	76.44 $\pm$ 1.22	74.70 $\pm$ 4.25	76.72 $\pm$ 3.78	75.63 $\pm$ 4.10	28.48 $\pm$ 9.45	75.33 $\pm$ 3.39	27.99 $\pm$ 0.00	75.69 $\pm$ 5.23	75.95 $\pm$ 2.54	81.86 $\pm$ 7.93
Strawberry	63.51 $\pm$ 2.99	64.22 $\pm$ 4.90	60.22 $\pm$ 9.83	58.49 $\pm$ 2.90	63.03 $\pm$ 7.09	61.89 $\pm$ 4.92	67.21 $\pm$ 4.90	59.82 $\pm$ 1.80	61.03 $\pm$ 3.64	65.77 $\pm$ 7.40	52.92 $\pm$ 6.34	58.05 $\pm$ 7.04
SwedishLeaf	79.81 $\pm$ 1.70	76.00 $\pm$ 4.97	79.81 $\pm$ 1.70	47.97 $\pm$ 1.68	65.66 $\pm$ 4.54	62.61 $\pm$ 4.53	49.23 $\pm$ 4.26	67.15 $\pm$ 4.00	56.93 $\pm$ 3.43	62.29 $\pm$ 1.68	15.90 $\pm$ 6.80	79.78 $\pm$ 2.56
Symbols	88.78 $\pm$ 7.99	88.74 $\pm$ 7.74	91.24 $\pm$ 0.96	83.44 $\pm$ 1.29	84.92 $\pm$ 3.31	84.29 $\pm$ 1.01	36.11 $\pm$ 0.23	86.40 $\pm$ 1.63	58.19 $\pm$ 7.40	83.75 $\pm$ 0.90	27.56 $\pm$ 4.43	88.04 $\pm$ 1.30
SyntheticControl	87.80 $\pm$ 2.38	96.33 $\pm$ 0.33	92.60 $\pm$ 1.01	48.53 $\pm$ 3.68	92.73 $\pm$ 2.94	73.22 $\pm$ 3.10	52.11 $\pm$ 3.20	66.22 $\pm$ 3.66	56.27 $\pm$ 3.45	72.78 $\pm$ 3.66	20.93 $\pm$ 4.87	95.07 $\pm$ 1.62
ToeSegmentation1	85.96 $\pm$ 7.05	77.37 $\pm$ 5.14	81.58 $\pm$ 5.14	54.74 $\pm$ 3.58	68.16 $\pm$ 3.79	55.56 $\pm$ 2.16	53.95 $\pm$ 2.74	51.61 $\pm$ 0.91	53.25 $\pm$ 3.38	54.82 $\pm$ 1.32	48.77 $\pm$ 4.31	70.53 $\pm$ 10.39
ToeSegmentation2	74.77 $\pm$ 12.53	<u>77.85 $\pm$ 7.33</u>	<u>74.77 $\pm$ 8.79</u>	50.77 $\pm$ 2.88	68.46 $\pm$ 6.25	58.97 $\pm$ 1.60	50.26 $\pm$ 0.44	59.49 $\pm$ 6.72	37.54 $\pm$ 3.14	64.10 $\pm$ 2.70	52.31 $\pm$ 9.12	78.00 $\pm$ 6.36
Trace	88.60 $\pm$ 7.37	<u>99.00 $\pm$ 1.00</u>	93.00 $\pm$ 8.00	53.00 $\pm$ 4.42	76.40 $\pm$ 5.77	51.67 $\pm$ 4.93	53.33 $\pm$ 2.08	49.67 $\pm$ 3.79	59.80 $\pm$ 2.59	55.67 $\pm$ 8.50	48.80 $\pm$ 10.71	100.00 $\pm$ 0.00
TwoLeadECG	97.98 $\pm$ 4.17	95.47 $\pm$ 5.89	75.84 $\pm$ 5.58	58.75 $\pm$ 2.38	82.55 $\pm$ 14.28	62.16 $\pm$ 1.52	60.81 $\pm$ 0.10	56.34 $\pm$ 2.59	61.88 $\pm$ 1.65	68.51 $\pm$ 9.61	52.10 $\pm$ 2.73	96.68 $\pm$ 5.58
TwoPatterns	64.62 $\pm$ 3.64	74.98 $\pm$ 5.04	74.94 $\pm$ 1.43	38.09 $\pm$ 3.07	51.91 $\pm$ 7.48	40.83 $\pm$ 3.89	26.00 $\pm$ 0.92	39.83 $\pm$ 3.80	26.50 $\pm$ 1.83	37.54 $\pm$ 1.63	27.50 $\pm$ 4.13	74.22 $\pm$ 2.81
UMD	87.08 $\pm$ 10.10	91.53 $\pm$ 5.28	96.67 $\pm$ 2.05	56.53 $\pm$ 3.69	74.44 $\pm$ 17.00	61.57 $\pm$ 1.45	62.96 $\pm$ 5.82	68.52 $\pm$ 10.79	71.81 $\pm$ 2.17	70.14 $\pm$ 10.69	48.47 $\pm$ 7.32	95.78 $\pm$ 2.33
UWaveGestureLibraryAll	42.53 $\pm$ 1.15	39.33 $\pm$ 5.11	44.53 $\pm$ 1.84	77.02 $\pm$ 5.21	57.17 $\pm$ 2.24	78.18 $\pm$ 3.39	26.42 $\pm$ 1.62	<u>77.47 $\pm$ 3.61</u>	35.67 $\pm$ 1.53	76.96 $\pm$ 4.32	65.94 $\pm$ 6.48	48.36 $\pm$ 0.30
UWaveGestureLibraryX	39.64 $\pm$ 3.12	46.31 $\pm$ 4.46	46.23 $\pm$ 4.40	56.88 $\pm$ 2.64	50.95 $\pm$ 2.22	58.42 $\pm$ 2.37	19.27 $\pm$ 3.18	58.58 $\pm$ 2.34	31.22 $\pm$ 3.55	57.86 $\pm$ 1.47	26.48 $\pm$ 5.59	47.36 $\pm$ 0.30
UWaveGestureLibraryY	37.00 $\pm$ 1.20	38.44 $\pm$ 2.57	40.23 $\pm$ 3.63	52.69 $\pm$ 2.24	48.41 $\pm$ 3.12	51.42 $\pm$ 3.10	21.22 $\pm$ 1.11	51.72 $\pm$ 2.98	30.92 $\pm$ 5.43	50.81 $\pm$ 4.53	19.79 $\pm$ 3.07	51.36 $\pm$ 0.31
UWaveGestureLibraryZ	38.74 $\pm$ 2.41	44.95 $\pm$ 4.51	40.85 $\pm$ 4.13	48.35 $\pm$ 3.41	47.76 $\pm$ 4.10	50.34 $\pm$ 2.37	20.23 $\pm$ 1.68	49.48 $\pm$ 1.66	31.23 $\pm$ 4.16	48.05 $\pm$ 3.36	25.20 $\pm$ 6.78	55.54 $\pm$ 2.61
Wafer	76.83 $\pm$ 9.15	69.99 $\pm$ 14.88	71.79 $\pm$ 15.46	55.60 $\pm$ 16.77	73.34 $\pm$ 19.66	80.42 $\pm$ 16.85	76.25 $\pm$ 9.34	83.86 $\pm$ 16.37	75.80 $\pm$ 12.54	83.26 $\pm$ 17.85	64.01 $\pm$ 22.29	71.65 $\pm$ 9.58
Wine	57.41 $\pm$ 13.03	48.15 $\pm$ 3.93	61.11 $\pm$ 5.40	55.93 $\pm$ 3.56	53.33 $\pm$ 5.62	47.53 $\pm$ 8.35	54.94 $\pm$ 8.55	50.00 $\pm$ 0.00	46.67 $\pm$ 6.47	37.65 $\pm$ 19.80	50.00 $\pm$ 0.00	51.48 $\pm$ 2.03
WordSynonyms	25.08 $\pm$ 1.54	21.22 $\pm$ 1.56	25.08 $\pm$ 1.54	29.56 $\pm$ 2.06	30.38 $\pm$ 2.74	45.87 $\pm$ 4.63	7.99 $\pm$ 1.24	44.67 $\pm$ 4.83	18.81 $\pm$ 2.07	41.33 $\pm$ 4.67	22.26 $\pm$ 2.64	47.05 $\pm$ 3.87
Worms	57.14 $\pm$ 9.76	55.58 $\pm$ 5.91	62.86 $\pm$ 4.55	24.68 $\pm$ 4.68	40.26 $\pm$ 5.11	22.94 $\pm$ 0.75	19.91 $\pm$ 3.27	25.97 $\pm$ 0.00	16.62 $\pm$ 2.32	22.08 $\pm$ 1.30	22.08 $\pm$ 5.11	44.42 $\pm$ 6.26
WormsTwoClass	68.05 $\pm$ 7.21	59.74 $\pm$ 6.69	62.08 $\pm$ 7.97	48.83 $\pm$ 3.13	51.17 $\pm$ 8.09	52.38 $\pm$ 2.70	42.86 $\pm$ 3.44	51.08 $\pm$ 5.25	48.83 $\pm$ 3.26	52.81 $\pm$ 3.27	48.83 $\pm$ 5.92	54.81 $\pm$ 7.59
Yoga	55.71 $\pm$ 3.37	53.57 $\pm$ 4.72	<u>55.19 $\pm$ 2.57</u>	50.13 $\pm$ 2.22	53.89 $\pm$ 3.78	49.79 $\pm$ 3.21	52.94 $\pm$ 1.57	47.50 $\pm$ 3.55	51.67 $\pm$ 1.99	49.68 $\pm$ 4.28	48.87 $\pm$ 2.05	51.31 $\pm$ 4.34
Avg. Acc.	61.27	59.47	<u>63.55</u>	52.30	56.94	57.86	43.20	57.20	47.30	58.35	42.94	64.14
Avg. Rank	4.77	5.34	<u>4.23</u>	7.73	6.60	5.69	9.55	5.94	8.70	5.79	9.48	4.19
#Best	21	13	<u>30</u>	6	4	8	4	7	2	8	1	36

Table 10: Per-dataset 10-shot classification results on 128 shared UCR datasets. Each value is reported as mean $\pm$ standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
ACSF1	85.40 $\pm$ 2.19	76.80 $\pm$ 3.35	80.80 $\pm$ 0.84	38.60 $\pm$ 0.89	58.80 $\pm$ 2.59	64.67 $\pm$ 0.58	42.67 $\pm$ 1.53	67.00 $\pm$ 2.00	53.60 $\pm$ 2.51	67.00 $\pm$ 2.00	36.80 $\pm$ 3.27	81.33 $\pm$ 1.53
Adiac	62.46 $\pm$ 4.53	56.98 $\pm$ 3.84	62.46 $\pm$ 4.53	45.32 $\pm$ 1.06	47.72 $\pm$ 5.48	56.01 $\pm$ 2.77	48.68 $\pm$ 0.78	47.91 $\pm$ 6.68	71.71 $\pm$ 1.94	48.00 $\pm$ 3.58	12.84 $\pm$ 1.42	76.42 $\pm$ 1.68
AllGestureWiimoteX	49.37 $\pm$ 3.15	43.11 $\pm$ 2.57	39.26 $\pm$ 2.50	23.91 $\pm$ 1.60	43.54 $\pm$ 2.85	37.14 $\pm$ 0.52	15.00 $\pm$ 0.87	35.19 $\pm$ 1.33	25.80 $\pm$ 2.24	31.00 $\pm$ 1.57	43.54 $\pm$ 2.85	52.31 $\pm$ 2.26
AllGestureWiimoteY	54.89 $\pm$ 2.09	50.74 $\pm$ 4.62	40.17 $\pm$ 2.24	27.26 $\pm$ 1.11	48.03 $\pm$ 1.58	42.57 $\pm$ 3.19	16.95 $\pm$ 2.22	42.57 $\pm$ 3.81	30.37 $\pm$ 4.65	41.24 $\pm$ 2.84	48.03 $\pm$ 1.58	56.34 $\pm$ 3.54
AllGestureWiimoteZ	49.34 $\pm$ 1.38	46.03 $\pm$ 2.76	39.80 $\pm$ 2.97	22.43 $\pm$ 1.80	34.54 $\pm$ 4.20	31.29 $\pm$ 1.13	16.67 $\pm$ 1.15	29.67 $\pm$ 3.32	21.71 $\pm$ 1.69	32.19 $\pm$ 0.59	34.54 $\pm$ 4.20	44.43 $\pm$ 2.91
ArrowHead	72.23 $\pm$ 5.22	63.89 $\pm$ 7.06	65.49 $\pm$ 9.09	64.34 $\pm$ 0.65	62.86 $\pm$ 7.57	60.38 $\pm$ 1.19	48.19 $\pm$ 2.58	59.62 $\pm$ 2.88	57.83 $\pm$ 5.80	63.43 $\pm$ 3.48	45.83 $\pm$ 13.23	81.49 $\pm$ 3.22
BME	64.27 $\pm$ 9.29	69.33 $\pm$ 1.56	78.67 $\pm$ 4.64	64.53 $\pm$ 0.30	97.60 $\pm$ 3.29	71.11 $\pm$ 2.34	78.89 $\pm$ 1.02	88.44 $\pm$ 1.02	91.33 $\pm$ 2.31	86.67 $\pm$ 0.00	46.40 $\pm$ 7.46	99.20 $\pm$ 0.30
Beef	51.33 $\pm$ 6.91	44.00 $\pm$ 6.83	52.00 $\pm$ 2.98	61.33 $\pm$ 4.47	46.00 $\pm$ 6.41	70.00 $\pm$ 3.33	51.11 $\pm$ 5.09	54.44 $\pm$ 3.85	64.00 $\pm$ 7.23	80.00 $\pm$ 0.00	37.33 $\pm$ 4.35	68.00 $\pm$ 1.83
BeetleFly	71.00 $\pm$ 8.94	74.00 $\pm$ 6.52	75.00 $\pm$ 6.12	86.00 $\pm$ 4.18	86.00 $\pm$ 2.24	81.67 $\pm$ 2.89	48.33 $\pm$ 2.89	83.33 $\pm$ 2.89	49.00 $\pm$ 11.94	75.00 $\pm$ 5.00	70.00 $\pm$ 12.25	96.67 $\pm$ 2.89
BirdChicken	84.00 $\pm$ 12.45	92.00 $\pm$ 5.70	94.00 $\pm$ 4.18	68.00 $\pm$ 2.74	81.00 $\pm$ 9.62	71.67 $\pm$ 7.64	41.67 $\pm$ 7.64	68.33 $\pm$ 16.07	56.00 $\pm$ 5.48	56.67 $\pm$ 7.64	56.00 $\pm$ 4.18	89.00 $\pm$ 5.48
CBF	98.96 $\pm$ 0.23	98.27 $\pm$ 0.96	99.24 $\pm$ 0.32	67.49 $\pm$ 2.09	95.98 $\pm$ 4.64	90.07 $\pm$ 1.09	39.37 $\pm$ 1.51	91.44 $\pm$ 1.93	51.24 $\pm$ 7.49	90.11 $\pm$ 0.88	49.31 $\pm$ 9.05	97.71 $\pm$ 0.69
Car	57.67 $\pm$ 10.84	58.67 $\pm$ 3.21	65.33 $\pm$ 2.17	73.00 $\pm$ 2.98	62.67 $\pm$ 5.22	70.00 $\pm$ 2.89	27.22 $\pm$ 5.85	65.56 $\pm$ 0.96	37.33 $\pm$ 3.84	72.22 $\pm$ 1.92	50.33 $\pm$ 8.28	76.33 $\pm$ 5.45
Chinatown	98.19 $\pm$ 0.13	98.13 $\pm$ 0.16	98.54 $\pm$ 0.21	54.58 $\pm$ 24.80	97.73 $\pm$ 0.75	94.75 $\pm$ 2.10	95.63 $\pm$ 1.01	97.28 $\pm$ 0.61	96.62 $\pm$ 0.60	96.60 $\pm$ 1.10	63.56 $\pm$ 20.21	94.56 $\pm$ 3.09
ChlorineConcentration	41.18 $\pm$ 3.66	34.41 $\pm$ 8.63	37.86 $\pm$ 5.20	34.42 $\pm$ 4.40	35.51 $\pm$ 6.32	35.06 $\pm$ 1.39	38.85 $\pm$ 1.62	32.99 $\pm$ 4.01	39.05 $\pm$ 4.08	39.50 $\pm$ 4.07	33.01 $\pm$ 11.68	34.77 $\pm$ 6.93
CinCECGTorso	54.03 $\pm$ 3.03	65.94 $\pm$ 5.62	64.64 $\pm$ 5.11	45.36 $\pm$ 1.09	80.65 $\pm$ 7.24	81.59 $\pm$ 1.76	35.82 $\pm$ 2.41	84.42 $\pm$ 2.94	63.93 $\pm$ 1.27	85.19 $\pm$ 1.24	52.77 $\pm$ 4.47	94.71 $\pm$ 4.65
Coffee	97.14 $\pm$ 6.39	97.14 $\pm$ 1.60	100.00 $\pm$ 0.00	96.43 $\pm$ 4.37	97.86 $\pm$ 1.96	91.67 $\pm$ 2.06	86.90 $\pm$ 16.50	92.86 $\pm$ 0.00	98.57 $\pm$ 3.19	90.48 $\pm$ 2.06	56.43 $\pm$ 7.74	95.71 $\pm$ 2.99
Computers	66.48 $\pm$ 3.03	63.20 $\pm$ 6.81	68.08 $\pm$ 4.62	48.88 $\pm$ 2.52	56.40 $\pm$ 10.98	52.13 $\pm$ 9.60	52.13 $\pm$ 6.02	50.93 $\pm$ 12.73	49.68 $\pm$ 5.04	53.87 $\pm$ 12.94	51.52 $\pm$ 3.69	65.60 $\pm$ 6.66
CricketX	60.72 $\pm$ 2.35	55.18 $\pm$ 4.36	51.28 $\pm$ 3.40	30.15 $\pm$ 2.84	42.92 $\pm$ 1.99	46.50 $\pm$ 1.21	14.27 $\pm$ 1.71	45.38 $\pm$ 2.45	20.87 $\pm$ 2.94	45.38 $\pm$ 1.94	19.33 $\pm$ 2.95	66.92 $\pm$ 2.71
CricketY	56.92 $\pm$ 2.81	49.44 $\pm$ 1.94	47.69 $\pm$ 2.43	32.82 $\pm$ 4.21	42.15 $\pm$ 5.17	44.62 $\pm$ 0.44	9.32 $\pm$ 1.04	44.02 $\pm$ 2.09	17.90 $\pm$ 2.44	41.11 $\pm$ 0.53	23.95 $\pm$ 3.97	57.95 $\pm$ 1.33
CricketZ	63.18 $\pm$ 1.66	57.38 $\pm$ 2.58	48.82 $\pm$ 4.98	30.56 $\pm$ 3.52	43.18 $\pm$ 5.27	46.32 $\pm$ 1.71	12.91 $\pm$ 2.15	47.78 $\pm$ 1.41	18.26 $\pm$ 2.08	45.30 $\pm$ 1.50	21.95 $\pm$ 3.02	65.30 $\pm$ 1.16
Crop	57.92 $\pm$ 1.22	54.52 $\pm$ 1.67	57.92 $\pm$ 1.22	42.46 $\pm$ 2.10	44.96 $\pm$ 1.35	54.14 $\pm$ 2.07	52.84 $\pm$ 0.15	54.82 $\pm$ 1.79	54.74 $\pm$ 0.79	54.60 $\pm$ 1.45	25.02 $\pm$ 2.10	57.97 $\pm$ 1.25
DiatomSizeReduction	87.32 $\pm$ 3.50	71.05 $\pm$ 13.58	96.80 $\pm$ 0.91	44.77 $\pm$ 1.16	62.68 $\pm$ 4.64	87.04 $\pm$ 1.80	66.88 $\pm$ 2.86	51.96 $\pm$ 5.03	48.89 $\pm$ 18.63	87.47 $\pm$ 0.19	30.13 $\pm$ 0.15	92.61 $\pm$ 2.00
DistalPhalanxOutlineAgeGroup	68.20 $\pm$ 2.98	68.78 $\pm$ 4.24	70.50 $\pm$ 1.61	69.50 $\pm$ 1.66	69.50 $\pm$ 2.81	66.67 $\pm$ 1.10	65.71 $\pm$ 3.62	70.50 $\pm$ 1.25	68.78 $\pm$ 2.72	66.91 $\pm$ 1.25	48.06 $\pm$ 11.76	70.65 $\pm$ 2.18
DistalPhalanxOutlineCorrect	64.78 $\pm$ 12.49	65.07 $\pm$ 11.29	63.62 $\pm$ 12.87	50.00 $\pm$ 13.55	60.14 $\pm$ 9.92	58.94 $\pm$ 4.04	67.15 $\pm$ 2.41	55.43 $\pm$ 14.84	62.39 $\pm$ 9.46	61.71 $\pm$ 6.72	47.61 $\pm$ 8.80	60.51 $\pm$ 11.85
DistalPhalanxTW	55.83 $\pm$ 3.65	59.14 $\pm$ 2.12	60.58 $\pm$ 3.58	47.34 $\pm$ 6.42	57.55 $\pm$ 6.08	53.96 $\pm$ 3.14	53.00 $\pm$ 2.91	54.92 $\pm$ 0.83	52.52 $\pm$ 4.35	54.92 $\pm$ 2.91	32.37 $\pm$ 14.50	54.10 $\pm$ 3.11
DodgerLoopDay	49.25 $\pm$ 3.71	47.25 $\pm$ 4.28	39.75 $\pm$ 3.11	41.50 $\pm$ 2.24	47.00 $\pm$ 4.11	49.58 $\pm$ 5.64	28.75 $\pm$ 2.50	45.42 $\pm$ 1.91	29.00 $\pm$ 2.40	48.75 $\pm$ 1.25	47.00 $\pm$ 4.11	48.75 $\pm$ 1.77
DodgerLoopGame	68.70 $\pm$ 6.09	83.33 $\pm$ 5.28	75.07 $\pm$ 2.64	79.28 $\pm$ 0.97	88.84 $\pm$ 1.67	76.81 $\pm$ 1.26	48.79 $\pm$ 3.42	81.40 $\pm$ 1.51	53.48 $\pm$ 2.83	78.99 $\pm$ 1.45	88.84 $\pm$ 1.67	83.62 $\pm$ 0.65
DodgerLoopWeekend	94.06 $\pm$ 1.39	94.20 $\pm$ 2.11	96.81 $\pm$ 0.83	97.83 $\pm$ 0.00	96.81 $\pm$ 0.65	96.86 $\pm$ 0.42	72.71 $\pm$ 1.82	97.58 $\pm$ 0.42	76.38 $\pm$ 1.82	97.10 $\pm$ 0.00	96.81 $\pm$ 0.65	97.83 $\pm$ 0.00
ECG200	80.00 $\pm$ 4.00	83.00 $\pm$ 2.00	80.40 $\pm$ 4.28	77.60 $\pm$ 4.34	79.80 $\pm$ 6.53	79.33 $\pm$ 3.21	78.00 $\pm$ 3.61	81.00 $\pm$ 5.00	82.00 $\pm$ 2.92	82.00 $\pm$ 3.61	63.00 $\pm$ 15.26	77.40 $\pm$ 4.88
ECG5000	83.56 $\pm$ 5.09	78.23 $\pm$ 2.20	80.72 $\pm$ 4.37	88.37 $\pm$ 1.22	76.31 $\pm$ 5.83	79.59 $\pm$ 2.75	73.64 $\pm$ 2.21	84.75 $\pm$ 1.16	79.28 $\pm$ 5.11	82.19 $\pm$ 1.79	24.56 $\pm$ 15.15	73.78 $\pm$ 6.18
ECGFiveDays	95.40 $\pm$ 1.88	88.99 $\pm$ 4.44	97.56 $\pm$ 1.20	74.87 $\pm$ 4.61	94.61 $\pm$ 2.00	83.82 $\pm$ 7.05	96.44 $\pm$ 1.42	91.64 $\pm$ 1.90	97.51 $\pm$ 1.12	86.18 $\pm$ 8.29	57.47 $\pm$ 4.38	75.28 $\pm$ 1.71
EOGHorizontalSignal	30.61 $\pm$ 3.99	43.15 $\pm$ 5.84	30.94 $\pm$ 3.48	29.89 $\pm$ 1.60	38.78 $\pm$ 6.14	39.59 $\pm$ 1.67	8.01 $\pm$ 1.27	41.62 $\pm$ 1.42	10.61 $\pm$ 3.44	40.06 $\pm$ 3.72	39.56 $\pm$ 4.17	49.17 $\pm$ 1.62
EOGVerticalSignal	21.71 $\pm$ 2.53	28.23 $\pm$ 2.47	19.83 $\pm$ 3.08	31.55 $\pm$ 2.19	40.06 $\pm$ 1.98	36.00 $\pm$ 1.12	8.47 $\pm$ 0.84	37.75 $\pm$ 1.67	11.99 $\pm$ 1.33	37.02 $\pm$ 1.99	30.55 $\pm$ 3.17	42.21 $\pm$ 2.26
Earthquakes	53.81 $\pm$ 8.62	53.96 $\pm$ 9.50	51.37 $\pm$ 3.47	52.52 $\pm$ 6.49	54.82 $\pm$ 2.85	60.91 $\pm$ 4.63	49.16 $\pm$ 8.34	59.71 $\pm$ 6.86	54.68 $\pm$ 5.26	61.87 $\pm$ 3.60	47.91 $\pm$ 6.66	54.68 $\pm$ 7.63
ElectricDevices	40.12 $\pm$ 4.42	38.65 $\pm$ 3.96	36.00 $\pm$ 4.64	28.04 $\pm$ 1.63	39.08 $\pm$ 8.26	48.59 $\pm$ 6.21	30.07 $\pm$ 3.26	33.74 $\pm$ 3.43	37.44 $\pm$ 1.49	48.14 $\pm$ 2.51	16.08 $\pm$ 5.57	49.42 $\pm$ 5.02
EthanolLevel	26.56 $\pm$ 1.96	25.76 $\pm$ 1.05	26.16 $\pm$ 0.71	29.28 $\pm$ 1.78	25.92 $\pm$ 1.70	28.07 $\pm$ 2.55	25.20 $\pm$ 3.34	26.80 $\pm$ 0.69	25.60 $\pm$ 1.13	29.87 $\pm$ 2.47	25.56 $\pm$ 1.05	28.24 $\pm$ 0.50
FaceAll	77.73 $\pm$ 3.48	80.30 $\pm$ 1.67	77.73 $\pm$ 3.48	67.09 $\pm$ 2.40	54.49 $\pm$ 5.41	67.46 $\pm$ 1.82	59.68 $\pm$ 1.83	67.32 $\pm$ 2.82	60.69 $\pm$ 2.63	64.06 $\pm$ 2.32	32.89 $\pm$ 2.71	63.01 $\pm$ 1.34
FaceFour	87.95 $\pm$ 3.07	59.32 $\pm$ 7.51	92.95 $\pm$ 2.03	74.77 $\pm$ 0.95	66.36 $\pm$ 10.77	85.98 $\pm$ 1.31	65.91 $\pm$ 2.27	88.26 $\pm$ 0.66	69.55 $\pm$ 1.69	83.71 $\pm$ 1.31	34.55 $\pm$ 3.82	79.09 $\pm$ 1.90
FacesUCR	87.80 $\pm$ 1.60	84.26 $\pm$ 2.06	82.41 $\pm$ 2.48	63.80 $\pm$ 1.79	54.40 $\pm$ 3.05	80.54 $\pm$ 1.15	70.11 $\pm$ 1.05	83.69 $\pm$ 1.13	72.08 $\pm$ 1.16	79.51 $\pm$ 1.53	34.67 $\pm$ 3.28	79.08 $\pm$ 1.41
FiftyWords	45.36 $\pm$ 2.39	36.70 $\pm$ 1.89	45.36 $\pm$ 2.39	53.67 $\pm$ 1.47	50.46 $\pm$ 2.83	64.32 $\pm$ 1.87	16.85 $\pm$ 0.67	67.99 $\pm$ 1.84	37.05 $\pm$ 3.62	57.36 $\pm$ 1.54	56.40 $\pm$ 1.98	71.34 $\pm$ 2.10
Fish	83.31 $\pm$ 3.78	56.00 $\pm$ 4.28	71.54 $\pm$ 6.28	65.60 $\pm$ 3.49	62.74 $\pm$ 2.15	68.95 $\pm$ 4.29	24.38 $\pm$ 3.80	71.05 $\pm$ 1.44	57.14 $\pm$ 3.45	75.05 $\pm$ 2.31	38.17 $\pm$ 5.05	78.29 $\pm$ 3.92
FordA	78.26 $\pm$ 5.14	75.61 $\pm$ 0.71	78.70 $\pm$ 3.26	49.09 $\pm$ 0.93	67.88 $\pm$ 3.90	51.99 $\pm$ 2.52	52.22 $\pm$ 0.95	48.81 $\pm$ 1.31	53.05 $\pm$ 1.09	48.21 $\pm$ 1.67	50.06 $\pm$ 2.65	57.15 $\pm$ 1.85
FordB	63.83 $\pm$ 5.74	62.17 $\pm$ 3.33	61.70 $\pm$ 3.19	48.54 $\pm$ 0.72	59.06 $\pm$ 1.85	46.75 $\pm$ 0.61	49.79 $\pm$ 1.45	47.24 $\pm$ 1.24	5			

Table 10: Per-dataset 10-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
GesturePebbleZ1	76.86 ± 5.40	75.47 ± 8.13	77.67 ± 2.20	66.63 ± 2.78	68.84 ± 4.61	75.00 ± 1.16	22.29 ± 3.87	77.33 ± 2.10	38.95 ± 10.62	81.40 ± 2.53	68.84 ± 4.61	75.12 ± 5.19
GesturePebbleZ2	73.16 ± 4.51	63.42 ± 5.15	73.16 ± 2.82	60.38 ± 3.46	67.59 ± 4.18	69.62 ± 2.90	19.20 ± 0.73	74.26 ± 2.56	28.86 ± 9.85	73.00 ± 1.59	67.59 ± 4.18	75.57 ± 4.64
GunPoint	85.73 ± 9.32	82.13 ± 8.02	91.73 ± 11.16	68.27 ± 5.09	77.60 ± 4.39	71.78 ± 3.79	64.00 ± 6.57	72.00 ± 1.15	77.73 ± 11.90	76.67 ± 6.67	62.00 ± 16.53	97.47 ± 2.23
GunPointAgeSpan	89.56 ± 7.42	72.47 ± 9.28	74.87 ± 11.59	83.54 ± 2.13	82.22 ± 4.04	85.44 ± 1.67	52.22 ± 3.12	85.13 ± 2.39	71.08 ± 9.43	83.54 ± 3.02	53.61 ± 10.43	92.72 ± 5.80
GunPointMaleVersusFemale	92.78 ± 8.49	93.48 ± 5.65	92.22 ± 9.65	67.09 ± 4.97	95.57 ± 2.50	91.14 ± 4.91	60.97 ± 6.06	93.57 ± 3.96	74.05 ± 7.11	94.41 ± 2.22	67.22 ± 11.33	98.54 ± 0.94
GunPointOldVersusYoung	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	76.19 ± 1.96	79.17 ± 3.15	78.20 ± 2.30	57.46 ± 1.38	77.25 ± 5.77	61.90 ± 5.09	79.26 ± 2.47	85.78 ± 19.21	86.79 ± 4.71
Ham	60.76 ± 5.19	66.48 ± 2.64	65.52 ± 4.83	71.24 ± 5.65	62.86 ± 4.86	66.03 ± 7.40	63.49 ± 10.09	68.89 ± 6.69	61.71 ± 8.23	68.89 ± 7.02	61.52 ± 3.79	62.86 ± 3.37
HandOutlines	50.76 ± 12.61	50.92 ± 12.38	57.46 ± 2.17	81.57 ± 4.57	82.65 ± 3.66	81.62 ± 2.31	38.65 ± 4.09	78.47 ± 5.74	35.95 ± 0.00	81.62 ± 1.95	76.97 ± 8.74	71.51 ± 5.35
Haptics	29.68 ± 4.36	29.16 ± 4.62	33.31 ± 5.65	37.73 ± 3.74	31.95 ± 2.95	37.66 ± 2.89	23.81 ± 5.22	37.66 ± 2.77	25.45 ± 4.31	38.20 ± 2.76	35.32 ± 3.26	39.22 ± 4.29
Herring	55.00 ± 9.20	55.62 ± 5.91	55.00 ± 7.28	50.31 ± 6.57	59.69 ± 3.39	52.60 ± 3.93	47.92 ± 2.39	53.65 ± 8.02	49.06 ± 4.50	50.52 ± 1.80	57.19 ± 6.68	55.31 ± 5.01
HouseTwenty	90.92 ± 1.10	88.91 ± 1.82	89.58 ± 3.18	65.55 ± 5.48	81.51 ± 3.71	81.23 ± 2.43	60.50 ± 3.36	74.79 ± 1.46	64.20 ± 2.95	81.79 ± 4.63	70.76 ± 6.04	89.24 ± 2.81
InlineSkate	30.04 ± 2.77	24.04 ± 1.78	21.27 ± 3.03	25.16 ± 1.55	21.67 ± 3.27	28.42 ± 1.51	16.91 ± 0.91	28.73 ± 0.83	18.22 ± 1.24	29.09 ± 2.20	22.62 ± 1.07	33.49 ± 2.77
InsectEPGRegularTrain	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	54.62 ± 4.82	79.12 ± 3.30	61.31 ± 3.90	42.70 ± 1.23	60.37 ± 1.29	55.18 ± 4.83	64.39 ± 5.02	96.63 ± 7.54	86.59 ± 4.06
InsectEPGSmallTrain	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	60.00 ± 1.05	81.69 ± 6.46	69.08 ± 1.20	55.82 ± 1.06	69.88 ± 0.80	58.71 ± 4.87	66.40 ± 1.81	83.13 ± 0.00	85.06 ± 2.08
InsectWingbeatSound	34.13 ± 0.95	34.52 ± 3.87	30.58 ± 3.83	60.48 ± 0.59	47.36 ± 1.34	60.99 ± 1.05	39.46 ± 3.03	61.67 ± 1.18	49.66 ± 3.26	61.20 ± 0.34	46.64 ± 2.91	51.36 ± 2.05
ItalyPowerDemand	92.21 ± 1.02	92.56 ± 1.30	92.75 ± 1.07	53.43 ± 4.80	93.97 ± 0.64	87.66 ± 8.42	94.69 ± 1.03	95.59 ± 0.11	95.82 ± 0.32	94.33 ± 0.49	49.23 ± 14.84	92.19 ± 2.62
LargeKitchenAppliances	64.05 ± 4.75	66.13 ± 5.93	66.08 ± 5.44	40.85 ± 3.41	67.95 ± 4.05	46.22 ± 6.49	39.82 ± 1.74	43.11 ± 5.30	40.53 ± 2.82	45.33 ± 5.07	41.71 ± 0.52	55.09 ± 5.30
Lightning2	70.49 ± 3.84	64.92 ± 7.48	65.57 ± 4.34	58.03 ± 2.74	69.18 ± 5.84	71.04 ± 1.89	51.37 ± 4.13	63.93 ± 3.28	58.36 ± 5.26	68.85 ± 4.34	60.00 ± 3.77	77.05 ± 4.18
Lightning7	64.38 ± 5.97	66.03 ± 3.41	60.00 ± 4.37	60.27 ± 3.36	61.10 ± 9.76	74.43 ± 1.58	20.09 ± 4.81	65.75 ± 1.37	36.71 ± 3.41	75.34 ± 2.37	26.30 ± 5.76	66.85 ± 7.34
Mallat	68.22 ± 8.96	69.46 ± 5.32	95.78 ± 0.57	90.38 ± 0.87	87.49 ± 9.31	92.79 ± 0.83	66.92 ± 5.03	85.20 ± 1.21	79.12 ± 6.08	89.69 ± 0.98	25.09 ± 0.73	89.72 ± 1.50
Meat	53.00 ± 18.31	59.33 ± 6.08	85.00 ± 4.86	87.00 ± 7.94	60.00 ± 5.27	61.11 ± 4.19	68.89 ± 7.52	55.56 ± 19.25	55.33 ± 20.66	86.11 ± 7.88	38.67 ± 15.92	64.00 ± 6.08
MedicalImages	56.61 ± 3.74	54.08 ± 3.55	52.39 ± 5.17	34.84 ± 5.66	48.24 ± 3.52	46.27 ± 5.02	40.00 ± 3.41	45.09 ± 2.04	47.45 ± 1.06	47.72 ± 6.48	18.58 ± 8.91	54.68 ± 2.10
MedbournePedestrian	84.60 ± 1.98	81.55 ± 1.62	80.47 ± 2.59	30.74 ± 5.50	68.14 ± 1.82	73.08 ± 1.13	75.36 ± 1.75	78.01 ± 1.38	77.48 ± 0.93	76.77 ± 0.49	68.14 ± 1.82	70.88 ± 1.49
MiddlePhalanxOutlineAgeGroup	44.29 ± 12.74	54.68 ± 6.95	55.19 ± 4.97	54.16 ± 10.18	53.51 ± 5.82	58.66 ± 1.35	50.43 ± 7.47	54.11 ± 4.79	50.13 ± 8.45	56.49 ± 1.12	41.82 ± 18.66	51.04 ± 7.11
MiddlePhalanxOutlineCorrect	60.00 ± 5.84	65.02 ± 2.98	60.62 ± 1.86	54.23 ± 4.23	56.36 ± 2.69	50.86 ± 7.18	64.72 ± 6.49	54.98 ± 7.77	58.97 ± 7.04	52.23 ± 9.34	52.44 ± 10.59	59.18 ± 5.65
MiddlePhalanxTW	49.74 ± 5.84	48.05 ± 4.11	49.48 ± 2.12	45.71 ± 2.70	46.23 ± 6.05	46.10 ± 6.40	37.23 ± 5.86	42.64 ± 7.61	35.97 ± 2.62	41.34 ± 8.07	23.64 ± 8.52	44.42 ± 6.27
MixedShapesRegularTrain	67.65 ± 7.57	70.16 ± 5.08	72.53 ± 6.92	70.49 ± 2.59	73.35 ± 4.70	73.87 ± 1.26	24.40 ± 1.33	74.08 ± 2.13	28.76 ± 6.79	73.65 ± 2.23	63.69 ± 2.98	79.60 ± 1.74
MixedShapesSmallTrain	70.61 ± 10.30	69.46 ± 11.49	75.93 ± 3.55	73.56 ± 2.63	73.16 ± 8.70	78.08 ± 2.18	23.04 ± 1.80	77.03 ± 1.02	27.06 ± 6.78	77.29 ± 1.92	65.95 ± 4.04	81.61 ± 2.47
MoteStrain	89.38 ± 0.88	89.14 ± 3.04	90.18 ± 0.68	86.26 ± 0.20	86.58 ± 1.30	87.19 ± 1.69	77.40 ± 1.04	87.43 ± 0.48	85.32 ± 0.75	90.65 ± 0.48	67.41 ± 11.72	89.23 ± 1.53
NonInvasiveFetalECGThorax1	55.91 ± 3.81	50.28 ± 5.45	55.91 ± 3.81	81.90 ± 0.59	63.56 ± 4.40	77.13 ± 1.08	36.83 ± 0.46	78.18 ± 0.43	46.96 ± 3.44	75.71 ± 1.08	58.08 ± 1.54	76.84 ± 1.40
NonInvasiveFetalECGThorax2	62.50 ± 1.19	67.13 ± 2.71	62.50 ± 1.19	86.79 ± 1.25	75.21 ± 2.50	83.55 ± 0.48	57.79 ± 2.45	83.27 ± 1.00	69.23 ± 2.65	80.68 ± 0.66	70.22 ± 2.79	82.53 ± 1.33
OSULeaf	72.64 ± 5.62	69.26 ± 4.17	69.01 ± 3.75	34.88 ± 3.54	39.34 ± 3.85	46.42 ± 2.52	19.83 ± 4.60	40.50 ± 2.19	29.83 ± 2.82	37.19 ± 2.07	28.10 ± 5.12	59.42 ± 3.58
OliveOil	56.00 ± 14.79	40.00 ± 0.00	66.67 ± 9.72	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	45.56 ± 9.62	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00
PLAID	28.04 ± 4.01	25.14 ± 3.42	22.46 ± 6.07	34.45 ± 1.13	54.53 ± 5.93	50.16 ± 2.34	32.59 ± 0.99	44.82 ± 3.36	37.77 ± 2.12	46.80 ± 0.84	54.53 ± 5.93	59.89 ± 3.28
PhalangesOutlinesCorrect	56.88 ± 6.98	56.74 ± 8.68	57.48 ± 7.09	54.27 ± 11.23	49.91 ± 6.62	58.78 ± 4.01	55.79 ± 3.74	55.36 ± 17.14	52.42 ± 4.71	54.23 ± 12.22	53.99 ± 12.03	55.73 ± 8.07
Phoneme	28.10 ± 1.26	25.96 ± 1.60	28.10 ± 1.26	8.58 ± 0.50	11.88 ± 0.54	13.82 ± 0.86	9.55 ± 1.16	11.71 ± 0.45	10.91 ± 1.85	10.88 ± 0.47	10.94 ± 0.56	21.95 ± 1.27
PickupGestureWiimoteZ	50.80 ± 15.01	68.40 ± 2.61	73.60 ± 2.19	51.60 ± 1.67	54.40 ± 6.84	56.67 ± 3.06	26.67 ± 1.15	54.67 ± 1.15	42.00 ± 5.10	60.67 ± 1.15	54.40 ± 6.84	70.40 ± 2.61
PigAirwayPressure	26.35 ± 2.05	12.88 ± 0.53	50.58 ± 0.92	4.81 ± 0.00	4.13 ± 0.43	4.81 ± 0.48	2.08 ± 1.00	6.25 ± 1.27	4.62 ± 0.87	6.73 ± 0.48	5.00 ± 0.65	9.62 ± 0.48
PigArtPressure	54.71 ± 1.57	25.19 ± 5.85	69.81 ± 3.29	8.27 ± 0.40	6.06 ± 2.62	8.97 ± 0.28	6.41 ± 1.00	9.78 ± 0.73	6.35 ± 0.92	8.65 ± 1.27	18.08 ± 2.06	31.73 ± 1.92
PigCVP	28.85 ± 2.68	19.62 ± 1.84	61.35 ± 2.42	7.12 ± 0.40	3.27 ± 1.61	7.37 ± 0.73	7.05 ± 1.21	8.33 ± 1.69	4.13 ± 0.80	5.93 ± 1.11	11.63 ± 1.24	19.52 ± 1.34
Plane	100.00 ± 0.00	99.43 ± 1.28	100.00 ± 0.00	96.57 ± 1.28	96.95 ± 1.24	98.10 ± 0.00	98.73 ± 0.55	97.14 ± 0.95	98.48 ± 0.52	97.14 ± 0.95	49.71 ± 5.36	99.24 ± 1.24
PowerCons	84.00 ± 3.74	85.67 ± 2.68	88.11 ± 1.91	73.22 ± 7.34	83.67 ± 3.78	85.93 ± 2.85	53.33 ± 0.56	86.30 ± 2.25	59.67 ± 5.39	84.63 ± 0.85	74.33 ± 14.06	82.33 ± 5.46
ProximalPhalanxOutlineAgeGroup	69.07 ± 5.10	80.49 ± 4.77	80.98 ± 2.76	75.90 ± 2.95	75.90 ± 8.78	81.30 ± 5.55	81.14 ± 3.66	81.63 ± 4.89	76.88 ± 9.31	77.56 ± 6.40	68.78 ± 16.36	79.80 ± 3.30
ProximalPhalanxOutlineCorrect	63.71 ± 11.74	64.60 ± 2.91	63.85 ± 3.21	56.01 ± 10.53	61.86 ± 4.47	64.60 ± 0.69	66.55 ± 3.51	64.26 ± 0.69	61.86 ± 6.47	65.98 ± 3.31	51.34 ± 15.85	60.07 ± 4.20
ProximalPhalanxTW	60.78 ± 11.91	70.05 ± 3.85	72.39 ± 1.88	51.12 ± 7.39	70.73 ± 1.42	64.07 ± 2.20	60.16 ± 2.69	65.53 ± 2.25	57.95 ± 5.18	59.84 ± 3.91	25.07 ± 15.12	66.66 ± 3.47
RefrigerationDevices	42.67 ± 7.00	44.27 ± 6.42	43.41 ± 5.31	34.72 ± 1.59	37.71 ± 6.85	44.00 ± 2.97	34.40 ± 1.33	34.76 ± 2.56	35.31 ± 4.05	44.62 ± 2.22	32.37 ± 1.67	41.23 ± 4.90
Rock	41.60 ± 6.84	34.40 ± 4.56	36.00 ± 1.41	81.60 ± 0.89	57.20 ± 9.01	82.67 ± 1.15	84.67 ± 6.11	88.00 ± 0.00	62.00 ± 6.16	86.67 ± 3.06	30.80 ± 4.82	76.40 ± 2.97
ScreenType	35.84 ± 5.22	36.48 ± 5.07	37.23 ± 5.64	36.37 ± 2.19	39.15 ± 5.32	40.53 ± 1.87	35.73 ± 2.12	36.09 ± 2.74	33.12 ± 2.51	39.82 ± 2.27	36.32 ± 3.91	37.12 ± 3.93
SemgHandGenderCh2	63.47 ± 5.58	74.93 ± 6.24	65.43 ± 6.18	67.83 ± 4.45	66.37 ± 2.41	62.89 ± 2.12	51.67 ± 5.43	65.78 ± 6.45	55.43 ± 2.94	60.72 ± 4.24	58.77 ± 7.13	63.80 ± 10.73
SemgHandMovementCh2	37.69 ± 5.41	36.27 ± 1.15	37.82 ± 1.58	37.42 ± 2.10	32.22 ± 2.74	30.67 ± 2.78	25.04 ± 0.56	34.52 ± 3.16	25.69 ± 1.85	33.04 ± 3.33	33.64 ± 4.19	45.56 ± 2.93

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Table 10: Per-dataset 10-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
SemgHandSubjectCh2	43.56 $\pm$ 2.78	45.20 $\pm$ 5.06	40.22 $\pm$ 5.31	62.80 $\pm$ 4.54	54.13 $\pm$ 4.47	60.37 $\pm$ 0.71	34.22 $\pm$ 3.27	68.07 $\pm$ 2.65	39.96 $\pm$ 2.48	60.96 $\pm$ 1.10	38.18 $\pm$ 2.42	66.71 $\pm$ 4.45
ShakeGestureWiimoteZ	82.00 $\pm$ 6.16	91.20 $\pm$ 2.28	88.80 $\pm$ 1.10	55.60 $\pm$ 2.97	68.40 $\pm$ 7.67	60.00 $\pm$ 4.00	33.33 $\pm$ 3.06	62.67 $\pm$ 3.06	45.20 $\pm$ 7.29	56.67 $\pm$ 1.15	68.40 $\pm$ 7.67	91.60 $\pm$ 2.97
ShapeletSim	86.33 $\pm$ 6.80	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	48.78 $\pm$ 0.82	95.22 $\pm$ 4.26	59.07 $\pm$ 5.56	48.70 $\pm$ 1.40	50.19 $\pm$ 2.80	51.00 $\pm$ 1.98	49.81 $\pm$ 1.95	52.33 $\pm$ 4.64	96.67 $\pm$ 1.96
ShapesAll	68.60 $\pm$ 2.31	66.63 $\pm$ 1.82	68.60 $\pm$ 2.31	62.47 $\pm$ 0.34	65.47 $\pm$ 2.49	72.11 $\pm$ 0.69	12.89 $\pm$ 4.16	73.28 $\pm$ 0.38	31.00 $\pm$ 5.80	65.56 $\pm$ 1.08	70.00 $\pm$ 1.20	83.13 $\pm$ 0.32
SmallKitchenAppliances	67.20 $\pm$ 8.07	61.87 $\pm$ 4.21	64.69 $\pm$ 7.08	36.59 $\pm$ 1.57	60.00 $\pm$ 3.85	55.11 $\pm$ 4.59	41.51 $\pm$ 2.40	38.76 $\pm$ 2.56	47.25 $\pm$ 3.08	51.02 $\pm$ 1.54	41.33 $\pm$ 5.03	57.60 $\pm$ 5.56
SmoothSubspace	92.00 $\pm$ 1.56	87.07 $\pm$ 3.67	92.00 $\pm$ 3.02	38.13 $\pm$ 6.59	67.73 $\pm$ 10.31	79.56 $\pm$ 2.78	80.67 $\pm$ 0.67	60.00 $\pm$ 11.02	86.13 $\pm$ 3.38	90.67 $\pm$ 4.00	43.60 $\pm$ 9.96	89.47 $\pm$ 2.60
SonyAIBORobotSurface1	95.11 $\pm$ 2.42	96.01 $\pm$ 0.87	92.45 $\pm$ 1.40	42.96 $\pm$ 0.07	87.95 $\pm$ 1.92	64.28 $\pm$ 1.83	67.05 $\pm$ 1.01	76.43 $\pm$ 0.69	64.86 $\pm$ 0.43	67.83 $\pm$ 1.06	45.09 $\pm$ 4.84	86.09 $\pm$ 3.97
SonyAIBORobotSurface2	96.73 $\pm$ 3.11	92.82 $\pm$ 3.91	94.67 $\pm$ 2.40	77.50 $\pm$ 0.71	90.95 $\pm$ 2.30	81.25 $\pm$ 2.70	83.07 $\pm$ 2.21	86.08 $\pm$ 2.22	82.71 $\pm$ 1.50	85.73 $\pm$ 2.18	61.15 $\pm$ 12.88	84.11 $\pm$ 2.29
StarLightCurves	71.47 $\pm$ 14.56	82.56 $\pm$ 3.87	75.10 $\pm$ 5.06	74.16 $\pm$ 3.22	77.51 $\pm$ 4.09	75.29 $\pm$ 4.53	29.60 $\pm$ 9.11	76.31 $\pm$ 3.24	30.20 $\pm$ 4.95	74.55 $\pm$ 5.89	70.71 $\pm$ 14.11	85.26 $\pm$ 3.54
Strawberry	79.73 $\pm$ 8.80	76.70 $\pm$ 3.26	76.70 $\pm$ 4.55	60.81 $\pm$ 1.94	79.89 $\pm$ 6.30	61.98 $\pm$ 1.58	80.63 $\pm$ 1.13	59.10 $\pm$ 1.09	77.51 $\pm$ 4.36	68.11 $\pm$ 2.15	50.54 $\pm$ 10.66	75.84 $\pm$ 0.99
SwedishLeaf	88.03 $\pm$ 1.39	83.68 $\pm$ 1.26	88.03 $\pm$ 1.39	66.11 $\pm$ 1.36	73.47 $\pm$ 2.54	75.95 $\pm$ 1.36	60.69 $\pm$ 4.66	81.39 $\pm$ 2.49	68.80 $\pm$ 1.87	74.72 $\pm$ 1.82	28.96 $\pm$ 3.20	85.89 $\pm$ 1.27
Symbols	93.83 $\pm$ 2.55	89.89 $\pm$ 7.08	92.52 $\pm$ 1.13	75.94 $\pm$ 0.93	85.87 $\pm$ 1.87	83.32 $\pm$ 0.52	34.77 $\pm$ 0.30	85.96 $\pm$ 1.03	61.21 $\pm$ 7.54	82.51 $\pm$ 0.70	21.11 $\pm$ 2.93	89.33 $\pm$ 0.82
SyntheticControl	95.60 $\pm$ 1.14	97.87 $\pm$ 0.61	97.60 $\pm$ 0.43	59.07 $\pm$ 3.98	94.20 $\pm$ 1.64	90.78 $\pm$ 1.64	59.44 $\pm$ 1.95	79.44 $\pm$ 4.86	68.40 $\pm$ 3.43	90.11 $\pm$ 1.35	22.60 $\pm$ 7.62	97.33 $\pm$ 0.85
ToeSegmentation1	93.25 $\pm$ 1.60	89.56 $\pm$ 3.33	91.58 $\pm$ 1.68	55.53 $\pm$ 3.14	68.95 $\pm$ 9.85	56.73 $\pm$ 5.97	50.29 $\pm$ 6.71	55.85 $\pm$ 3.55	49.39 $\pm$ 2.18	55.70 $\pm$ 4.98	50.53 $\pm$ 1.40	80.53 $\pm$ 2.75
ToeSegmentation2	81.38 $\pm$ 10.25	87.08 $\pm$ 4.53	85.85 $\pm$ 2.64	50.15 $\pm$ 3.62	73.38 $\pm$ 4.70	68.21 $\pm$ 10.44	56.67 $\pm$ 7.86	58.46 $\pm$ 6.30	46.77 $\pm$ 3.41	61.54 $\pm$ 3.35	43.54 $\pm$ 14.75	80.31 $\pm$ 4.70
Trace	92.20 $\pm$ 4.60	100.00 $\pm$ 0.00	95.20 $\pm$ 5.07	55.80 $\pm$ 5.72	86.20 $\pm$ 10.35	72.67 $\pm$ 1.53	55.67 $\pm$ 4.51	61.33 $\pm$ 4.62	74.80 $\pm$ 3.27	78.00 $\pm$ 4.36	46.80 $\pm$ 8.17	100.00 $\pm$ 0.00
TwoLeadECG	99.86 $\pm$ 0.08	98.79 $\pm$ 1.25	97.56 $\pm$ 2.37	58.05 $\pm$ 1.68	91.89 $\pm$ 5.00	70.85 $\pm$ 3.05	75.62 $\pm$ 1.63	65.26 $\pm$ 1.91	78.86 $\pm$ 1.56	78.99 $\pm$ 2.23	54.01 $\pm$ 3.73	99.68 $\pm$ 0.24
TwoPatterns	70.89 $\pm$ 4.77	77.56 $\pm$ 2.96	78.86 $\pm$ 3.59	42.28 $\pm$ 1.08	74.73 $\pm$ 6.48	42.55 $\pm$ 1.37	26.22 $\pm$ 0.85	42.35 $\pm$ 1.82	29.82 $\pm$ 3.35	37.95 $\pm$ 1.72	25.60 $\pm$ 4.32	70.06 $\pm$ 0.70
UMD	92.78 $\pm$ 3.35	98.33 $\pm$ 1.05	98.75 $\pm$ 0.58	64.17 $\pm$ 3.05	91.53 $\pm$ 4.77	76.62 $\pm$ 2.44	75.93 $\pm$ 7.23	89.58 $\pm$ 4.22	93.06 $\pm$ 1.47	87.27 $\pm$ 3.21	47.50 $\pm$ 8.96	98.67 $\pm$ 7.42
UWaveGestureLibraryAll	48.45 $\pm$ 3.31	48.15 $\pm$ 2.07	49.06 $\pm$ 3.81	80.57 $\pm$ 3.71	68.35 $\pm$ 4.39	84.19 $\pm$ 2.90	27.54 $\pm$ 1.68	84.63 $\pm$ 3.19	42.77 $\pm$ 5.45	83.32 $\pm$ 3.39	81.20 $\pm$ 2.71	52.41 $\pm$ 1.70
UWaveGestureLibraryX	51.98 $\pm$ 1.26	57.08 $\pm$ 2.26	52.99 $\pm$ 2.93	60.99 $\pm$ 2.04	57.69 $\pm$ 1.76	62.37 $\pm$ 1.85	17.30 $\pm$ 0.19	62.30 $\pm$ 0.55	36.29 $\pm$ 8.88	61.26 $\pm$ 1.95	56.81 $\pm$ 2.90	53.50 $\pm$ 1.91
UWaveGestureLibraryY	45.58 $\pm$ 1.39	47.42 $\pm$ 0.67	46.02 $\pm$ 2.85	55.44 $\pm$ 1.29	53.52 $\pm$ 3.23	56.83 $\pm$ 0.99	23.04 $\pm$ 3.42	56.65 $\pm$ 1.12	32.16 $\pm$ 4.98	55.22 $\pm$ 1.56	46.61 $\pm$ 1.40	52.03 $\pm$ 0.97
UWaveGestureLibraryZ	48.34 $\pm$ 2.78	51.60 $\pm$ 2.75	44.60 $\pm$ 1.67	53.18 $\pm$ 1.58	53.45 $\pm$ 2.14	54.99 $\pm$ 0.25	20.01 $\pm$ 2.19	54.36 $\pm$ 2.47	34.40 $\pm$ 4.42	51.61 $\pm$ 2.72	46.80 $\pm$ 1.81	60.65 $\pm$ 2.84
Wafer	84.60 $\pm$ 6.19	78.03 $\pm$ 11.28	81.53 $\pm$ 4.78	69.44 $\pm$ 4.07	84.77 $\pm$ 11.10	82.09 $\pm$ 7.52	76.30 $\pm$ 12.19	83.57 $\pm$ 7.70	81.81 $\pm$ 10.33	81.67 $\pm$ 9.12	78.96 $\pm$ 10.53	82.53 $\pm$ 8.58
Wine	68.15 $\pm$ 6.33	52.59 $\pm$ 5.80	64.07 $\pm$ 8.03	50.74 $\pm$ 1.66	50.00 $\pm$ 0.00	58.64 $\pm$ 5.35	50.62 $\pm$ 1.07	50.00 $\pm$ 3.70	50.00 $\pm$ 0.00	62.96 $\pm$ 7.41	47.78 $\pm$ 4.97	49.26 $\pm$ 2.11
WordSynonyms	35.14 $\pm$ 2.03	30.72 $\pm$ 3.50	35.14 $\pm$ 2.03	35.86 $\pm$ 2.07	39.44 $\pm$ 4.12	51.78 $\pm$ 3.11	15.15 $\pm$ 0.48	51.10 $\pm$ 4.21	25.64 $\pm$ 4.61	49.01 $\pm$ 4.47	33.92 $\pm$ 1.82	58.90 $\pm$ 2.04
Worms	50.39 $\pm$ 8.29	51.95 $\pm$ 5.35	45.71 $\pm$ 7.71	23.38 $\pm$ 5.11	39.74 $\pm$ 4.27	23.38 $\pm$ 6.49	26.84 $\pm$ 3.97	27.27 $\pm$ 9.09	25.45 $\pm$ 3.74	25.97 $\pm$ 9.09	20.78 $\pm$ 5.35	48.57 $\pm$ 4.91
WormsTwoClass	64.94 $\pm$ 5.51	63.12 $\pm$ 4.37	61.30 $\pm$ 7.20	49.09 $\pm$ 4.63	47.27 $\pm$ 1.97	50.65 $\pm$ 6.49	48.05 $\pm$ 4.50	48.05 $\pm$ 4.68	48.05 $\pm$ 3.56	51.95 $\pm$ 4.50	46.49 $\pm$ 5.15	56.36 $\pm$ 3.85
Yoga	59.91 $\pm$ 2.90	59.97 $\pm$ 4.98	59.88 $\pm$ 3.11	55.72 $\pm$ 1.71	57.07 $\pm$ 1.49	58.79 $\pm$ 3.68	53.03 $\pm$ 3.90	59.02 $\pm$ 1.75	51.43 $\pm$ 2.65	59.87 $\pm$ 3.99	51.89 $\pm$ 2.29	57.73 $\pm$ 2.05
Avg. Acc.	66.24	64.93	<u>67.18</u>	55.43	62.72	62.79	46.04	61.94	51.50	63.13	46.88	69.16
Avg. Rank	4.83	5.03	<u>4.55</u>	8.00	6.23	5.75	9.87	5.73	8.87	5.64	9.58	3.91
#Best	25	15	<u>28</u>	6	5	7	3	5	1	8	1	40

Table 11: Key target-adaptation settings for TimeMorph.

Component	Setting
Base LLM	meta-llama/Llama-3.2-1B
Vision backbone	facebook/dinov2-base
Classification prompt suffix / answer	Class: / one class token, e.g., $\langle \text{cls}_y \rangle$ followed by EOS
Morphology-map construction	Overlapping-window grid transform with per-sequence median/IQR normalization
Temporal patch length / stride	16 / 8
Temporal hidden size / heads	128 / 8
Branch projectors	LayerNorm-Linear-GELU-Dropout MLPs when dimensions differ
Language projector	MLP projector into the LLM hidden dimension followed by output LayerNorm
Default UCR checkpoint	Stage II (TSQA transfer)
Warm-up / joint epochs	5 / 55
Batch size / eval batch size	8 / 8
Gradient accumulation	4
LR (encoder / projector / LoRA)	$2 \times 10^{-4} / 1 \times 10^{-4} / 1 \times 10^{-4}$
Language-side adaptation	LoRA on $q/k/v/o_proj$, $gate_proj$, up_proj , and $down_proj$ ($r = 16$, $\alpha = 32$)
Tokenizer training mode	class_rows
Anonymous class-token initialization	pretrained vocabulary-row mean plus small random perturbation
Prediction rule	anonymous class-token scoring; semantic-prior class-token scoring when label text is meaningful
Freeze vision backbone	true
Downstream data augmentation	none

B Additional Experimental Details

We summarize the key implementation choices of TimeMorph for the few-shot benchmark in Table 11. These settings are fixed across the main benchmark reported in Section 4.

Morphology-map transform. The operator Γ_{img} in Eq. 5 applies per-sample min-max normalization over the two-dimensional grid and a fixed contrast adjustment $u \mapsto u^{0.8}$. In the default morphology mode, the effective window length is $\max(1, \lfloor \sqrt{T} \rfloor)$, the stride length is $\max(1, \lfloor 0.1q \rfloor)$, incomplete windows are completed by replicate padding on the left, and the resulting grid is resized to the DINOv2 image resolution with nearest-neighbor interpolation before RGB replication. We use DINOv2 patch tokens from the configured intermediate layer and exclude the class token.

Stage I pretraining hyperparameters. Stage I uses mask ratio 0.4, branch-dropout probability 0.1, jitter standard deviation 0.03, scaling range $[0.8, 1.2]$, and time-mask ratio 0.1 for the two augmented views. The Stage I loss weights are $\lambda_{\text{rec}} = 1.0$, $\lambda_t = 0.25$, $\lambda_v = 0.25$, and $\lambda_f = 0.5$. These values are fixed before the target few-shot runs.

B.1 Pretraining Stages and Data Sources

Table 12 summarizes the three training stages discussed in the main paper and clarifies which checkpoint is used by the UCR and external semantic-label evaluations.

After staged pretraining, target adaptation uses only the sampled support labels of the target dataset. For UCR targets contained in the Stage I pool, unlabeled TRAIN sequences may already have contributed to the encoder pretraining.

B.2 Code Release and Assets

We will release an anonymized code package with the submission. The release includes the TimeMorph implementation, target-adaptation and evaluation scripts, baseline launchers, reporting scripts for the tables in this paper, environment files, and README instructions for preparing the public datasets used in our experiments. The code is released under the MIT license. The package does not redistribute raw third-party datasets or gated pretrained model weights; users should obtain those assets from the original providers and comply with their licenses and terms of use. The released scripts document which experiments are directly reproducible from the package and which require separately downloaded datasets or pretrained checkpoints.

B.3 Asset and License Table

Table 13 summarizes the main existing assets used in this paper. We list the original source, role, version when available, license or stated terms, and whether the asset is redistributed in our release.

Table 12: Stage data sources and roles for TimeMorph.

Stage	Training data	Validation / monitoring data	Role in reported experiments
Stage I: Representation Pretraining	Raw TSQA-train series, raw M4-train series, and official TRAIN-split raw series from a 98-dataset UCR training pool.	Raw TSQA validation/test series and raw M4 validation/test series. No UCR official TEST split is used.	Learns dual-view representations before the LLM interface; corresponds to <code>stage0_encoder_ssl</code> in the implementation.
Stage II: Embedding-Space Alignment Pretraining	TSQA train split only, with answer-token language-model training on time-series question answering.	TSQA validation/test splits.	Provides the default checkpoint used by all main UCR few-shot results in Section 4; corresponds to <code>stage1_tsqa_transfer</code> .
Stage III: Optional Label-Semantic Calibration	Mixture of TSQA retention examples, M4 captioning examples, and synthetic attribute supervision derived from TSQA/M4 raw series.	Corresponding validation/test splits of the same mixture.	Provides the calibrated checkpoint used by the external semantic-label evaluations; not used for the default UCR benchmark. This corresponds to <code>stage2_semantic_bridge</code> and later caption specialization in code.

545 For assets whose official source does not state a separate dataset license, we cite the official repository
546 or archive page and do not redistribute the raw data.

Table 13: Existing assets used in this paper and their licenses or stated terms. URLs were checked during manuscript preparation.

Asset	Role	Source / version	License or stated terms	Redistribution in our release
UCR 2018	Archive benchmark and unlabeled Stage I source pool	TSC website; Zenodo v1	Zenodo record lists Creative Commons Attribution 4.0 International (CC BY 4.0); archive should be cited via Dau et al.	Not redistributed; scripts expect users to download from the original archive or cited mirror.
TSQA / ChatTime	Stage I raw series and Stage II time-series question answering	Hugging Face dataset card	Apache-2.0 according to the Hugging Face dataset metadata	Not redistributed; scripts load from the original dataset source.
M4 Competition dataset	Stage I raw series and Stage III captioning / label-semantic calibration data	Official M4 repository	The official repository publicly provides the dataset and asks users to cite the M4 competition resources; no separate dataset LICENSE file was found in the repository.	Not redistributed; users download from the official repository.
MIT-BIH Arrhythmia Database	External semantic-label evaluation	PhysioNet v1.0.0	Open Data Commons Attribution License v1.0; PhysioNet citation required	Not redistributed; pre-processing scripts require a local PhysioNet download.
Sleep-EDF Database Expanded	External semantic-label evaluation	PhysioNet v1.0.0	Open Data Commons Attribution License v1.0; PhysioNet citation required	Not redistributed; pre-processing scripts require a local PhysioNet download.
PhysioNet/CinC 2017 AF dataset	External semantic-label evaluation	PhysioNet v1.0.0	Open Data Commons Attribution License v1.0; PhysioNet and challenge citations required	Not redistributed; pre-processing scripts require a local PhysioNet download.
Llama 3.2 1B	Pretrained causal LLM backbone	Hugging Face model card	Llama 3.2 Community License and Acceptable Use Policy	Not redistributed; users must obtain access from Meta / Hugging Face.

Table 14: Compute-resource budget for the reported experiment groups. Downstream jobs use single RTX 3090 workers unless noted otherwise.

Experiment group	Scope	Compute workers	Mean per run	Total hours	GPU-	Notes
Stage I–II pretraining	Staged pretraining checkpoint used to initialize target adaptation	1 RTX 3090 worker	–	19.81		Stage I encoder SSL used 9.57 GPU-hours; Stage I–II together used 19.81 GPU-hours.
Optional Stage III calibration	Label-semantic calibration and caption-specialization training used outside the default UCR results	1 RTX 3090 worker	–	29.03		Not used for the default UCR benchmark.
TimeMorph main UCR benchmark	128 UCR datasets, 1/2/5/10-shot, five runs per setting; 2,507 completed downstream runs	6 single-GPU workers	0.67 GPU-h/run	1,677		Batch size 8, gradient accumulation 4, 60 downstream epochs.
UCR baseline runs	ResNet, TapNet, COSCO-ResNet, PatchTST, GPT4TS, and TSLib-family baselines under the same few-shot protocol	6 single-GPU workers	0.40–0.49 GPU-h/run	10,937		Includes the retained main UCR baseline logs; GPT4TS coverage is reported for the completed 60-dataset run.
External-dataset TimeMorph runs	MIT-BIH, SleepEDF, and CinC2017 under 10/20/30-shot protocols	single-GPU workers	1.69 GPU-h/run	153		Same target-adaptation recipe as the external few-shot experiments.
External-dataset baseline runs	Reported baseline families on MIT-BIH, SleepEDF, and CinC2017	single-GPU workers	0.68 GPU-h/job	22		Launcher jobs cover all three external datasets and baseline families.
Top-level ablations	No-pretraining, temporal-only, visual-only, class-token decoding, and no-LLM backbone ablations	6 single-GPU workers	1.67 GPU-h/run	3,895		Includes the two retained no-LLM linear runs added together.
Semantic-prior ablation	External semantic-label interface runs	single-GPU workers	1.69 GPU-h/run	31		Uses the same 10/20/30-shot external evaluation protocol.
Efficiency probe	Representative UCR runtime subset with parameter, memory, latency, and throughput measurements	one RTX 3090	–	–		Detailed runtime and memory values are reported in Tables 4, 17, and 19.

Table 13: Existing assets used in this paper and their licenses or stated terms (continued).

Asset	Role	Source / version	License or stated terms	Redistribution in our release
DINOv2-base	Frozen visual backbone	Hugging Face model card	Apache-2.0	Not redistributed; loaded from the original model repository.
OpenTSLM base code	Starting code- base time-series language-model components	Project repository and re-lease package	MIT license in the repository license file	Modified code is released in anonymized form under MIT, preserving applicable notices.
Python libraries, including PyTorch, Transformers, PEFT, NumPy, pandas, scikit-learn, and related dependencies	ML Training, adaptation, evaluation, and reporting	Version constraints listed in the released environment files	Governed by their respective open-source licenses	Not vendored; installed by the environment files.

B.4 Computational Resource Reporting

All reported downstream experiments were run on a local server with six NVIDIA GeForce RTX 3090 GPUs (24GB each), two Intel Xeon Silver 4314 CPUs, and 251GiB system memory. Unless otherwise stated, each training job used one GPU, and the batch launchers executed up to six single-GPU workers in parallel. Table 14 summarizes the compute used by the main experiment groups. GPU-hours are summed over single-GPU jobs; wall-clock time can be lower when multiple workers run concurrently. The efficiency measurements in Appendix B.7 additionally report peak memory, train-step latency, inference latency, throughput, total parameters, and downstream-updated parameters on the representative runtime subset.

Table 15: Few-shot classification accuracy (%) on three external datasets. Each cell reports mean $\pm$ standard deviation over 10 runs under the same few-shot protocol. Best is bold and second-best is underlined within each dataset-shot column.

Model	MIT-BIH			SleepEDF			CinC2017		
	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot
ChronoMorph	55.03 $\pm$ 9.40	63.95 $\pm$ 13.80	64.15 $\pm$ 5.32	37.98 $\pm$ 3.69	48.80 $\pm$ 2.31	54.69 $\pm$ 3.67	32.80 $\pm$ 9.08	36.98 $\pm$ 4.27	34.13 $\pm$ 9.73
ResNet	37.97 $\pm$ 14.13	55.66 $\pm$ 8.58	57.07 $\pm$ 9.05	60.27 $\pm$ 7.78	70.59 $\pm$ 4.14	<u>69.64 $\pm$ 3.64</u>	30.95 $\pm$ 12.41	<u>33.73 $\pm$ 7.43</u>	36.20 $\pm$ 4.84
TapNet	31.94 $\pm$ 7.18	40.15 $\pm$ 10.28	44.11 $\pm$ 11.14	<u>58.75 $\pm$ 9.67</u>	<u>70.50 $\pm$ 5.64</u>	70.86 $\pm$ 5.23	30.05 $\pm$ 6.73	32.80 $\pm$ 8.81	31.01 $\pm$ 11.54
COSCO-ResNet	32.33 $\pm$ 14.25	35.08 $\pm$ 13.75	41.17 $\pm$ 8.94	57.86 $\pm$ 11.29	63.04 $\pm$ 8.19	67.32 $\pm$ 3.13	<u>31.41 $\pm$ 4.48</u>	31.04 $\pm$ 6.97	31.98 $\pm$ 8.37
PatchTST	38.94 $\pm$ 14.35	55.36 $\pm$ 8.84	57.08 $\pm$ 9.40	25.21 $\pm$ 11.89	29.47 $\pm$ 9.76	32.08 $\pm$ 8.05	<u>26.07 $\pm$ 6.24</u>	30.35 $\pm$ 7.51	31.21 $\pm$ 3.36
OneFitsAll	32.92 $\pm$ 16.63	42.48 $\pm$ 10.82	54.87 $\pm$ 10.04	26.98 $\pm$ 3.35	27.57 $\pm$ 3.81	26.60 $\pm$ 3.59	30.80 $\pm$ 6.07	27.56 $\pm$ 4.33	29.23 $\pm$ 3.03
TimesNet	51.01 $\pm$ 5.84	54.21 $\pm$ 7.81	56.15 $\pm$ 8.77	21.11 $\pm$ 2.91	23.42 $\pm$ 2.34	25.17 $\pm$ 1.82	27.25 $\pm$ 2.81	28.41 $\pm$ 4.24	29.43 $\pm$ 2.44
DLinear	<u>55.57 $\pm$ 9.85</u>	57.45 $\pm$ 7.95	<u>58.84 $\pm$ 8.69</u>	19.57 $\pm$ 1.32	19.94 $\pm$ 1.73	19.92 $\pm$ 1.69	24.92 $\pm$ 1.16	25.17 $\pm$ 1.22	24.53 $\pm$ 1.70
Autoformer	38.05 $\pm$ 5.65	42.49 $\pm$ 8.58	44.39 $\pm$ 7.25	26.25 $\pm$ 7.52	24.55 $\pm$ 4.24	28.64 $\pm$ 8.05	28.37 $\pm$ 5.44	27.67 $\pm$ 2.76	26.30 $\pm$ 4.36
Crossformer	53.48 $\pm$ 9.36	<u>58.55 $\pm$ 10.43</u>	58.24 $\pm$ 9.71	23.75 $\pm$ 1.74	24.53 $\pm$ 3.34	26.49 $\pm$ 3.02	24.02 $\pm$ 6.10	25.50 $\pm$ 2.86	27.35 $\pm$ 2.35
FEDformer	58.88 $\pm$ 5.77	56.80 $\pm$ 9.98	55.12 $\pm$ 10.09	15.15 $\pm$ 8.43	24.73 $\pm$ 6.55	25.47 $\pm$ 5.56	20.35 $\pm$ 8.32	26.90 $\pm$ 3.54	27.91 $\pm$ 4.87
Informer	53.19 $\pm$ 6.03	54.20 $\pm$ 7.61	52.35 $\pm$ 10.80	20.35 $\pm$ 12.77	22.18 $\pm$ 5.40	24.81 $\pm$ 8.94	26.93 $\pm$ 9.30	28.14 $\pm$ 4.01	30.12 $\pm$ 5.95

Table 16: Semantic-prior class-token ablation on three external datasets (accuracy, %). Each cell reports mean $\pm$ standard deviation over 10 runs under the same few-shot protocol. The better variant within each dataset-shot column is bold. Δ denotes Semantic prior minus Anonymous labels.

Variant	MIT-BIH			SleepEDF			CinC2017		
	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot
Anonymous labels	55.03 $\pm$ 9.40	63.95 $\pm$ 13.80	64.15 $\pm$ 5.32	37.98 $\pm$ 3.69	48.80 $\pm$ 2.31	54.69 $\pm$ 3.67	32.80 $\pm$ 9.08	36.98 $\pm$ 4.27	34.13 $\pm$ 9.73
Semantic prior	57.05 $\pm$ 10.96	66.62 $\pm$ 6.54	68.87 $\pm$ 11.23	42.56 $\pm$ 3.24	52.69 $\pm$ 5.05	57.24 $\pm$ 4.34	34.72 $\pm$ 3.15	35.24 $\pm$ 1.24	38.86 $\pm$ 7.00
Semantic prior – Anonymous	+2.02	+2.67	+4.72	+4.58	+3.89	+2.55	+1.92	-1.74	+4.73

Retained launcher summaries also include failed or exploratory jobs. These runs account for less than 2 additional GPU-hours in the preserved logs, and dry-run checks add a negligible amount of compute.

B.5 External Dataset Results

Table 15 reports the full 10-, 20-, and 30-shot results on MIT-BIH, SleepEDF, and CinC2017. All methods use the same external few-shot settings and reported run count; we use these datasets as transfer checks beyond the primary UCR benchmark rather than as a claim of uniform dominance across external domains.

B.6 Semantic-Prior Decision Ablation

Table 16 expands the semantic-prior comparison summarized in Table 6. Both variants use one-pass class-token scoring; the semantic-prior variant additionally initializes and regularizes class-token rows from label-verbalizer prototypes. The semantic prior improves the averaged performance on all three datasets, while individual shot-level deltas remain dataset-dependent.

B.7 Efficiency Summary

Table 17 reports computational efficiency and adaptation cost under the 10-shot UCR protocol. Runtime and memory are measured on the predefined representative UCR subset with batch size 8 on a single RTX 3090, while parameter counts are reported separately for total deployed parameters and downstream-updated parameters.

B.8 Baseline Training Recipes and Hyperparameters

Table 18 summarizes the training rules used by the reported baseline families. The goal of this table is to make the few-shot comparison protocol explicit: all methods share the same sampled support/query split, but they do not all use the same checkpoint-selection rule or the same epoch schedule.

B.9 Computational Efficiency Probe Details

The efficiency probe is implemented by the repository’s `efficiency_probe.py` script. It uses the same model families as the main benchmark and omits TimeMorph-only ablations. The default runtime subset contains COFFEE, GUNPOINT, ECG5000, ELECTRICDEVICES, CROP, ACSF1, PHONEME, and NONINVASIVEFETALECGTHORAX1, covering short, medium, and long UCR

Table 17: Computational efficiency and adaptation cost under the 10-shot UCR protocol. Total parameters count the full deployed model, while updated parameters count only the parameters optimized during downstream adaptation. Adapter size estimates the per-dataset trainable-state footprint. Runtime is measured on representative UCR datasets with batch size 8 on a single RTX 3090.

Model	Total params	Updated params	Updated %	Adapter MB	Peak GB	Train ms/step	Infer ms/ex.	Infer ex./s
ChronoMorph	1.28B	12.3M	0.96	46.7	13.75	1207.8	156.85	6.4
ResNet	629.1K	629.1K	100.00	2.4	0.33	13.2	0.26	4273.3
TapNet	912.1K	912.1K	100.00	3.5	0.22	15.0	0.47	3345.0
COSCO-ResNet	662.1K	662.1K	100.00	2.5	0.58	50.8	0.25	4315.1
PatchTST	665.4K	598.3K	90.80	2.3	1.54	32.6	0.75	2149.8
DLinear	1.2M	1.2M	100.00	4.6	0.04	2.7	0.07	14389.5
TimesNet	2.5M	2.5M	100.00	9.7	0.14	66.7	2.46	614.5
Autoformer	1.2M	1.2M	100.00	4.5	0.40	23.1	11.10	91.3
Crossformer	2.8M	2.8M	100.00	10.8	0.06	24.9	0.59	1709.6
FEDformer	1.8M	1.8M	100.00	6.7	0.26	244.1	5.17	201.9
Informer	1.4M	1.4M	100.00	5.3	0.22	18.0	0.62	1690.1
GPT4TS	82.2M	1.1M	1.34	4.2	0.56	22.6	0.86	1181.2

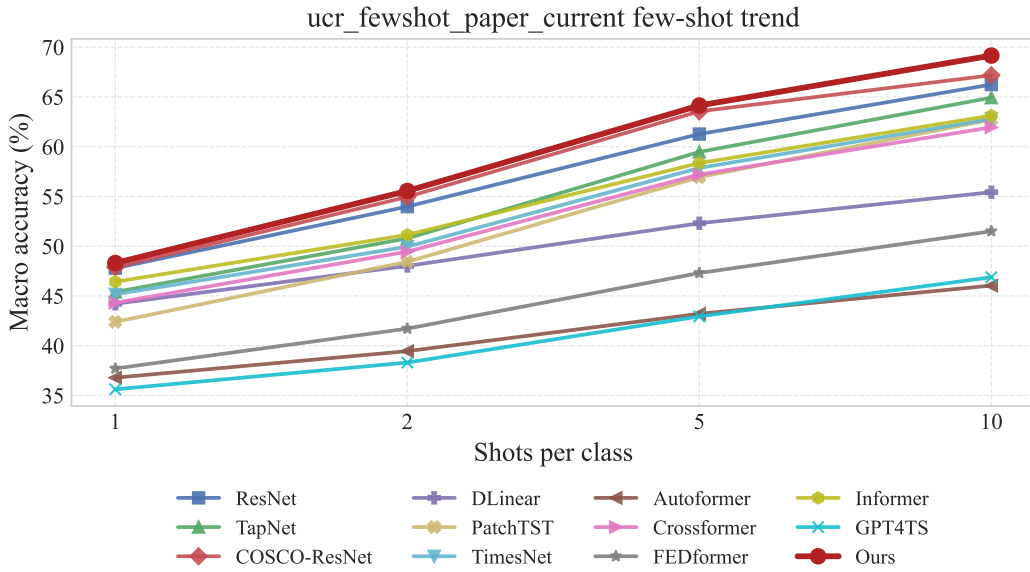


Figure 2: Shot-scaling trend on the 128 UCR datasets. Each point reports the mean accuracy over five runs. TimeMorph remains the strongest method across all four shot settings.

series as well as different class counts. The script writes the LaTeX tables, CSV summary, and accuracy-versus-updated-parameters plot to the efficiency-report directory.

C Additional Plot

Figure 2 visualizes the shot-scaling trend over the 128 UCR datasets. TimeMorph remains the strongest method on the averaged benchmark curve across the entire few-shot range, with the largest absolute advantage appearing at 10-shot.

Table 18: Few-shot training recipes for TimeMorph and the reported baseline families. All methods use the same sampled support sets within each run and evaluate on all official TEST examples from the target dataset.

Method / family	Optimizer	Epochs	Batch / LR	Model selection rule	Val	ES	Few-shot notes
TimeMorph	AdamW	5 warm-up + 55 joint	$8 / 2 \times 10^{-4}$ (encoder), 1×10^{-4} (projector, LoRA)	Report the last joint-adaptation checkpoint	No	No	Class-token rows remain trainable; language-side adaptation uses LoRA in the joint adaptation phase.
ResNet, TapNet	AdamW	Fixed schedule in the few-shot runner (60 in the benchmark configs)	model-specific / benchmark configs use 10^{-4}	Best support-set accuracy, tie-broken by support loss	No	No	Trained only on the sampled support set; the support set doubles as the model-selection set.
COSCO-ResNet	SAM with SGD base optimizer + prototypical loss	Fixed schedule (100 in the benchmark configs)	128 / 10^{-2} by default	Final trained model; support centroids computed after training	No	No	Query predictions are made by distances to support centroids rather than a learned classifier head.
PatchTST	AdamW + linear warmup	Two-step fixed schedule (head warm-up + joint fine-tuning; 60 total epochs in the benchmark configs)	16 / separate backbone-head LRs	Report the last joint fine-tuning checkpoint	No	No	Uses the same support/query split and full dataset class set as the main protocol.
TSLib family (DLINEAR, AUTO-FORMER, CROSS-FORMER, FED-FORMER, INFORMER, TIMESNET)	RAdam + cosine schedule	Fixed model-specific schedule from the TSLib profile or benchmark override	usually 16 / 10^{-3}	Report the last checkpoint	No	No	Single-stage training; evaluation masks logits to the target dataset's class set. TimesNet uses a shorter profile than the other TSLib baselines.
GPT4TS	RAdam (default runner setting)	Fixed schedule (60 in the benchmark configs)	64 / 10^{-4} in the benchmark configs	Report the last checkpoint	No	No	Implemented with the One-Fits-All classification runner under the same sampled support/query protocol.

Table 19: Available per-dataset runtime details for the computational-efficiency probe. The full machine-readable runtime file is produced by the efficiency script; rows with missing measurements are omitted from the paper appendix.

Dataset	Model	Peak GB	Train ms/step	Infer ms/ex.	Infer ex./s
ACSF1	ChronoMorph	13.75	1262.4	173.87	5.8
ACSF1	ResNet	1.01	26.0	0.32	3167.0
ACSF1	TapNet	0.59	33.7	1.45	688.2
ACSF1	COSCO-ResNet	1.64	119.2	0.43	2303.7
ACSF1	PatchTST	6.50	107.6	2.23	447.7
ACSF1	DLinear	0.10	4.5	0.07	13898.9
ACSF1	TimesNet	0.27	121.8	4.54	220.5
ACSF1	Autoformer	1.19	40.5	12.53	79.8
ACSF1	Crossformer	0.12	25.0	0.58	1723.7
ACSF1	FEDformer	0.61	277.0	7.03	142.3
ACSF1	Informer	0.56	23.4	1.09	917.6
ACSF1	GPT4TS	1.00	29.1	0.89	1121.4

Table 20: Ablation of aggregate curriculum pretraining on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full ChronoMorph model.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
w/o Curriculum Pretraining	44.70	51.81	60.65	66.60	55.94	3.35
ChronoMorph	48.31	55.55	64.14	69.16	59.29	–

589 D Ablation Tables

590 Table 5 in the main paper reports the compact ablation summary. Tables 20–22 expand that summary
591 into three focused views under the same few-shot protocol. Table 23 summarizes the LLM-backbone
592 ablation, and Table 24 reports its full per-dataset comparison on all 128 UCR datasets. Because UCR
593 class identifiers are anonymous, the label-space ablation should be interpreted as testing closed-set
594 class-token scoring; the semantic-prior class-token ablation is reported on external datasets with
595 meaningful class names in Appendix B.6.

Table 21: Ablation of dual-view time-series-to-language modeling on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. These comparisons assess the effect of removing one top-level view at a time. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full ChronoMorph model.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
Temporal only	43.51	50.55	58.40	64.39	54.21	5.08
Visual only	45.88	50.02	60.86	63.58	55.09	4.20
ChronoMorph	48.31	55.55	64.14	69.16	59.29	–

Table 22: Ablation of closed-set class-token label-space scoring on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full ChronoMorph model. Since UCR labels are anonymous, this ablation does not evaluate semantic label priors.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
w/o Class-Token Label-Space Scoring	44.46	50.44	60.08	65.59	55.14	4.15
ChronoMorph	48.31	55.55	64.14	69.16	59.29	–

Table 23: Ablation of the LLM backbone on the 120 shared UCR datasets covered by both runs. Each value is benchmark-level mean accuracy (%) over the available runs recorded in the result files. Drop denotes the average accuracy decrease relative to full ChronoMorph.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
ChronoMorph	47.93	54.98	63.27	68.48	58.67	–
w/o LLM Backbone	42.95	48.20	57.06	61.01	52.31	6.36

Table 24: Per-dataset LLM-backbone ablation on 120 shared UCR datasets. Each value is mean $\pm$ standard deviation accuracy (%) over the available runs recorded in the result files.

Dataset	ChronoMorph				w/o LLM Backbone			
	1-shot	2-shot	5-shot	10-shot	1-shot	2-shot	5-shot	10-shot
ACSF1	41.40 $\pm$ 10.01	54.60 $\pm$ 3.58	62.80 $\pm$ 4.32	71.80 $\pm$ 2.17	36.00 $\pm$ 10.37	42.80 $\pm$ 3.96	51.60 $\pm$ 5.46	62.60 $\pm$ 3.44
Adiac	45.17 $\pm$ 5.28	56.01 $\pm$ 1.25	69.51 $\pm$ 2.42	76.42 $\pm$ 1.68	26.85 $\pm$ 1.68	21.53 $\pm$ 2.11	36.37 $\pm$ 4.24	33.66 $\pm$ 2.69
AllGestureWiimoteX	19.91 $\pm$ 1.48	26.51 $\pm$ 4.36	39.40 $\pm$ 2.86	52.31 $\pm$ 2.26	20.23 $\pm$ 4.10	23.34 $\pm$ 2.88	31.00 $\pm$ 1.71	35.34 $\pm$ 1.51
AllGestureWiimoteY	25.54 $\pm$ 4.42	29.43 $\pm$ 1.69	48.11 $\pm$ 2.89	56.34 $\pm$ 3.54	23.11 $\pm$ 3.11	26.46 $\pm$ 2.53	38.97 $\pm$ 1.87	44.00 $\pm$ 2.17
AllGestureWiimoteZ	22.26 $\pm$ 3.96	27.91 $\pm$ 5.21	36.14 $\pm$ 0.97	44.43 $\pm$ 2.91	22.26 $\pm$ 4.37	25.74 $\pm$ 5.60	31.91 $\pm$ 0.57	36.06 $\pm$ 1.75
ArrowHead	48.80 $\pm$ 4.79	57.03 $\pm$ 2.81	73.60 $\pm$ 3.71	81.49 $\pm$ 3.22	44.57 $\pm$ 6.72	47.20 $\pm$ 4.58	61.03 $\pm$ 6.78	58.74 $\pm$ 3.82
BME	66.40 $\pm$ 3.88	79.47 $\pm$ 11.40	97.73 $\pm$ 1.53	99.20 $\pm$ 0.30	66.80 $\pm$ 2.92	74.53 $\pm$ 11.00	93.47 $\pm$ 5.34	93.60 $\pm$ 0.37
Beef	30.00 $\pm$ 8.16	43.33 $\pm$ 5.27	51.33 $\pm$ 6.91	68.00 $\pm$ 1.83	31.33 $\pm$ 8.69	38.67 $\pm$ 6.06	36.67 $\pm$ 7.07	48.00 $\pm$ 3.80
BeetleFly	55.00 $\pm$ 14.58	61.00 $\pm$ 12.45	57.00 $\pm$ 11.51	59.00 $\pm$ 4.18	56.00 $\pm$ 5.48	58.00 $\pm$ 4.47	51.00 $\pm$ 9.62	59.00 $\pm$ 4.18
BirdChicken	68.00 $\pm$ 14.83	69.00 $\pm$ 22.19	85.00 $\pm$ 11.73	89.00 $\pm$ 5.48	74.00 $\pm$ 12.94	76.00 $\pm$ 9.62	86.00 $\pm$ 8.94	88.00 $\pm$ 5.70
CBF	67.51 $\pm$ 4.85	85.91 $\pm$ 4.23	93.76 $\pm$ 3.90	97.71 $\pm$ 0.69	69.18 $\pm$ 4.62	78.13 $\pm$ 11.07	87.40 $\pm$ 4.28	90.04 $\pm$ 1.83
Car	48.00 $\pm$ 5.06	58.67 $\pm$ 3.80	69.67 $\pm$ 3.98	76.33 $\pm$ 5.45	43.33 $\pm$ 12.86	53.00 $\pm$ 7.21	63.33 $\pm$ 8.50	66.33 $\pm$ 3.80
Chinatown	67.17 $\pm$ 13.59	85.31 $\pm$ 4.44	82.45 $\pm$ 9.21	88.05 $\pm$ 3.11	57.26 $\pm$ 10.98	72.36 $\pm$ 12.11	78.02 $\pm$ 7.25	75.39 $\pm$ 3.06
ChlorineConcentration	34.73 $\pm$ 3.81	34.25 $\pm$ 5.73	36.45 $\pm$ 2.05	33.98 $\pm$ 7.65	34.41 $\pm$ 3.06	34.99 $\pm$ 5.96	36.05 $\pm$ 1.89	36.44 $\pm$ 3.56
CinCECGTorso	32.32 $\pm$ 4.12	44.68 $\pm$ 5.88	49.61 $\pm$ 6.52	71.09 $\pm$ 3.22	31.30 $\pm$ 5.37	40.86 $\pm$ 7.99	37.54 $\pm$ 6.84	49.80 $\pm$ 0.39
Coffee	75.71 $\pm$ 13.69	88.57 $\pm$ 4.66	94.29 $\pm$ 1.96	95.71 $\pm$ 2.99	68.57 $\pm$ 11.68	82.14 $\pm$ 15.15	72.86 $\pm$ 20.45	87.86 $\pm$ 9.65
Computers	46.00 $\pm$ 10.27	52.72 $\pm$ 10.03	55.92 $\pm$ 7.37	59.36 $\pm$ 8.46	53.20 $\pm$ 7.38	58.96 $\pm$ 7.32	58.00 $\pm$ 7.81	64.00 $\pm$ 4.03
CricketX	26.26 $\pm$ 5.29	37.33 $\pm$ 3.78	45.74 $\pm$ 2.60	56.77 $\pm$ 3.51	25.59 $\pm$ 5.37	31.90 $\pm$ 2.09	38.97 $\pm$ 3.38	47.74 $\pm$ 2.29
CricketY	25.54 $\pm$ 2.45	34.51 $\pm$ 6.10	45.64 $\pm$ 1.99	54.31 $\pm$ 1.68	25.13 $\pm$ 3.56	33.28 $\pm$ 2.21	36.62 $\pm$ 1.65	41.95 $\pm$ 1.59
CricketZ	29.28 $\pm$ 4.82	33.59 $\pm$ 5.51	49.59 $\pm$ 2.62	57.38 $\pm$ 4.03	28.36 $\pm$ 1.29	29.44 $\pm$ 4.90	42.87 $\pm$ 1.47	46.62 $\pm$ 1.96
DiatomSizeReduction	85.82 $\pm$ 8.55	90.39 $\pm$ 5.62	93.66 $\pm$ 1.97	92.61 $\pm$ 2.00	86.73 $\pm$ 8.56	85.62 $\pm$ 7.19	77.78 $\pm$ 3.00	75.23 $\pm$ 1.32
DistalPhalanxOutlineAgeGroup	40.72 $\pm$ 17.98	67.77 $\pm$ 3.39	69.35 $\pm$ 1.20	70.65 $\pm$ 2.18	43.31 $\pm$ 18.16	70.22 $\pm$ 4.60	70.65 $\pm$ 1.56	71.80 $\pm$ 1.79
DistalPhalanxOutlineCorrect	53.91 $\pm$ 13.35	55.80 $\pm$ 14.95	63.77 $\pm$ 11.63	65.51 $\pm$ 11.85	60.58 $\pm$ 6.43	51.23 $\pm$ 12.26	64.49 $\pm$ 7.76	59.06 $\pm$ 13.65
DistalPhalanxTW	54.53 $\pm$ 1.72	51.37 $\pm$ 8.97	56.40 $\pm$ 4.87	54.10 $\pm$ 3.11	51.51 $\pm$ 3.83	47.48 $\pm$ 13.90	57.27 $\pm$ 3.58	55.11 $\pm$ 1.40
DodgerLoopGame	56.96 $\pm$ 11.69	61.01 $\pm$ 13.03	72.61 $\pm$ 10.25	83.62 $\pm$ 0.65	50.29 $\pm$ 8.64	54.93 $\pm$ 10.24	60.58 $\pm$ 4.46	73.19 $\pm$ 1.36
DodgerLoopWeekend	85.36 $\pm$ 4.30	84.64 $\pm$ 2.73	93.62 $\pm$ 2.37	97.83 $\pm$ 0.00	87.68 $\pm$ 3.80	84.35 $\pm$ 2.83	88.26 $\pm$ 2.14	92.32 $\pm$ 0.40
ECG200	60.60 $\pm$ 6.91	73.00 $\pm$ 3.46	67.40 $\pm$ 8.91	77.40 $\pm$ 4.88	62.80 $\pm$ 12.30	67.80 $\pm$ 5.22	68.60 $\pm$ 2.97	77.60 $\pm$ 2.70
ECG5000	67.18 $\pm$ 10.57	67.68 $\pm$ 9.76	73.89 $\pm$ 8.21	73.78 $\pm$ 6.18	64.82 $\pm$ 0.00	39.33 $\pm$ 0.00	85.18 $\pm$ 0.00	80.11 $\pm$ 0.00
ECGFiveDays	61.77 $\pm$ 6.61	59.35 $\pm$ 9.08	70.96 $\pm$ 3.91	75.28 $\pm$ 1.71	58.07 $\pm$ 0.00	71.31 $\pm$ 0.00	70.85 $\pm$ 0.00	76.42 $\pm$ 0.00
EOGHorizontalSignal	29.06 $\pm$ 4.16	37.35 $\pm$ 4.07	46.35 $\pm$ 2.27	49.17 $\pm$ 1.62	31.22 $\pm$ 0.00	33.15 $\pm$ 0.00	45.30 $\pm$ 0.00	44.20 $\pm$ 0.00
EOGVerticalSignal	27.35 $\pm$ 1.90	31.82 $\pm$ 3.47	36.74 $\pm$ 2.24	42.21 $\pm$ 2.26	22.10 $\pm$ 0.00	33.70 $\pm$ 0.00	34.53 $\pm$ 0.00	38.95 $\pm$ 0.00
Earthquakes	55.54 $\pm$ 11.57	55.25 $\pm$ 14.33	54.68 $\pm$ 9.31	54.68 $\pm$ 7.63	50.36 $\pm$ 0.00	61.87 $\pm$ 0.00	39.57 $\pm$ 0.00	42.45 $\pm$ 0.00
ElectricDevices	33.14 $\pm$ 10.96	47.14 $\pm$ 3.99	47.06 $\pm$ 4.18	49.42 $\pm$ 5.02	17.31 $\pm$ 0.00	46.54 $\pm$ 0.00	47.30 $\pm$ 0.00	47.89 $\pm$ 0.00
EthanolLevel	26.68 $\pm$ 1.86	27.00 $\pm$ 3.11	25.92 $\pm$ 1.40	28.24 $\pm$ 0.50	26.20 $\pm$ 0.00	27.20 $\pm$ 0.00	27.00 $\pm$ 0.00	24.00 $\pm$ 0.00
FaceAll	32.84 $\pm$ 6.86	39.09 $\pm$ 1.28	54.54 $\pm$ 1.26	63.01 $\pm$ 1.34	32.49 $\pm$ 0.00	28.58 $\pm$ 0.00	40.30 $\pm$ 0.00	44.02 $\pm$ 0.00
FaceFour	54.77 $\pm$ 8.25	64.55 $\pm$ 4.91	75.68 $\pm$ 4.37	79.09 $\pm$ 1.90	46.59 $\pm$ 0.00	53.41 $\pm$ 0.00	65.91 $\pm$ 0.00	73.86 $\pm$ 0.00
FacesUCR	50.37 $\pm$ 2.98	55.08 $\pm$ 2.32	68.58 $\pm$ 2.57	79.08 $\pm$ 1.41	46.83 $\pm$ 0.00	46.68 $\pm$ 0.00	54.83 $\pm$ 0.00	58.29 $\pm$ 0.00
FiftyWords	36.48 $\pm$ 2.66	47.30 $\pm$ 2.89	61.58 $\pm$ 2.55	71.34 $\pm$ 2.10	31.87 $\pm$ 0.00	33.41 $\pm$ 0.00	45.05 $\pm$ 0.00	54.51 $\pm$ 0.00
Fish	34.74 $\pm$ 4.91	48.91 $\pm$ 10.25	64.57 $\pm$ 5.92	78.29 $\pm$ 3.92	36.00 $\pm$ 0.00	34.29 $\pm$ 0.00	54.86 $\pm$ 0.00	61.71 $\pm$ 0.00
FordA	50.71 $\pm$ 0.98	53.35 $\pm$ 2.84	54.00 $\pm$ 2.22	57.15 $\pm$ 1.85	51.44 $\pm$ 0.00	51.36 $\pm$ 0.00	50.61 $\pm$ 0.00	48.26 $\pm$ 0.00
FordB	49.75 $\pm$ 2.56	50.42 $\pm$ 2.72	52.32 $\pm$ 2.78	55.60 $\pm$ 1.91	52.84 $\pm$ 0.00	49.75 $\pm$ 0.00	51.60 $\pm$ 0.00	52.84 $\pm$ 0.00
FreezerRegularTrain	61.90 $\pm$ 22.25	79.14 $\pm$ 7.94	85.11 $\pm$ 3.83	89.20 $\pm$ 5.65	76.35 $\pm$ 0.00	69.68 $\pm$ 0.00	81.09 $\pm$ 0.00	78.95 $\pm$ 0.00
FreezerSmallTrain	66.53 $\pm$ 10.64	64.30 $\pm$ 13.28	75.05 $\pm$ 3.10	82.86 $\pm$ 3.49	67.30 $\pm$ 0.00	75.09 $\pm$ 0.00	77.82 $\pm$ 0.00	82.63 $\pm$ 0.00

Table 24: Per-dataset LLM-backbone ablation on 120 shared UCR datasets (continued).

Dataset	ChronoMorph				w/o LLM Backbone			
	1-shot	2-shot	5-shot	10-shot	1-shot	2-shot	5-shot	10-shot
Fungi	79.35 ± 5.30	80.97 ± 3.24	81.18 ± 3.16	79.57 ± 1.20	40.32 ± 0.00	37.10 ± 0.00	40.86 ± 0.00	51.61 ± 0.00
GestureMidAirD1	32.00 ± 3.94	44.92 ± 6.31	60.31 ± 2.35	67.08 ± 1.00	22.31 ± 0.00	36.92 ± 0.00	44.62 ± 0.00	50.00 ± 0.00
GestureMidAirD2	33.85 ± 3.03	43.85 ± 4.03	54.62 ± 1.96	55.69 ± 1.85	25.38 ± 0.00	34.62 ± 0.00	44.62 ± 0.00	42.31 ± 0.00
GestureMidAirD3	18.46 ± 2.61	21.08 ± 1.85	32.15 ± 3.33	33.69 ± 2.63	15.38 ± 0.00	20.00 ± 0.00	20.00 ± 0.00	23.08 ± 0.00
GesturePebbleZ1	35.81 ± 4.48	37.91 ± 6.35	63.84 ± 8.99	75.12 ± 5.19	40.70 ± 0.00	36.05 ± 0.00	49.42 ± 0.00	64.53 ± 0.00
GesturePebbleZ2	33.29 ± 5.85	41.39 ± 4.24	60.89 ± 8.06	75.57 ± 4.64	24.68 ± 0.00	47.47 ± 0.00	54.43 ± 0.00	67.72 ± 0.00
GunPoint	58.00 ± 7.79	79.33 ± 16.25	90.13 ± 10.49	97.47 ± 2.23	60.67 ± 0.00	60.67 ± 0.00	85.33 ± 0.00	90.00 ± 0.00
GunPointAgeSpan	60.51 ± 10.32	64.49 ± 7.90	81.52 ± 9.45	92.72 ± 5.80	61.71 ± 0.00	52.22 ± 0.00	71.52 ± 0.00	71.84 ± 0.00
GunPointMaleVersusFemale	81.01 ± 10.12	87.66 ± 5.66	94.62 ± 6.24	98.54 ± 0.94	64.56 ± 0.00	77.53 ± 0.00	98.10 ± 0.00	97.15 ± 0.00
GunPointOldVersusYoung	57.97 ± 10.78	70.92 ± 5.05	80.44 ± 8.63	86.79 ± 4.71	54.29 ± 0.00	71.11 ± 0.00	76.83 ± 0.00	80.32 ± 0.00
Ham	52.76 ± 9.30	50.86 ± 8.07	60.57 ± 9.91	62.86 ± 3.37	61.90 ± 0.00	64.76 ± 0.00	65.71 ± 0.00	58.10 ± 0.00
HandOutlines	61.73 ± 6.31	64.59 ± 10.14	68.54 ± 6.21	71.51 ± 5.35	62.43 ± 0.00	66.49 ± 0.00	72.16 ± 0.00	55.68 ± 0.00
Haptics	25.65 ± 4.99	26.62 ± 4.64	26.95 ± 4.88	39.22 ± 4.29	19.81 ± 0.00	27.92 ± 0.00	32.79 ± 0.00	33.77 ± 0.00
Herring	50.00 ± 10.00	55.00 ± 4.34	45.00 ± 4.87	55.31 ± 5.01	53.12 ± 0.00	59.38 ± 0.00	45.31 ± 0.00	62.50 ± 0.00
HouseTwenty	62.18 ± 8.04	71.93 ± 10.30	84.37 ± 7.51	89.24 ± 2.81	65.55 ± 0.00	56.30 ± 0.00	62.18 ± 0.00	85.71 ± 0.00
InlineSkate	17.13 ± 2.24	18.98 ± 2.54	28.07 ± 1.19	33.49 ± 2.77	13.45 ± 0.00	19.64 ± 0.00	23.27 ± 0.00	25.45 ± 0.00
InsectEPGRegularTrain	55.34 ± 14.47	68.76 ± 9.60	80.24 ± 3.80	86.59 ± 4.06	54.22 ± 0.00	57.03 ± 0.00	75.90 ± 0.00	77.51 ± 0.00
InsectEPGSmallTrain	50.76 ± 7.86	67.79 ± 2.45	83.05 ± 2.80	85.06 ± 2.08	34.14 ± 0.00	67.47 ± 0.00	71.49 ± 0.00	77.51 ± 0.00
InsectWingbeatSound	25.27 ± 2.20	31.49 ± 3.72	41.44 ± 2.25	51.36 ± 2.05	25.86 ± 0.00	30.00 ± 0.00	35.30 ± 0.00	38.43 ± 0.00
ItalyPowerDemand	58.52 ± 15.27	65.03 ± 13.70	84.26 ± 5.94	92.19 ± 2.62	86.98 ± 0.00	53.35 ± 0.00	85.33 ± 0.00	92.03 ± 0.00
LargeKitchenAppliances	45.44 ± 5.60	42.56 ± 5.00	50.13 ± 3.07	55.09 ± 5.30	44.53 ± 0.00	43.20 ± 0.00	50.93 ± 0.00	50.67 ± 0.00
Lightning2	59.67 ± 8.87	63.28 ± 8.25	69.51 ± 10.47	77.05 ± 4.18	55.74 ± 0.00	63.93 ± 0.00	62.30 ± 0.00	68.85 ± 0.00
Lightning7	41.64 ± 7.60	46.03 ± 6.17	66.30 ± 3.83	66.85 ± 7.34	47.95 ± 0.00	58.90 ± 0.00	63.01 ± 0.00	69.86 ± 0.00
Mallat	68.24 ± 7.54	80.67 ± 4.06	90.88 ± 1.15	89.72 ± 1.50	58.68 ± 0.00	70.02 ± 0.00	66.10 ± 0.00	60.00 ± 0.00
Meat	55.67 ± 9.62	57.67 ± 17.30	64.00 ± 11.82	64.00 ± 6.08	51.67 ± 0.00	56.67 ± 0.00	75.00 ± 0.00	73.33 ± 0.00
MedicalImages	27.66 ± 4.41	31.63 ± 5.59	42.03 ± 2.75	54.68 ± 2.10	25.53 ± 0.00	35.39 ± 0.00	38.68 ± 0.00	53.29 ± 0.00
MelbournePedestrian	39.16 ± 5.22	49.78 ± 4.68	65.81 ± 3.77	70.88 ± 1.49	37.15 ± 0.00	41.94 ± 0.00	52.85 ± 0.00	65.23 ± 0.00
MiddlePhalanxOutlineAgeGroup	39.48 ± 7.40	42.99 ± 12.95	49.74 ± 4.72	51.04 ± 7.11	43.51 ± 0.00	31.17 ± 0.00	60.39 ± 0.00	52.60 ± 0.00
MiddlePhalanxOutlineCorrect	51.68 ± 5.00	48.87 ± 3.56	48.45 ± 7.93	59.18 ± 5.65	41.92 ± 0.00	54.98 ± 0.00	62.20 ± 0.00	64.26 ± 0.00
MiddlePhalanxTW	37.14 ± 16.84	44.81 ± 7.78	43.64 ± 5.71	44.42 ± 6.27	27.27 ± 0.00	46.75 ± 0.00	22.08 ± 0.00	48.05 ± 0.00
MixedShapesRegularTrain	56.06 ± 5.34	67.98 ± 4.06	76.68 ± 5.11	79.60 ± 1.74	48.62 ± 0.00	69.44 ± 0.00	68.49 ± 0.00	74.68 ± 0.00
MixedShapesSmallTrain	63.30 ± 3.34	68.35 ± 3.26	78.71 ± 1.70	81.61 ± 2.47	64.54 ± 0.00	65.81 ± 0.00	61.90 ± 0.00	71.88 ± 0.00
MoteStrain	54.36 ± 12.77	58.56 ± 15.12	80.37 ± 4.50	89.23 ± 1.53	68.61 ± 0.00	50.00 ± 0.00	82.27 ± 0.00	83.39 ± 0.00
NonInvasiveFetalECGThorax1	39.16 ± 2.28	50.58 ± 2.13	67.42 ± 0.56	76.84 ± 1.40	24.53 ± 0.00	36.49 ± 0.00	46.41 ± 0.00	51.30 ± 0.00
NonInvasiveFetalECGThorax2	47.48 ± 2.06	59.72 ± 2.76	75.31 ± 0.82	82.53 ± 1.33	39.64 ± 0.00	46.87 ± 0.00	55.73 ± 0.00	66.77 ± 0.00
OSULeaf	29.83 ± 5.82	36.28 ± 6.97	49.92 ± 6.46	59.42 ± 3.58	33.47 ± 0.00	29.75 ± 0.00	43.80 ± 0.00	49.17 ± 0.00
OliveOil	28.00 ± 14.26	23.33 ± 9.72	28.67 ± 11.69	40.00 ± 0.00	16.67 ± 0.00	40.00 ± 0.00	16.67 ± 0.00	40.00 ± 0.00
PLAID	32.10 ± 5.72	40.19 ± 5.06	52.96 ± 5.39	59.89 ± 3.28	21.60 ± 0.00	36.13 ± 0.00	53.63 ± 0.00	58.66 ± 0.00
PhalangesOutlinesCorrect	51.93 ± 9.54	50.84 ± 2.93	53.61 ± 8.69	55.73 ± 8.07	61.89 ± 0.00	54.55 ± 0.00	36.60 ± 0.00	65.27 ± 0.00
Phoneme	6.87 ± 1.15	10.02 ± 0.59	15.63 ± 1.56	21.95 ± 1.27	7.12 ± 0.00	10.50 ± 0.00	15.82 ± 0.00	17.25 ± 0.00
PickupGestureWiimoteZ	47.20 ± 7.29	58.80 ± 3.03	68.00 ± 2.45	70.40 ± 2.61	26.00 ± 0.00	54.00 ± 0.00	60.00 ± 0.00	52.00 ± 0.00
PigAirwayPressure	5.48 ± 2.19	9.81 ± 0.43	9.81 ± 0.73	9.62 ± 0.48	5.77 ± 0.00	7.21 ± 0.00	7.69 ± 0.00	4.81 ± 0.00
PigArtPressure	15.38 ± 4.07	30.67 ± 1.38	33.08 ± 1.33	31.73 ± 1.92	17.79 ± 0.00	25.00 ± 0.00	23.56 ± 0.00	24.52 ± 0.00
PigCVP	11.44 ± 2.08	19.33 ± 0.99	19.52 ± 0.55	19.52 ± 1.34	10.58 ± 0.00	16.83 ± 0.00	15.87 ± 0.00	18.27 ± 0.00

Table 24: Per-dataset LLM-backbone ablation on 120 shared UCR datasets (continued).

Dataset	ChronoMorph				w/o LLM Backbone			
	1-shot	2-shot	5-shot	10-shot	1-shot	2-shot	5-shot	10-shot
Plane	93.90 $\pm$ 6.92	99.24 $\pm$ 0.80	99.24 $\pm$ 1.24	99.24 $\pm$ 1.24	91.43 $\pm$ 0.00	90.48 $\pm$ 0.00	100.00 $\pm$ 0.00	99.05 $\pm$ 0.00
PowerCons	60.89 $\pm$ 11.09	66.78 $\pm$ 15.52	76.11 $\pm$ 6.55	82.33 $\pm$ 5.46	77.78 $\pm$ 0.00	48.89 $\pm$ 0.00	52.22 $\pm$ 0.00	78.33 $\pm$ 0.00
ProximalPhalanxOutlineAgeGroup	48.98 $\pm$ 23.27	66.24 $\pm$ 17.89	76.88 $\pm$ 5.79	79.80 $\pm$ 3.30	78.54 $\pm$ 0.00	81.46 $\pm$ 0.00	83.90 $\pm$ 0.00	76.10 $\pm$ 0.00
ProximalPhalanxOutlineCorrect	52.23 $\pm$ 12.03	59.93 $\pm$ 12.82	65.84 $\pm$ 4.70	60.07 $\pm$ 4.20	59.79 $\pm$ 0.00	64.26 $\pm$ 0.00	62.89 $\pm$ 0.00	62.89 $\pm$ 0.00
ProximalPhalanxTW	55.22 $\pm$ 6.63	57.76 $\pm$ 7.59	66.24 $\pm$ 3.19	69.66 $\pm$ 3.47	60.98 $\pm$ 0.00	39.02 $\pm$ 0.00	73.17 $\pm$ 0.00	45.37 $\pm$ 0.00
RefrigerationDevices	29.76 $\pm$ 11.02	33.39 $\pm$ 7.13	40.96 $\pm$ 5.59	41.23 $\pm$ 4.90	31.20 $\pm$ 0.00	26.40 $\pm$ 0.00	41.60 $\pm$ 0.00	40.00 $\pm$ 0.00
Rock	39.20 $\pm$ 13.75	64.00 $\pm$ 7.07	76.00 $\pm$ 3.16	76.40 $\pm$ 2.97	20.00 $\pm$ 0.00	66.00 $\pm$ 0.00	56.00 $\pm$ 0.00	64.00 $\pm$ 0.00
ScreenType	32.53 $\pm$ 4.84	35.63 $\pm$ 4.40	34.51 $\pm$ 4.14	37.12 $\pm$ 3.93	33.60 $\pm$ 0.00	33.60 $\pm$ 0.00	35.73 $\pm$ 0.00	33.33 $\pm$ 0.00
SemgHandGenderCh2	51.90 $\pm$ 14.92	59.73 $\pm$ 10.87	66.80 $\pm$ 7.25	63.80 $\pm$ 10.73	42.00 $\pm$ 0.00	68.17 $\pm$ 0.00	60.33 $\pm$ 0.00	70.67 $\pm$ 0.00
SemgHandMovementCh2	28.13 $\pm$ 4.95	31.16 $\pm$ 4.70	36.31 $\pm$ 5.92	45.56 $\pm$ 2.93	28.89 $\pm$ 0.00	26.00 $\pm$ 0.00	32.00 $\pm$ 0.00	37.78 $\pm$ 0.00
SemgHandSubjectCh2	37.42 $\pm$ 8.82	43.20 $\pm$ 6.72	54.31 $\pm$ 3.37	66.71 $\pm$ 4.45	41.56 $\pm$ 0.00	44.22 $\pm$ 0.00	37.33 $\pm$ 0.00	56.89 $\pm$ 0.00
ShakeGestureWiimoteZ	60.40 $\pm$ 7.92	81.60 $\pm$ 2.61	92.00 $\pm$ 0.00	91.60 $\pm$ 2.97	70.00 $\pm$ 0.00	72.00 $\pm$ 0.00	82.00 $\pm$ 0.00	84.00 $\pm$ 0.00
ShapeletSim	62.44 $\pm$ 7.44	67.44 $\pm$ 10.92	92.00 $\pm$ 3.88	96.67 $\pm$ 1.96	52.78 $\pm$ 0.00	61.11 $\pm$ 0.00	86.67 $\pm$ 0.00	96.67 $\pm$ 0.00
ShapesAll	50.20 $\pm$ 3.03	63.03 $\pm$ 2.22	76.20 $\pm$ 1.83	83.13 $\pm$ 0.32	46.50 $\pm$ 0.00	52.67 $\pm$ 0.00	64.00 $\pm$ 0.00	71.83 $\pm$ 0.00
SmallKitchenAppliances	42.88 $\pm$ 15.27	55.68 $\pm$ 8.74	51.63 $\pm$ 12.25	57.60 $\pm$ 5.56	54.93 $\pm$ 0.00	56.00 $\pm$ 0.00	45.07 $\pm$ 0.00	54.40 $\pm$ 0.00
SmoothSubspace	68.53 $\pm$ 8.95	85.07 $\pm$ 5.20	88.00 $\pm$ 2.87	89.47 $\pm$ 2.60	58.67 $\pm$ 0.00	77.33 $\pm$ 0.00	76.00 $\pm$ 0.00	87.33 $\pm$ 0.00
SonyAIBORobotSurface1	57.20 $\pm$ 8.62	71.85 $\pm$ 8.28	84.86 $\pm$ 2.67	86.09 $\pm$ 3.97	67.22 $\pm$ 0.00	69.72 $\pm$ 0.00	70.88 $\pm$ 0.00	78.04 $\pm$ 0.00
SonyAIBORobotSurface2	66.09 $\pm$ 10.78	68.21 $\pm$ 5.28	77.21 $\pm$ 3.67	84.11 $\pm$ 2.29	62.12 $\pm$ 0.00	67.68 $\pm$ 0.00	73.87 $\pm$ 0.00	75.87 $\pm$ 0.00
StarLightCurves	63.03 $\pm$ 14.23	74.47 $\pm$ 15.08	81.86 $\pm$ 7.93	85.26 $\pm$ 3.54	64.24 $\pm$ 0.00	77.25 $\pm$ 0.00	85.65 $\pm$ 0.00	82.39 $\pm$ 0.00
Strawberry	51.46 $\pm$ 5.21	59.41 $\pm$ 6.38	58.05 $\pm$ 7.04	75.84 $\pm$ 0.99	54.05 $\pm$ 0.00	54.59 $\pm$ 0.00	52.97 $\pm$ 0.00	59.73 $\pm$ 0.00
SwedishLeaf	57.98 $\pm$ 2.72	64.99 $\pm$ 3.83	79.78 $\pm$ 2.56	85.89 $\pm$ 1.27	52.96 $\pm$ 0.00	56.48 $\pm$ 0.00	72.64 $\pm$ 0.00	74.24 $\pm$ 0.00
Symbols	80.94 $\pm$ 7.54	82.69 $\pm$ 3.43	88.04 $\pm$ 1.30	89.33 $\pm$ 0.82	86.03 $\pm$ 0.00	80.10 $\pm$ 0.00	91.16 $\pm$ 0.00	76.98 $\pm$ 0.00
SyntheticControl	84.20 $\pm$ 6.54	87.93 $\pm$ 3.95	95.07 $\pm$ 1.62	97.33 $\pm$ 0.85	75.00 $\pm$ 0.00	80.67 $\pm$ 0.00	90.33 $\pm$ 0.00	90.67 $\pm$ 0.00
ToeSegmentation1	58.16 $\pm$ 3.70	60.26 $\pm$ 8.54	70.53 $\pm$ 10.39	80.53 $\pm$ 2.75	67.11 $\pm$ 0.00	67.98 $\pm$ 0.00	67.54 $\pm$ 0.00	76.75 $\pm$ 0.00
ToeSegmentation2	75.23 $\pm$ 12.13	76.15 $\pm$ 12.44	78.00 $\pm$ 6.36	80.31 $\pm$ 4.70	69.23 $\pm$ 0.00	70.00 $\pm$ 0.00	66.92 $\pm$ 0.00	70.77 $\pm$ 0.00
Trace	99.20 $\pm$ 1.79	98.40 $\pm$ 3.05	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	97.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
UWaveGestureLibraryZ	46.44 $\pm$ 2.61	52.88 $\pm$ 2.03	55.54 $\pm$ 2.61	60.65 $\pm$ 2.84	38.11 $\pm$ 0.00	41.57 $\pm$ 0.00	51.20 $\pm$ 0.00	56.31 $\pm$ 0.00
Wafer	46.27 $\pm$ 7.11	48.01 $\pm$ 15.62	71.65 $\pm$ 9.58	82.53 $\pm$ 8.58	68.14 $\pm$ 0.00	67.07 $\pm$ 0.00	86.02 $\pm$ 0.00	80.50 $\pm$ 0.00
Wine	51.11 $\pm$ 7.48	48.89 $\pm$ 5.34	51.48 $\pm$ 2.03	49.26 $\pm$ 2.11	61.11 $\pm$ 0.00	57.41 $\pm$ 0.00	50.00 $\pm$ 0.00	48.15 $\pm$ 0.00
WordSynonyms	20.91 $\pm$ 2.73	32.92 $\pm$ 2.64	47.05 $\pm$ 3.87	58.90 $\pm$ 2.04	13.64 $\pm$ 0.00	21.79 $\pm$ 0.00	25.24 $\pm$ 0.00	30.09 $\pm$ 0.00
Worms	24.68 $\pm$ 8.95	38.70 $\pm$ 4.63	44.42 $\pm$ 6.26	48.57 $\pm$ 4.91	42.86 $\pm$ 0.00	35.06 $\pm$ 0.00	42.86 $\pm$ 0.00	44.16 $\pm$ 0.00
WormsTwoClass	47.53 $\pm$ 10.69	54.03 $\pm$ 7.72	54.81 $\pm$ 7.59	56.36 $\pm$ 3.85	54.55 $\pm$ 0.00	45.45 $\pm$ 0.00	51.95 $\pm$ 0.00	55.84 $\pm$ 0.00
Yoga	50.42 $\pm$ 4.51	50.64 $\pm$ 4.90	51.31 $\pm$ 4.34	57.73 $\pm$ 2.05	51.50 $\pm$ 0.00	55.20 $\pm$ 0.00	53.20 $\pm$ 0.00	53.63 $\pm$ 0.00
Average	47.93	54.98	63.27	68.48	46.95	51.20	57.06	61.01

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